BuzzCalculus

完整題庫題本・Complete Problem Set

總題數	**1095** 題
極限	159 題
微分	284 題
積分	471 題
級數	181 題
編譯	XeLaTeX + XeCJK

姓名 ______________ 日期 ______________ 分數 ______________

本題本由 BuzzCalculus 題庫自動產生，含全部題目、答案速查與解法提示。題目保留原始 LaTeX prompt，答案採 WebWork 輸入格式。

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1 **lim-001**
R1 Warm-up • 25s • rank-1 / beginner-friendly

$$\lim_{x\to 0}\frac{\sin x}{x}$$

2 **lim-002**
R1 Warm-up • 30s • rank-1 / beginner-friendly

$$\lim_{x\to 0}\frac{1-\cos x}{x^2}$$

3 **lim-003**
R1 Warm-up • 25s • rank-1 / beginner-friendly

$$\lim_{x\to 2}\frac{x^2-4}{x-2}$$

4 **lim-004**
R1 Warm-up • 30s • rank-1 / beginner-friendly

$$\lim_{x\to\infty}\frac{3x^2-x+7}{2x^2+5}$$

5 **lim-005**
R2 Basic • 30s • rank-2 / beginner-friendly

$$\lim_{x\to 0}\frac{e^x-1}{x}$$

6 **lim-006**
R2 Basic • 35s • rank-2 / beginner-friendly

$$\lim_{x\to 0}\frac{\ln(1+3x)}{x}$$

7 **lim-007**
R2 Basic • 35s • rank-2 / beginner-friendly

$$\lim_{x\to 0}\frac{\tan(4x)}{\sin(2x)}$$

8 **lim-008**
R2 Basic • 45s • rank-2 / beginner-friendly

$$\lim_{x\to\infty} x\left(\sqrt{x^2+1}-x\right)$$

9 **lim-009**
R3 Standard • 45s • rank-3

$$\lim_{x\to 0}\frac{\sin x-x}{x^3}$$

10 **lim-010**
R3 Standard • 50s • rank-3

$$\lim_{x\to 0}\frac{e^{2x}-1-2x}{x^2}$$

11 **lim-011**
R3 Standard • 45s • rank-3

$$\lim_{x\to 0+} x\ln x$$

12 **lim-012**
R3 Standard • 65s • rank-3

$$\lim_{x\to 0}\frac{\ln(1+x)-x+\frac{x^2}{2}}{x^3}$$

13 **lim-013**
R3 Standard • 75s • rank-3

$$\lim_{x\to 0}\frac{\tan x-\sin x}{x^3}$$

14 **lim-014**
R4 Advanced • 75s • rank-4

$$\lim_{x\to 0}\frac{\ln(1+x)-\sin x}{x^2}$$

15 **lim-015**
R3 Standard • 70s • rank-3

$$\lim_{x\to 0}\frac{\sec x-1}{x^2}$$

16 **lim-016**
R4 Advanced • 90s • rank-4

$$\lim_{x\to\infty} x\left(\sqrt{x^2+x}-x-\frac{1}{2}\right)$$

17 **lim-017**
R3 Standard • 75s • rank-3

$$\lim_{x\to 0}\frac{e^x-e^{-x}-2x}{x^3}$$

18 **lim-018**
R3 Standard • 70s • rank-3

$$\lim_{x\to 0}\frac{\sin x-x\cos x}{x^3}$$

19 **lim-019**
R4 Advanced • 95s • rank-4

$$\lim_{x\to 0}\frac{(1+x)^{1/x}-e}{x}$$

20 **lim-020**
R3 Standard • 70s • rank-3

$$\lim_{x\to 0}\frac{\arctan x-x}{x^3}$$

21 **td-lim-001**

R6 Boss+ • 90s • rank-6 / boss-rank

$$\lim_{n\to\infty} \prod_{k=1}^{n} \left(1 + \frac{k}{n^2}\right)$$

22 **td-lim-002**

R6 Boss+ • 120s • rank-6 / boss-rank

$$\lim_{n\to\infty} n \left(\prod_{k=1}^{n} \left(1 + \frac{k}{n^2}\right) - \sqrt{e} \right)$$

23 **td-lim-003**

R6 Boss+ • 110s • rank-6 / boss-rank

$$\lim_{n\to\infty} n^2 \left(\left(1 + \frac{1}{n}\right)^n - e + \frac{e}{2n} \right)$$

24 **td-lim-004**

R4 Advanced • 80s • rank-4

$$\lim_{x\to 0} \frac{(1+x)^x - 1 - x^2}{x^3}$$

25 **lim-021**

R2 Basic • 55s • taylor / trig-limit

$$\lim_{x\to 0} \frac{\tan x - x}{x^3}$$

26 **lim-022**

R2 Basic • 55s • rationalize / infinity

$$\lim_{x\to\infty} \left(\sqrt{x^2 + 3x} - x\right)$$

27 **lim-023**

R2 Basic • 50s • taylor / exponential

$$\lim_{x\to 0} \frac{e^{2x} - 1 - 2x}{x^2}$$

28 **lim-024**

R2 Basic • 50s • taylor / log

$$\lim_{x\to 0} \frac{\log(1+x) - x}{x^2}$$

29 **lim-025**

R4 Advanced • 55s • multivariable / standard-limit

$$\lim_{(x,y)\to(0,0)} \frac{\sin(xy)}{xy}$$

30 **lim-026**

R4 Advanced • 70s • multivariable / squeeze

$$\lim_{(x,y)\to(0,0)} \frac{x^2 y}{x^2 + y^2}$$

31 **lim-027**

R4 Advanced • 70s • multivariable / path-test

$$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2 + y^2}$$

32 **lim-028**

R4 Advanced • 75s • multivariable / rationalize

$$\lim_{(x,y)\to(0,0)} \frac{\sqrt{1 + x^2 + y^2} - 1}{x^2 + y^2}$$

33 **lim-029**

R4 Advanced • 60s • multivariable / squeeze

$$\lim_{(x,y)\to(0,0)} \frac{x^4 + y^4}{x^2 + y^2}$$

34 **lim-030**

R2 Basic • 50s • standard-limit / trig-limit

$$\lim_{x\to 0} \frac{1 - \cos(3x)}{x^2}$$

35 **lim-031**

R4 Advanced • 70s • multivariable / squeeze

$$\lim_{(x,y)\to(0,0)} \frac{x^2 y^2}{x^2 + y^2}$$

36 **lim-032**

R4 Advanced • 70s • multivariable / path-test

$$\lim_{(x,y)\to(0,0)} \frac{x^2 - y^2}{x^2 + y^2}$$

37 **lim-033**

R4 Advanced • 60s • multivariable / standard-limit

$$\lim_{(x,y)\to(0,0)} \frac{e^{x+y} - 1}{x + y}$$

38 **lim-034**

R4 Advanced • 70s • multivariable / squeeze

$$\lim_{(x,y)\to(0,0)} \frac{x^3}{x^2 + y^2}$$

39 **lim-035**

R2 Basic • 50s • standard-limit / trig-limit

$$\lim_{x\to 0} \frac{\sin(5x)}{\sin(2x)}$$

40 **lim-036**

R3 Standard • 70s • taylor / log

$$\lim_{x\to 0} \frac{\log(1+x+x^2)-x}{x^2}$$

41 **lim-037**

R2 Basic • 50s • rationalize / rank-2

$$\lim_{x\to 0} \frac{\sqrt{1+4x}-1}{x}$$

42 **lim-038**

R3 Standard • 70s • taylor / trig-limit

$$\lim_{x\to 0} \frac{\sin x - x\cos x}{x^3}$$

43 **lim-039**

R4 Advanced • 80s • multivariable / path-test

$$\lim_{(x,y)\to(0,0)} \frac{x^2y}{x^4+y^2}$$

44 **lim-040**

R2 Basic • 55s • standard-limit / log

$$\lim_{x\to\infty} x\log\left(1+\frac{2}{x}\right)$$

45 **gap-tech-001**

R2 Basic • 45s • technique-recognition / rationalize

Technique for $\lim\limits_{x\to\infty}\left(\sqrt{x^2+3x}-x\right)$

46 **gap-tech-002**

R2 Basic • 45s • technique-recognition / taylor

Technique for $\lim\limits_{x\to 0}\dfrac{e^{2x}-1-2x}{x^2}$

47 **gap-tech-008**

R2 Basic • 55s • technique-recognition / multivariable-limit

Technique for $\lim\limits_{(x,y)\to(0,0)}\dfrac{x^2y}{x^4+y^2}$

48 **mob-tech-001**

R2 Basic • 45s • technique-sprint / technique-recognition

$\lim\limits_{x\to\infty}(\sqrt{x^2+3x}-x)$ Use which technique?

49 **mob-tech-002**

R2 Basic • 45s • technique-sprint / technique-recognition

$\lim\limits_{x\to 0}\dfrac{e^{2x}-1-2x}{x^2}$ Use which technique?

50 **mob-tech-009**

R2 Basic • 45s • technique-sprint / technique-recognition

$\lim\limits_{(x,y)\to(0,0)}\dfrac{xy}{x^2+y^2}$ Use which technique?

51 **mob-trap-009**

R2 Basic • 45s • trap-drill / rank-2

$\dfrac{0}{0}$ limit form Main trap?

52 **mob-limtrap-001**

R3 Standard • 70s • limit-trap / taylor

$$\lim_{x\to 0} \frac{\sin x - x}{x^3}$$

53 **mob-limtrap-002**

R3 Standard • 70s • limit-trap / taylor

$$\lim_{x\to 0} \frac{\tan x - x}{x^3}$$

54 **mob-limtrap-003**

R3 Standard • 70s • limit-trap / taylor

$$\lim_{x\to 0} \frac{1-\cos x - x^2/2}{x^4}$$

55 **mob-limtrap-004**

R3 Standard • 70s • limit-trap / taylor

$$\lim_{x\to 0} \frac{\log(1+x)-x+x^2/2}{x^3}$$

56 **mob-limtrap-005**

R3 Standard • 70s • limit-trap / taylor

$$\lim_{x\to 0} \frac{e^x-1-x-x^2/2}{x^3}$$

57 **mob-limtrap-006**

R3 Standard • 70s • limit-trap / taylor

$$\lim_{x\to 0} \frac{\sqrt{1+x}-1-x/2}{x^2}$$

58 **mob-limtrap-007**

R3 Standard • 70s • limit-trap / taylor

$$\lim_{(x,y)\to(0,0)} \frac{x^2y^2}{x^2+y^2}$$

59 **mob-limtrap-008**

R3 Standard • 70s • limit-trap / taylor

$$\lim_{(x,y)\to(0,0)} \sqrt{x^2+y^2}\sin\frac{1}{\sqrt{x^2+y^2}}$$

60 **mob-limtrap-009**

R3 Standard • 45s • limit-trap / path-test

$$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2}$$

61 **mob-limtrap-010**

R3 Standard • 45s • limit-trap / path-test

$$\lim_{(x,y)\to(0,0)} \frac{x^2-y^2}{x^2+y^2}$$

62 **rel-basic-001**

R1 Warm-up • 45s • standard-limit / beginner-foundation

$$\lim_{x\to 0} \frac{\sin(5x)}{x}$$

63 **rel-basic-002**

R1 Warm-up • 45s • standard-limit / beginner-foundation

$$\lim_{x\to 0} \frac{1-\cos(3x)}{x^2}$$

64 **rel-basic-003**

R1 Warm-up • 45s • standard-limit / beginner-foundation

$$\lim_{x\to 0} \frac{e^{4x}-1}{x}$$

65 **rel-basic-004**

R1 Warm-up • 45s • standard-limit / beginner-foundation

$$\lim_{x\to 0} \frac{\log(1+2x)}{x}$$

66 **rel-basic-005**

R1 Warm-up • 45s • rationalize / beginner-foundation

$$\lim_{x\to 0} \frac{\sqrt{1+x}-1}{x}$$

67 **rel-basic-006**

R1 Warm-up • 45s • factorization / beginner-foundation

$$\lim_{x\to 2} \frac{x^2-4}{x-2}$$

68 **rel-adv-001**

R3 Standard • 75s • taylor / limit-trap

$$\lim_{x\to 0} \frac{\sin x - x + x^3/6}{x^5}$$

69 **rel-adv-002**

R3 Standard • 75s • taylor / limit-trap

$$\lim_{x\to 0} \frac{e^x - 1 - x - x^2/2 - x^3/6}{x^4}$$

70 **rel-adv-003**

R3 Standard • 75s • taylor / limit-trap

$$\lim_{x\to 0} \frac{\log(1+x) - x + x^2/2 - x^3/3}{x^4}$$

71 **rel-adv-004**

R3 Standard • 75s • taylor / limit-trap

$$\lim_{x\to 0} \frac{\tan x - x - x^3/3}{x^5}$$

72 **rel-boss-019**

R5 Boss • 110s • asymptotic / limit-trap

$$\lim_{n\to \infty} n\left(\log(n+1) - \log n\right)$$

73 **rel-hard-lim-001**

R5 Boss • 95s • taylor / limit-trap

$$\lim_{x\to 0} \frac{\sin x - x + x^3/6 - x^5/120}{x^7}$$

74 **rel-hard-lim-002**

R5 Boss • 95s • taylor / limit-trap

$$\lim_{x\to 0} \frac{\tan x - x - x^3/3 - 2x^5/15}{x^7}$$

75 **rel-hard-lim-003**

R5 Boss • 95s • taylor / limit-trap

$$\lim_{x\to 0} \frac{\log(1+x) - x + x^2/2 - x^3/3 + x^4/4}{x^5}$$

76 **rel-hard-lim-004**

R5 Boss • 95s • taylor / limit-trap

$$\lim_{x\to 0} \frac{e^x + e^{-x} - 2 - x^2 - x^4/12}{x^6}$$

77 **rel-hard-lim-005**

R5 Boss • 95s • taylor / limit-trap

$$\lim_{x\to 0} \frac{(1+x)^{1/x} - e}{x}$$

78 **rel-hard-lim-007**

R5 Boss • 95s • taylor / limit-trap

$$\lim_{(x,y)\to(0,0)} \frac{1-\cos(xy)}{x^2y^2}$$

79 **rel-hard-lim-008**

R5 Boss • 95s • taylor / limit-trap

$$\lim_{(x,y)\to(0,0)} \frac{e^{x^2+y^2}-1}{x^2+y^2}$$

80 **rel-hard-lim-009**

R5 Boss • 95s • taylor / limit-trap

$$\lim_{x\to 0} \frac{\arctan x - x + x^3/3}{x^5}$$

81 **rel-hard-lim-010**

R5 Boss • 95s • taylor / limit-trap

$$\lim_{n\to\infty} n^2\left(\left(1+\frac{1}{n}\right)^n - e + \frac{e}{2n}\right)$$

82 **rel-hard-lim-006**

R5 Boss • 80s • path-test / multivariable

$$\lim_{(x,y)\to(0,0)} \frac{x^2y^2}{(x^2+y^2)^2}$$

83 **exam-lim-001**

R5 Boss • 180s • taylor / trig-limit

$$\lim_{x\to 0} \frac{\sin(3x) - 3x + \frac{9}{2}x^3}{x^5}$$

84 **exam-lim-002**

R5 Boss • 180s • taylor / log

$$\lim_{x\to 0} \frac{\log(1+2x) - 2x + 2x^2}{x^3}$$

85 **exam-lim-003**

R5 Boss • 180s • taylor / exponential

$$\lim_{x\to 0} \frac{e^x - \cos x - \sin x}{x^2}$$

86 **exam-lim-004**

R5 Boss • 180s • taylor / trig-limit

$$\lim_{x\to 0} \frac{\tan x - x}{x^3}$$

87 **exam-lim-005**

R5 Boss • 180s • trig-limit / exam-style

$$\lim_{x\to 0} \frac{1-\cos(2x)}{x^2}$$

88 **exam-lim-006**

R5 Boss • 180s • taylor / radical

$$\lim_{x\to 0} \frac{\sqrt{1+4x} - 1 - 2x}{x^2}$$

89 **exam-lim-007**

R5 Boss • 180s • taylor / infinity-limit

$$\lim_{x\to\infty} x^2\left(\sqrt{1+\frac{3}{x}} - 1 - \frac{3}{2x}\right)$$

90 **exam-lim-008**

R5 Boss • 180s • rational / exam-style

$$\lim_{x\to\infty} \frac{x^2+1}{3x^2-2x}$$

91 **exam-lim-009**

R5 Boss • 180s • taylor / trig-limit

$$\lim_{x\to 0} \frac{x - \sin x}{x(1-\cos x)}$$

92 **exam-lim-010**

R5 Boss • 180s • taylor / inverse-trig

$$\lim_{x\to 0} \frac{\arctan(2x) - 2x}{x^3}$$

93 **exam-lim-011**

R5 Boss • 180s • log-limit / exam-style

$$\lim_{x\to 0^+} x^x$$

94 **exam-lim-012**
R5 Boss • 180s • exponential-limit / exam-style

$$\lim_{x\to\infty}\left(1+\frac{2}{x}\right)^{3x}$$

95 **exam-lim-013**
R5 Boss • 180s • taylor / exponential-limit

$$\lim_{x\to 0}\left(\frac{\sin x}{x}\right)^{1/x^2}$$

96 **exam-lim-014**
R5 Boss • 180s • taylor / log

$$\lim_{x\to 0}\frac{\log(\cos x)}{x^2}$$

97 **exam-lim-015**
R5 Boss • 180s • chain-rule / exponential-limit

$$\lim_{x\to 0}\frac{e^{\sin x}-1}{x}$$

98 **exam-lim-016**
R5 Boss • 180s • taylor / exponential-limit

$$\lim_{x\to 0}\frac{e^{\sin x}-1-x}{x^2}$$

99 **exam-lim-017**
R5 Boss • 180s • taylor / exam-style

$$\lim_{x\to 0}\frac{\cos x-e^{-x^2/2}}{x^4}$$

100 **exam-lim-018**
R5 Boss • 180s • taylor / trig-limit

$$\lim_{x\to 0}\frac{\tan x-\sin x}{x^3}$$

101 **exam-lim-019**
R5 Boss • 180s • log-limit / exam-style

$$\lim_{x\to\infty} x\left(\log(x+1)-\log x\right)$$

102 **exam-lim-020**
R5 Boss • 180s • taylor / inverse-trig

$$\lim_{x\to 0}\frac{\arcsin x-x}{x^3}$$

103 **uni-lim-001**
R6 Boss+ • 210s • taylor / trig-limit

$$\lim_{x\to 0}\frac{\sin x-x+\frac{x^3}{6}}{x^5}$$

104 **uni-lim-002**
R6 Boss+ • 210s • taylor / log

$$\lim_{x\to 0}\frac{\log(1+x)-x+\frac{x^2}{2}-\frac{x^3}{3}}{x^4}$$

105 **uni-lim-003**
R6 Boss+ • 210s • taylor / exponential

$$\lim_{x\to 0}\frac{e^{x^2}-\cos x}{x^2}$$

106 **uni-lim-004**
R6 Boss+ • 210s • taylor / trig-limit

$$\lim_{x\to 0}\frac{\tan x-\sin x}{x^3}$$

107 **uni-lim-005**
R6 Boss+ • 210s • taylor / radical

$$\lim_{x\to 0}\frac{\sqrt{1+x}-\sqrt{1-x}-x}{x^3}$$

108 **uni-lim-006**
R6 Boss+ • 210s • log-limit / infinity-limit

$$\lim_{x\to\infty} x\left((x+1)\log\left(1+\frac{1}{x}\right)-1\right)$$

109 **uni-lim-007**

R6 Boss+ • 210s • taylor / trig-limit

$$\lim_{x\to 0}\frac{\sin(2x)-2\sin x}{x^3}$$

110 **uni-lim-008**

R6 Boss+ • 210s • log-limit / exponential-limit

$$\lim_{x\to 0}\frac{(1+3x)^{1/x}-e^3}{x}$$

111 **uni-lim-009**

R6 Boss+ • 210s • taylor / hyperbolic

$$\lim_{x\to 0}\frac{\cosh x-\cos x}{x^2}$$

112 **uni-lim-010**

R6 Boss+ • 210s • taylor / inverse-trig

$$\lim_{x\to 0}\frac{\arctan x-\arcsin x}{x^3}$$

113 **uni-lim-011**

R6 Boss+ • 210s • taylor / log

$$\lim_{x\to 0}\frac{\log(1+\sin x)-x}{x^2}$$

114 **uni-lim-012**

R6 Boss+ • 210s • taylor / exponential-limit

$$\lim_{x\to 0}\frac{e^x-1-x}{1-\cos x}$$

115 **uni-lim-013**

R6 Boss+ • 210s • taylor / exponential-limit

$$\lim_{x\to 0}\left(\frac{x}{\sin x}\right)^{1/x^2}$$

116 **uni-lim-014**

R6 Boss+ • 210s • taylor / radical

$$\lim_{x\to 0}\frac{\frac{1}{\sqrt{1+x^2}}-\cos x}{x^4}$$

117 **uni-lim-015**

R6 Boss+ • 210s • taylor / log

$$\lim_{x\to 0}\frac{\log(1+x+x^2)-x}{x^2}$$

118 **depth-lim-001**

R6 Boss+ • 270s • nested-taylor / taylor

$$\lim_{x\to 0}\frac{\sin x-x+\frac{x^3}{6}-\frac{x^5}{120}}{x^7}$$

119 **depth-lim-002**

R6 Boss+ • 270s • nested-taylor / taylor

$$\lim_{x\to 0}\frac{\log(1+x)-x+\frac{x^2}{2}-\frac{x^3}{3}+\frac{x^4}{4}}{x^5}$$

120 **depth-lim-003**

R5 Boss • 220s • composite-taylor / taylor

$$\lim_{x\to 0}\frac{e^{\sin x}-1-x}{x^2}$$

121 **depth-lim-004**

R6 Boss+ • 270s • asymptotic-balance / taylor

$$\lim_{x\to 0}\frac{\cos x-\sqrt{1-x^2}}{x^4}$$

122 **depth-lim-005**

R5 Boss • 220s • asymptotic-balance / taylor

$$\lim_{x\to 0}\frac{\tan x-x}{x(1-\cos x)}$$

123 **depth-lim-006**

R6 Boss+ • 270s • infinity-limit / log-limit

$$\lim_{x\to\infty}x^2\left(\log\left(1+\frac{1}{x}\right)-\frac{1}{x+1}\right)$$

124 **depth-lim-007**
R6 Boss+ • 270s • exponential-limit / log-limit
$$\lim_{x\to 0}\frac{(1+x)^{1/x}-e}{x}$$

125 **depth-lim-008**
R5 Boss • 220s • inverse-trig / taylor
$$\lim_{x\to 0}\frac{\arcsin x-\arctan x}{x^3}$$

126 **depth-lim-009**
R5 Boss • 220s • hyperbolic / taylor
$$\lim_{x\to 0}\frac{\sinh x-\sin x}{x^3}$$

127 **depth-lim-010**
R5 Boss • 220s • trig-limit / asymptotic-balance
$$\lim_{x\to 0}\frac{1-\cos x}{x\sin x}$$

128 **depth-lim-011**
R5 Boss • 220s • log / taylor
$$\lim_{x\to 0}\frac{\log(\cos x)}{x^2}$$

129 **depth-lim-012**
R6 Boss+ • 270s • exponential-limit / log-limit
$$\lim_{x\to 0}\frac{(1+x+x^2)^{1/x}-e}{x}$$

130 **burst-boss-lim-001**
R5 Boss • 240s • true-boss / boss-rank
$$\lim_{x\to 0}\frac{\tan x-x-x^3/3}{x^5}$$

131 **burst-boss-lim-002**
R5 Boss • 240s • true-boss / boss-rank
$$\lim_{x\to 0}\frac{\sin x-x+x^3/6}{x^5}$$

132 **burst-boss-lim-003**
R5 Boss • 240s • true-boss / boss-rank
$$\lim_{x\to 0}\frac{\log(1+x)-x+x^2/2-x^3/3}{x^4}$$

133 **burst-boss-lim-004**
R6 Boss+ • 300s • true-boss / boss-rank
$$\lim_{x\to 0}\frac{e^{\sin x}-1-x-x^2/2}{x^4}$$

134 **burst-boss-lim-005**
R6 Boss+ • 300s • true-boss / boss-rank
$$\lim_{x\to 0}\frac{\cos(\sin x)-\cos x}{x^4}$$

135 **burst-boss-lim-006**
R6 Boss+ • 300s • true-boss / boss-rank
$$\lim_{x\to 0}\frac{\arctan x-x+x^3/3-x^5/5}{x^7}$$

136 **burst-boss-lim-007**
R6 Boss+ • 300s • true-boss / boss-rank
$$\lim_{x\to 0}\frac{(1+x)^{1/x}-e}{x}$$

137 **burst-boss-lim-008**
R6 Boss+ • 300s • true-boss / boss-rank
$$\lim_{x\to\infty} x\left(\left(1+\frac{1}{x}\right)^x-e\right)$$

138 **burst-boss2-lim-001**
R6 Boss+ • 420s • true-boss / boss-rank
$$\lim_{x\to 0}\left(\frac{\sin x}{x}\right)^{1/x^2}$$

139 **burst-boss2-lim-002**

R6 Boss+ • 420s • true-boss / boss-rank

$$\lim_{x\to 0} \frac{x/\tan x - 1 + x^2/3}{x^4}$$

140 **burst-boss2-lim-003**

R6 Boss+ • 420s • true-boss / boss-rank

$$\lim_{x\to 0} \frac{1/\sin x - 1/x - x/6}{x^3}$$

141 **burst-boss2-lim-004**

R6 Boss+ • 420s • true-boss / boss-rank

$$\lim_{x\to 0} \frac{\log(\cos x) + x^2/2}{x^4}$$

142 **burst-boss2-lim-005**

R6 Boss+ • 420s • true-boss / boss-rank

$$\lim_{x\to 0} \frac{\int_0^x e^{-t^2}\,dt - x + x^3/3}{x^5}$$

143 **burst-boss2-lim-006**

R6 Boss+ • 420s • true-boss / boss-rank

$$\lim_{x\to 0} \frac{\int_0^x \log(1+t^2)\,dt - x^3/3 + x^5/10}{x^7}$$

144 **burst-boss2-lim-007**

R6 Boss+ • 420s • true-boss / boss-rank

$$\lim_{n\to\infty} n\left(\sum_{k=1}^{n} \frac{1}{n+k} - \log 2\right)$$

145 **burst-boss2-lim-008**

R6 Boss+ • 420s • true-boss / boss-rank

$$\lim_{n\to\infty} n^2\left(\sum_{k=1}^{n} \frac{1}{n^2+k^2} - \frac{\pi}{4n}\right)$$

146 **burst-boss2-lim-009**

R6 Boss+ • 420s • true-boss / boss-rank

$$\lim_{n\to\infty} n\left(\sum_{k=1}^{n} \frac{1}{\sqrt{n^2+k^2}} - \log(1+\sqrt{2})\right)$$

147 **burst-boss2-lim-010**

R6 Boss+ • 420s • true-boss / boss-rank

$$\lim_{n\to\infty} n\left(\int_0^1 \left(1+\frac{x}{n}\right)^n dx - (e-1)\right)$$

148 **burst-boss3-lim-001**

R6 Boss+ • 480s • true-boss / boss-rank

$$\lim_{n\to\infty} n \sup_{0\le x\le 1} x^n(1-x)$$

149 **burst-boss3-lim-002**

R6 Boss+ • 480s • true-boss / boss-rank

$$\lim_{n\to\infty} n^2 \sup_{0\le x\le 1} x^n(1-x)^2$$

150 **burst-boss3-lim-003**

R6 Boss+ • 480s • true-boss / boss-rank

$$\lim_{n\to\infty} n \int_0^1 \frac{x^n}{1+x}\,dx$$

151 **burst-boss3-lim-004**

R6 Boss+ • 480s • true-boss / boss-rank

$$\lim_{n\to\infty} n^2\left(\frac{1}{n}\sum_{k=1}^{n} e^{k/n} - (e-1) - \frac{e-1}{2n}\right)$$

152 **burst-boss3-lim-005**

R6 Boss+ • 480s • true-boss / boss-rank

$$\lim_{n\to\infty} n^2\left(\frac{1}{n}\sum_{k=1}^{n} \log\left(1+\frac{k}{n}\right) - (2\log 2 - 1) - \frac{\log 2}{2n}\right)$$

153 **burst-boss3-lim-006**

R6 Boss+ • 480s • true-boss / boss-rank

$$\lim_{n\to\infty} n^2\left(\frac{1}{n}\sum_{k=1}^{n} \frac{1}{1+(k/n)^2} - \frac{\pi}{4} + \frac{1}{4n}\right)$$

154 **burst-boss3-lim-007**
R6 Boss+ • 480s • true-boss / boss-rank

$$\lim_{x\to 0}\frac{\int_0^x \frac{e^{t^2}-\cos t}{t^2}\,dt-\frac{3}{2}x}{x^3}$$

155 **burst-boss3-lim-008**
R6 Boss+ • 480s • true-boss / boss-rank

$$\lim_{x\to 0}\frac{\frac{1}{x^3}\int_0^x \sin(t^2)\,dt-\frac{1}{3}}{x^4}$$

156 **burst-boss3-lim-009**
R6 Boss+ • 480s • true-boss / boss-rank

$$\lim_{x\to 0}\frac{\log(\sin x/x)+x^2/6}{x^4}$$

157 **burst-boss3-lim-010**
R6 Boss+ • 480s • true-boss / boss-rank

$$\lim_{x\to 0}\frac{\log(\cosh x)-x^2/2}{x^4}$$

158 **burst-boss3-lim-011**
R6 Boss+ • 480s • true-boss / boss-rank

$$\lim_{x\to 0}\frac{\sqrt{1+x}-1-x/2+x^2/8}{x^3}$$

159 **burst-boss3-lim-012**
R6 Boss+ • 480s • true-boss / boss-rank

$$\lim_{x\to 0}\left(\frac{x}{\sin x}\right)^{1/x^2}$$

微分 Derivatives

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160 **der-001**
R1 Warm-up • 30s • rank-1 / beginner-friendly

$$\frac{d}{dx}\left(x^5-3x^2+7\right)$$

161 **der-002**
R1 Warm-up • 30s • rank-1 / beginner-friendly

$$\frac{d}{dx}\left(\sin x+\cos x\right)$$

162 **der-003**
R1 Warm-up • 40s • rank-1 / beginner-friendly

$$\frac{d}{dx}\left(e^x\ln x\right)$$

163 **der-004**
R1 Warm-up • 30s • rank-1 / beginner-friendly

$$\frac{d}{dx}\left(\sqrt{x}\right)$$

164 **der-005**
R2 Basic • 40s • rank-2 / beginner-friendly

$$\frac{d}{dx}\left(\ln(1+x^2)\right)$$

165 **der-006**
R2 Basic • 45s • rank-2 / beginner-friendly

$$\frac{d}{dx}\left(x^2\sin x\right)$$

166 **der-007**
R2 Basic • 35s • rank-2 / beginner-friendly

$$\frac{d}{dx}\left(\arctan x\right)$$

167 **der-008**
R2 Basic • 45s • rank-2 / beginner-friendly

$$\frac{d}{dx}\left(\frac{x}{x+1}\right)$$

168 **der-009**
R3 Standard • 55s • rank-3

$$\frac{d}{dx}\left(x^x\right)$$

169 **der-010**
R3 Standard • 60s • rank-3

$$\frac{d}{dx}\left(\sin(x^2)e^{-x}\right)$$

170 **der-011**
R3 Standard • 45s • rank-3

$$\frac{d}{dx}\left(\ln(\sin x)\right)$$

171 **der-012**

R4 Advanced • 70s • rank-4

$$\frac{d}{dx}\left(\frac{\ln x}{x^2+1}\right)$$

172 **der-013**

R4 Advanced • 90s • rank-4

$$\frac{d}{dx}\left(x^{\sin x}\right)$$

173 **der-014**

R3 Standard • 85s • rank-3

$$\frac{d}{dx}\left(\arctan\frac{1+x}{1-x}\right)$$

174 **der-015**

R3 Standard • 75s • rank-3

$$\frac{d}{dx}\left(\ln\left(x+\sqrt{1+x^2}\right)\right)$$

175 **der-016**

R3 Standard • 80s • rank-3

$$\frac{d}{dx}\left(\frac{\sin x}{1+\cos x}\right)$$

176 **der-017**

R3 Standard • 80s • rank-3

$$\frac{d}{dx}\left(e^{x^2}\cos(3x)\right)$$

177 **der-018**

R3 Standard • 80s • rank-3

$$\frac{d}{dx}\left(\ln\left(x+\sqrt{x^2-1}\right)\right)$$

178 **der-019**

R2 Basic • 55s • rank-2 / beginner-friendly

$$\frac{d}{dx}\left((x^2+1)^5\right)$$

179 **der-020**

R3 Standard • 75s • rank-3

$$\frac{d}{dx}\left(\arctan\sqrt{x}\right)$$

180 **td-der-001**

R4 Advanced • 100s • rank-4

$$\frac{d^3}{dx^3}\arctan\frac{1}{x}$$

181 **td-der-002**

R2 Basic • 60s • rank-2 / beginner-friendly

$$\frac{d}{dx}\left[\frac{1}{8}(35x^4-30x^2+3)\right]$$

182 **td-der-003**

R3 Standard • 55s • rank-3

$$\frac{d}{dx}\left(x\log(1+x)\right)$$

183 **td-der-004**

R3 Standard • 50s • rank-3

$$\frac{d}{dx}\left(\log(5+3\cos x)\right)$$

184 **der-021**

R4 Advanced • 65s • multivariable / partial-derivative

$$\frac{\partial}{\partial x}\left(x^2y+\sin(xy)\right)$$

185 **der-022**

R4 Advanced • 65s • multivariable / partial-derivative

$$\frac{\partial}{\partial y}\left(xe^{xy}+y^3\right)$$

186 **der-023**

R4 Advanced • 70s • multivariable / mixed-partial

$$\frac{\partial^2}{\partial x\partial y}e^{xy}$$

187 **der-024**

R4 Advanced • 75s • multivariable / directional-derivative

Directional derivative of $f = x^2 + y^2$ at $(1,2)$ in direction $(3,4)/5$

188 **der-025**

R4 Advanced • 60s • multivariable / partial-derivative

$$\frac{\partial}{\partial x}\log(x^2+y^2)$$

189 **der-026**

R2 Basic • 55s • product-rule / chain-rule

$$\frac{d}{dx}\left(x^2\sin(3x)\right)$$

190 **der-027**

R2 Basic • 55s • log / chain-rule

$$\frac{d}{dx}\log\sqrt{1+x^2}$$

191 **der-028**

R2 Basic • 60s • inverse-trig / chain-rule

$$\frac{d}{dx}\arctan(e^x)$$

192 **der-029**

R2 Basic • 60s • second-derivative / product-rule

$$\frac{d^2}{dx^2}(xe^x)$$

193 **der-030**

R3 Standard • 80s • implicit-differentiation / rank-3

Find y' from $x^2+xy+y^2=7$

194 **der-031**

R4 Advanced • 65s • multivariable / partial-derivative

$$\frac{\partial}{\partial x}\left(e^{x^2+y^2}\right)$$

195 **der-032**

R4 Advanced • 75s • multivariable / partial-derivative

$$\frac{\partial}{\partial y}\arctan\left(\frac{y}{x}\right)$$

196 **der-033**

R4 Advanced • 65s • multivariable / second-derivative

$$\frac{\partial^2}{\partial x^2}(x^2y+\sin x)$$

197 **der-034**

R4 Advanced • 75s • multivariable / jacobian

Jacobian determinant of $u=x+y,\ v=xy$

198 **der-035**

R4 Advanced • 80s • multivariable / chain-rule

If $z=x^2y,\ x=t^2,\ y=e^t,$ find dz/dt

199 **der-036**

R4 Advanced • 55s • multivariable / second-derivative

Laplacian of $f=x^2+y^2$

200 **der-037**

R4 Advanced • 90s • multivariable / hessian

Hessian determinant of $f=x^3+y^3-3xy$ at $(1,1)$

201 **der-038**

R2 Basic • 65s • quotient-rule / rank-2

$$\frac{d}{dx}\left(\frac{x^2+1}{x-1}\right)$$

202 **der-039**

R2 Basic • 45s • chain-rule / trig

$$\frac{d}{dx}(\sin^2 x)$$

203 **der-040**

R3 Standard • 80s • logarithmic-differentiation / rank-3

$$\frac{d}{dx}\left(x^{\log x}\right)$$

204 **der-041**

R4 Advanced • 80s • total-differential / linearization

Use total differential to estimate Δf for $f=x^2y$ at $(2,1),\ dx=0.01,\ dy=-0.02$

205 **der-042**

R4 Advanced • 85s • total-differential / linearization

Use total differential for $f=\sqrt{x^2+y^2}$ at $(3,4),\ dx=0.06,\ dy=-0.08$

206 **der-043**

R4 Advanced • 75s • total-differential / linearization

Estimate Δz for $z=\log(xy)$ at $(1,2),\ dx=0.02,\ dy=-0.04$

207 **der-044**

R4 Advanced • 90s • total-differential / linearization

Linear part df of $f=e^{xy}$

208 **der-045**

R4 Advanced • 100s • total-differential / linearization

Linear part df of $f=\arctan(y/x)$

209 **der-046**

R4 Advanced • 85s • total-differential / linearization

Linear estimate of x^2+y^2 at $(1.02,1.97)$ using base $(1,2)$

210 **der-047**

R5 Boss • 100s • total-differential-min / optimization

$$\min_{x,y}\left(x^2+2xy+3y^2-4x-6y\right)$$

211 **der-048**

R5 Boss • 100s • total-differential-min / optimization

$$\min_{x,y}\left(3x^2+2xy+2y^2-8x-6y\right)$$

212 **der-049**
R5 Boss • 70s • total-differential-min / optimization
x-coordinate of the minimum of $x^2 + y^2 + 2x - 4y + 7$

213 **der-050**
R5 Boss • 95s • total-differential-min / optimization
$\min\,(x^2 + y^2) \quad \text{subject to } x + 2y = 5$

214 **der-051**
R4 Advanced • 75s • hessian / second-derivative
Hessian determinant of $f = x^2 + xy + y^2$

215 **der-052**
R4 Advanced • 75s • hessian / second-derivative
Hessian determinant of $f = x^4 + y^4$ at $(1, 1)$

216 **der-053**
R4 Advanced • 95s • hessian / second-derivative
Hessian determinant of $f = e^{xy}$ at $(0, 0)$

217 **der-054**
R4 Advanced • 90s • hessian / second-derivative
Hessian determinant of $f = x^4 + y^4 - 4xy$ at $(1, 1)$

218 **der-055**
R4 Advanced • 70s • hessian / critical-point
Classify the critical point $(0, 0)$ of $f = x^2 + y^2$

219 **der-056**
R4 Advanced • 70s • hessian / critical-point
Classify the critical point $(0, 0)$ of $f = x^2 - y^2$

220 **der-057**
R4 Advanced • 70s • hessian / degenerate
Hessian determinant of $f = x^4 + y^2$ at $(0, 0)$

221 **der-058**
R4 Advanced • 100s • hessian / log
Hessian determinant of $f = \log(x^2 + y^2)$ at $(1, 0)$

222 **der-059**
R4 Advanced • 65s • wronskian / linear-independence
Wronskian $W(\sin x, \cos x)$

223 **der-060**
R4 Advanced • 65s • wronskian / linear-independence
Wronskian $W(\cos x, \sin x)$

224 **der-061**
R4 Advanced • 70s • wronskian / linear-independence
Wronskian $W(e^x, e^{2x})$

225 **der-062**
R4 Advanced • 60s • wronskian / polynomial
Wronskian $W(x, x^2)$

226 **der-063**
R4 Advanced • 85s • wronskian / polynomial
Wronskian $W(1, x, x^2)$

227 **der-064**
R4 Advanced • 75s • wronskian / exponential
Wronskian $W(e^x, xe^x)$

228 **der-065**
R4 Advanced • 65s • wronskian / log
Wronskian $W(1, \log x)$

229 **der-066**
R4 Advanced • 65s • wronskian / polynomial
Wronskian $W(x^2, x^3)$

230 **der-067**
R5 Boss • 80s • jacobian-chain / chain-rule
If $u = x + y,\ v = x - y,\ F = u^2 + v^2,$ find dF/dx

231 **der-068**
R5 Boss • 95s • jacobian-chain / chain-rule
If $u = x^2,\ v = \sin x,\ F = ue^v,$ find dF/dx

232 **der-069**

R6 Boss+ • 80s • jacobian-chain / jacobian

Jacobian determinant of $u = x^2 - y^2,\ v = 2xy$

233 **der-070**

R5 Boss • 100s • jacobian-chain / jacobian

Let $r = u + v,\ s = uv,\ u = x + y,\ v = x - y$. Find $\dfrac{\partial(r,s)}{\partial(x,y)}$

234 **der-071**

R5 Boss • 75s • jacobian-chain / jacobian

Jacobian $\dfrac{\partial(x,y)}{\partial(r,t)}$ for $x = r\cos t,\ y = r\sin t$

235 **der-072**

R5 Boss • 90s • jacobian-chain / jacobian

Jacobian $\dfrac{\partial(u,v)}{\partial(x,y)}$ for $u = xy,\ v = x/y$

236 **der-073**

R6 Boss+ • 115s • jacobian-chain / jacobian

Let $u = 2x + y,\ v = x + 3y,\ p = u^2 - v^2,\ q = 2uv$. Find $\dfrac{\partial(p,q)}{\partial(x,y)}$

237 **der-074**

R5 Boss • 95s • jacobian-chain / jacobian

If $f(x,y) = (x, xy),\ g(u,v) = (u+v, u-v),$ find $\det D(g \circ f)$ at $(2,3)$

238 **der-075**

R6 Boss+ • 75s • complex / real-part

Real part of $(x + iy)^3$

239 **der-076**

R6 Boss+ • 75s • complex / imaginary-part

Imaginary part of $(x + iy)^3$

240 **der-077**

R6 Boss+ • 45s • complex / modulus

$|1 + i|$

241 **der-078**

R6 Boss+ • 55s • complex / power

Real part of $(1 + i)^4$

242 **der-079**

R6 Boss+ • 70s • complex / cauchy-riemann

Is $u = x^2 - y^2,\ v = 2xy$ holomorphic by Cauchy-Riemann?

243 **der-080**

R6 Boss+ • 70s • complex / cauchy-riemann

Is $u = x^2 + y^2,\ v = 0$ holomorphic on an open set?

244 **der-081**

R6 Boss+ • 80s • complex / residue

$\left|\operatorname{Res}_{z=i} \dfrac{1}{z^2+1}\right|$

245 **der-082**

R6 Boss+ • 60s • complex / residue

$\operatorname{Res}_{z=2} \dfrac{1}{(z-2)^3}$

246 **der-083**

R6 Boss+ • 75s • complex / laurent

Coefficient of z^{-1} in e^z/z^3

247 **der-084**

R6 Boss+ • 70s • complex / contour-integral

$\dfrac{1}{2\pi i}\displaystyle\int_{|z|=2} \dfrac{1}{z-1}\,dz$

248 **der-085**
R6 Boss+ • 75s • complex / complex-derivative

Real part of $f'(1+i)$ for $f(z)=z^4$

249 **der-086**
R6 Boss+ • 80s • complex / jacobian

Jacobian determinant of the complex map $z\mapsto z^2$ at $z=x+iy$

250 **der-087**
R6 Boss+ • 70s • complex / harmonic

$\Delta\ \mathrm{Re}(z^3)\quad(z=x+iy)$

251 **der-088**
R6 Boss+ • 65s • complex / harmonic

Is $u=x^3-3xy^2$ harmonic?

252 **gap-der-param-001**
R2 Basic • 65s • parametric / parametric-differentiation

$x=t^2,\ y=t^3.\ \dfrac{dy}{dx}=?$

253 **gap-der-param-002**
R3 Standard • 70s • parametric / parametric-differentiation

$x=\cos t,\ y=\sin t.\ \dfrac{dy}{dx}=?$

254 **gap-der-param-003**
R2 Basic • 60s • parametric / second-derivative

$x=t,\ y=t^2.\ \dfrac{d^2y}{dx^2}=?$

255 **gap-der-param-004**
R2 Basic • 65s • parametric / parametric-differentiation

$x=t^2+1,\ y=t^3.\ \left.\dfrac{dy}{dx}\right|_{t=2}$

256 **gap-der-polar-001**
R4 Advanced • 95s • polar-curve / parametric

$r=1+\cos\theta.\ \left.\dfrac{dy}{dx}\right|_{\theta=\pi/2}$

257 **gap-der-param-005**
R2 Basic • 55s • parametric / speed

$x=3t,\ y=4t.$ Speed $\sqrt{(dx/dt)^2+(dy/dt)^2}$

258 **gap-der-app-001**
R3 Standard • 80s • related-rates / volume

A sphere has $r=3,\ dr/dt=2.\ \dfrac{dV}{dt}=?$

259 **gap-der-app-002**
R2 Basic • 70s • related-rates / area

A circle has $r=5,\ dr/dt=0.2.\ \dfrac{dA}{dt}=?$

260 **gap-der-app-003**
R3 Standard • 85s • related-rates / implicit-differentiation

Ladder length $10,\ x=6,\ dx/dt=2.\ dy/dt=?$

261 **gap-der-app-004**
R2 Basic • 60s • tangent-normal / derivative

Normal slope to $y=x^2$ at $x=2$

262 **gap-der-app-005**
R2 Basic • 65s • linear-approximation / application

Linear approximation of $\sqrt{4.04}$

263 **gap-der-app-006**
R2 Basic • 70s • newton-method / application

One Newton step for $f(x)=x^2-2$ from $x_0=1$

264 **gap-der-app-007**
R3 Standard • 80s • curvature / application

Curvature of $y=x^2$ at $x=0$

265 **gap-der-ode-001**
R2 Basic • 75s • ode-intro / separable

$y'+y=0,\ y(0)=2.\ y=?$

266 **gap-der-ode-002**
R2 Basic • 70s • ode-intro / separable

$y'=2y,\ y(0)=3.\ y=?$

267 **gap-der-ode-003**

R3 Standard • 95s • ode-intro / integrating-factor

$y' + y = e^x,\ y(0) = 0.\ y = ?$

268 **gap-der-ode-004**

R3 Standard • 90s • ode-intro / second-order

$y'' + y = 0,\ y(0) = 0,\ y'(0) = 1.\ y = ?$

269 **gap-der-ode-005**

R4 Advanced • 105s • ode-intro / logistic

$y' = y(1-y),\ y(0) = 1/2.\ y = ?$

270 **gap-tech-007**

R2 Basic • 45s • technique-recognition / parametric

$x = t^2,\ y = t^3$. Which differentiation technique?

271 **mob-tech-012**

R2 Basic • 45s • technique-sprint / technique-recognition

Optimize $f(x, y)$ subject to $g(x, y) = 0$ Use which technique?

272 **mob-tech-014**

R2 Basic • 45s • technique-sprint / technique-recognition

$\nabla \cdot \mathbf{F}$ and $\nabla \times \mathbf{F}$ Use which technique?

273 **mob-tech-018**

R2 Basic • 45s • technique-sprint / technique-recognition

$x^2y'' + xy' + (x^2 - \nu^2)y = 0$ Use which technique?

274 **mob-tech-020**

R2 Basic • 45s • technique-sprint / technique-recognition

$y = x^x$ Use which technique?

275 **mob-trap-001**

R2 Basic • 45s • trap-drill / rank-2

$\frac{d}{dx}\sin(3x)$ Main trap?

276 **mob-trap-007**

R2 Basic • 45s • trap-drill / rank-2

$\frac{\partial}{\partial x}y^2$ Main trap?

277 **mob-trap-011**

R2 Basic • 45s • trap-drill / rank-2

$\nabla \times (\nabla f)$ Main trap?

278 **mob-lm-001**

R4 Advanced • 85s • lagrange-multiplier / optimization

$\max xy$ subject to $x + y = 10\ (x, y > 0)$

279 **mob-lm-002**

R4 Advanced • 85s • lagrange-multiplier / optimization

$\min x^2 + y^2$ subject to $x + y = 6$

280 **mob-lm-003**

R4 Advanced • 85s • lagrange-multiplier / optimization

$\max xyz$ subject to $x + y + z = 12\ (x, y, z > 0)$

281 **mob-lm-004**

R4 Advanced • 85s • lagrange-multiplier / optimization

$\min x^2 + y^2$ subject to $xy = 8\ (x, y > 0)$

282 **mob-lm-005**

R4 Advanced • 85s • lagrange-multiplier / optimization

$\max x^2y$ subject to $x + 2y = 6\ (x, y > 0)$

283 **mob-lm-006**

R4 Advanced • 85s • lagrange-multiplier / optimization

$\min x^2 + y^2 + z^2$ subject to $x + y + z = 3$

284 **mob-lm-007**

R4 Advanced • 85s • lagrange-multiplier / optimization

$\max A$ rectangle with perimeter 20

285 **mob-lm-008**

R4 Advanced • 85s • lagrange-multiplier / optimization

$\min x^2 + y^2 + z^2$ subject to $x + y + z = 1$

286 **mob-lm-009**

R4 Advanced • 85s • lagrange-multiplier / optimization

$\max xy$ subject to $x^2 + y^2 = 50$

287 **mob-lm-010**

R4 Advanced • 85s • lagrange-multiplier / optimization

$\max(x + y)$ subject to $x^2 + y^2 = 2$

288 **mob-bessel-001**

R6 Boss+ • 70s • bessel / special-functions

$J_0(0)$

289 **mob-bessel-002**
R6 Boss+ • 70s • bessel / special-functions
$J_1(0)$

290 **mob-bessel-003**
R6 Boss+ • 70s • bessel / special-functions
coefficient of x^2 in $J_0(x) = 1 - x^2/4 + \cdots$

291 **mob-bessel-004**
R6 Boss+ • 70s • bessel / special-functions
coefficient of x in $J_1(x) = x/2 + \cdots$

292 **mob-bessel-005**
R6 Boss+ • 45s • bessel / special-functions
$J_0'(x)$

293 **mob-bessel-006**
R6 Boss+ • 45s • bessel / special-functions
$x^2y'' + xy' + (x^2 - \nu^2)y = 0$

294 **mob-bessel-007**
R6 Boss+ • 45s • bessel / special-functions
Singularity of Bessel equation at $x = 0$

295 **mob-bessel-008**
R6 Boss+ • 45s • bessel / special-functions
$J_{-2}(x)$ equals

296 **mob-bessel-009**
R6 Boss+ • 45s • bessel / special-functions
$J_{-1}(x)$ equals

297 **mob-bessel-010**
R6 Boss+ • 45s • bessel / special-functions
Bessel functions most often appear after which ODE type?

298 **mob-nabla-001**
R4 Advanced • 75s • nabla / vector-calculus
$\nabla \cdot \langle x, y, z\rangle$

299 **mob-nabla-002**
R4 Advanced • 75s • nabla / vector-calculus
z-component of $\nabla \times \langle y, -x, 0\rangle$

300 **mob-nabla-003**
R4 Advanced • 75s • nabla / vector-calculus
$\Delta(x^2 + y^2 + z^2)$

301 **mob-nabla-004**
R4 Advanced • 75s • nabla / vector-calculus
$D_{(1,1)/\sqrt{2}}(x^2 + y^2)$ at $(1, 1)$

302 **mob-nabla-005**
R4 Advanced • 75s • nabla / vector-calculus
$\nabla \times (\nabla f)$

303 **mob-nabla-006**
R4 Advanced • 75s • nabla / vector-calculus
$\nabla \cdot (\nabla \times \mathbf{F})$

304 **mob-nabla-007**
R4 Advanced • 75s • nabla / vector-calculus
$\int_C \nabla(x^2 + y^2) \cdot d\mathbf{r}$ from $(0, 0)$ to $(1, 1)$

305 **mob-nabla-008**
R4 Advanced • 45s • nabla / vector-calculus
$\nabla \cdot \nabla f$ is called

306 **mob-nabla-009**
R4 Advanced • 45s • nabla / vector-calculus
$\nabla \times \mathbf{F} = 0$ usually suggests

307 **mob-nabla-010**

R4 Advanced • 45s • nabla / vector-calculus

Green theorem circulation uses which scalar derivative?

308 **rel-basic-007**

R1 Warm-up • 45s • basic-derivative / beginner-foundation

$$\left.\frac{d}{dx}x^3\right|_{x=2}$$

309 **rel-basic-008**

R1 Warm-up • 45s • basic-derivative / beginner-foundation

$$\left.\frac{d}{dx}\sin x\right|_{x=0}$$

310 **rel-basic-009**

R1 Warm-up • 45s • chain-rule / beginner-foundation

$$\left.\frac{d}{dx}e^{2x}\right|_{x=0}$$

311 **rel-basic-010**

R1 Warm-up • 45s • basic-derivative / beginner-foundation

$$\left.\frac{d}{dx}\sqrt{x}\right|_{x=4}$$

312 **rel-basic-011**

R1 Warm-up • 45s • basic-derivative / beginner-foundation

$$\left.\frac{d}{dx}\log x\right|_{x=e}$$

313 **rel-basic-012**

R1 Warm-up • 45s • product-rule / beginner-foundation

$$\left.\frac{d}{dx}(x\log x)\right|_{x=1}$$

314 **rel-adv-005**

R4 Advanced • 75s • jacobian / multivariable

$$\frac{\partial(u,v)}{\partial(x,y)} \text{ at } (1,2), \quad u=x^2-y^2,\ v=2xy$$

315 **rel-adv-006**

R4 Advanced • 75s • directional-derivative / multivariable

$$D_{(2,-1,2)/3}(x^2y+yz) \text{ at } (1,2,3)$$

316 **rel-adv-007**

R4 Advanced • 75s • hessian / optimization

$$\det H_f(1,1), \quad f=x^4+y^4-4xy$$

317 **rel-adv-008**

R4 Advanced • 75s • lagrange-multiplier / optimization

$$\min(x^2+y^2+z^2) \quad \text{subject to } x+2y+2z=9$$

318 **rel-boss-007**

R6 Boss+ • 110s • bessel / special-functions

coefficient of x^4 in $J_0(x)$

319 **rel-hard-mv-005**

R5 Boss • 120s • jacobian / multivariable

$$\frac{\partial(u,v)}{\partial(x,y)} \text{ at } (2,1), \quad u=\frac{x}{y},\ v=xy$$

320 **rel-hard-mv-006**

R5 Boss • 120s • lagrange-multiplier / optimization

$$\max xyz \quad \text{subject to } x+y+z=1,\ x,y,z>0$$

321 **rel-hard-mv-007**

R5 Boss • 120s • lagrange-multiplier / optimization

$$\min(x^2+y^2) \quad \text{subject to } xy=1,\ x,y>0$$

322 **rel-hard-der-001**

R5 Boss • 95s • wronskian / boss-rank

$W(e^x, xe^x)$ at $x=0$

323 **rel-hard-der-002**

R5 Boss • 95s • wronskian / boss-rank

$W(\sin x, \cos x)$ at $x=0$

324 **rel-hard-der-003**

R5 Boss • 95s • hessian / boss-rank

$$\det H_f(1,0), \quad f=\log(x^2+y^2)$$

325 **rel-hard-der-004**

R5 Boss • 95s • curvature / boss-rank

κ for $y=\log x$ at $x=1$

326 **rel-hard-der-005**

R5 Boss • 95s • parametric-differentiation / boss-rank

$\frac{dy}{dx}$ at $t = 2,\quad x = t + \frac{1}{t},\ y = t - \frac{1}{t}$

327 **rel-hard-der-006**

R5 Boss • 95s • parametric-differentiation / boss-rank

$\frac{d^2y}{dx^2}$ at $t = 2,\quad x = t^2,\ y = t^3$

328 **rel-hard-der-007**

R5 Boss • 95s • directional-derivative / nabla

$D_{(1,2,2)/3}(xyz)$ at $(1, 1, 1)$

329 **rel-hard-der-008**

R5 Boss • 95s • nabla / vector-calculus

$\nabla \cdot \langle xy, yz, zx\rangle$ at $(1, 2, 3)$

330 **rel-hard-der-009**

R5 Boss • 95s • nabla / vector-calculus

z-component of $\nabla \times \left\langle \frac{-y}{x^2+y^2}, \frac{x}{x^2+y^2}, 0 \right\rangle$ at $(1, 0, 0)$

331 **rel-hard-der-010**

R5 Boss • 95s • nabla / laplacian

Δe^{x+y} at $(0, 0)$

332 **hc-der-001**

R6 Boss+ • 135s • log-differentiation / chain-rule

$\frac{d}{dx}\left(x^{\sin x}\right)$

333 **hc-der-002**

R6 Boss+ • 135s • log-differentiation / chain-rule

$\frac{d}{dx}\left((\sin x)^{\cos x}\right)$

334 **hc-der-003**

R6 Boss+ • 135s • exponential / chain-rule

$\frac{d}{dx}\left(e^{e^{2x}+x^2}\right)$

335 **hc-der-004**

R6 Boss+ • 150s • higher-derivative / exponential

$\frac{d^5}{dx^5}\left(x^4 e^{3x}\right)\Big|_{x=0}$

336 **hc-der-005**

R6 Boss+ • 150s • higher-derivative / taylor

$\frac{d^7}{dx^7}\left(x^2 \sin x\right)\Big|_{x=0}$

337 **hc-der-006**

R6 Boss+ • 150s • higher-derivative / taylor

$\frac{d^8}{dx^8} e^{x^2}\Big|_{x=0}$

338 **hc-der-007**

R6 Boss+ • 150s • higher-derivative / taylor

$\frac{d^{10}}{dx^{10}}\left(\frac{\sin x}{x}\right)\Big|_{x=0}$

339 **hc-der-008**

R6 Boss+ • 150s • implicit-differentiation / higher-derivative

$y = e^{xy},\ y(0) = 1.$ Find $y'''(0)$

340 **hc-der-009**

R6 Boss+ • 150s • higher-derivative / trig

$\frac{d^{12}}{dx^{12}} \cos(2x)\Big|_{x=0}$

341 **hc-der-010**

R6 Boss+ • 150s • higher-derivative / exponential

$$\frac{d^9}{dx^9}\left(e^x \sin x\right)\Big|_{x=0}$$

342 **exam-der-001**

R5 Boss • 180s • log-differentiation / exam-style

$$\frac{d}{dx}\left(x^x\right)$$

343 **exam-der-002**

R5 Boss • 180s • log-differentiation / chain-rule

$$\frac{d}{dx}\left(x^{\sin x}\right)$$

344 **exam-der-003**

R5 Boss • 180s • quotient-rule / log

$$\frac{d}{dx}\left(\frac{\log x}{x}\right)$$

345 **exam-der-004**

R5 Boss • 180s • chain-rule / inverse-trig

$$\frac{d}{dx}\arctan\left(\frac{1+x}{1-x}\right)$$

346 **exam-der-005**

R5 Boss • 180s • product-rule / exponential

$$\frac{d}{dx}\left(x^2 e^{-x}\sin x\right)$$

347 **exam-der-006**

R5 Boss • 180s • log / radical

$$\frac{d}{dx}\log\left(\sqrt{1+x^2}+x\right)$$

348 **exam-der-007**

R5 Boss • 180s • higher-derivative / chain-rule

$$\frac{d^2}{dx^2}\left(e^{x^2}\right)$$

349 **exam-der-008**

R5 Boss • 180s • quotient-rule / trig

$$\frac{d}{dx}\left(\frac{\sin x}{1+\cos x}\right)$$

350 **exam-der-009**

R5 Boss • 180s • quotient-rule / trig

$$\frac{d}{dx}\left(\frac{\cos x}{1+\sin x}\right)$$

351 **exam-der-010**

R5 Boss • 180s • product-rule / log

$$\frac{d}{dx}\left(x\log(1+x^2)\right)$$

352 **exam-der-011**

R5 Boss • 180s • quotient-rule / chain-rule

$$\frac{d}{dx}\left(\frac{(x^2+1)^3}{x-1}\right)$$

353 **exam-der-012**

R5 Boss • 180s • log-differentiation / exam-style

若$y=x^2e^x$，求$\dfrac{y'}{y}$

354 **exam-der-013**

R5 Boss • 180s • parametric / exam-style

$x=t^2+1,\ y=t^3-t.$ 求$\dfrac{dy}{dx}\Big|_{t=2}$

355 **exam-der-014**

R5 Boss • 180s • fundamental-theorem / chain-rule

$$\frac{d}{dx}\left(\int_0^{x^2} e^{t^2}\,dt\right)$$

356 **exam-der-015**

R5 Boss • 180s • fundamental-theorem / chain-rule

$$\frac{d}{dx}\left(\int_x^{x^2}\log(1+t^2)\,dt\right)$$

357 exam-der-016

R5 Boss • 180s • inverse-function / exam-style

$f(x)=x+e^x$. 若$f(0)=1$，求$(f^{-1})'(1)$

358 exam-der-017

R5 Boss • 180s • chain-rule / inverse-trig

$$\frac{d}{dx}\arctan(\sqrt{x})$$

359 exam-der-018

R5 Boss • 180s • trig / log

$$\frac{d}{dx}\log(\sec x+\tan x)$$

360 exam-der-019

R5 Boss • 180s • quotient-rule / radical

$$\frac{d}{dx}\left(\frac{x}{\sqrt{1+x^2}}\right)$$

361 exam-der-020

R5 Boss • 180s • quotient-rule / exponential

$$\frac{d}{dx}\left(\frac{e^x}{1+e^x}\right)$$

362 exam-multi-001

R5 Boss • 180s • multivariable / partial-derivative

$$\frac{\partial}{\partial x}\left(x^2y+\sin(xy)\right)$$

363 exam-multi-002

R5 Boss • 180s • multivariable / higher-derivative

$$\frac{\partial^2}{\partial x\partial y}\left(x^2y^3+e^{xy}\right)$$

364 exam-multi-003

R5 Boss • 180s • multivariable / gradient

若$f(x,y)=x^2+y^2$，求$|\nabla f(1,2)|^2$

365 exam-multi-004

R5 Boss • 180s • multivariable / hessian

求$f(x,y)=x^3+y^3-3xy$ 在$(1,1)$ 的 Hessian determinant

366 exam-multi-005

R5 Boss • 180s • multivariable / directional-derivative

求$f(x,y)=x^2y$ 在$(1,2)$ 沿$(3/5,4/5)$ 的方向導數

367 exam-multi-008

R5 Boss • 180s • multivariable / jacobian

若$u=x^2-y^2$，$v=2xy$，求$\dfrac{\partial(u,v)}{\partial(x,y)}$

368 exam-multi-009

R5 Boss • 180s • multivariable / nabla

$$\nabla\cdot(x^2y,\ y^2z,\ z^2x)$$

369 exam-multi-010

R5 Boss • 180s • multivariable / nabla

求$\nabla\times(y^2,\ z^2,\ x^2)$ 的z-component

370 uni-der-001

R6 Boss+ • 210s • log-differentiation / trig

$$\frac{d}{dx}\left(x^{\cos x}\right)$$

371 uni-der-002

R6 Boss+ • 210s • log-differentiation / trig

$$\frac{d}{dx}\left((\sin x)^x\right)$$

372 **uni-der-003**

R6 Boss+ • 210s • higher-derivative / exponential

$$\frac{d^3}{dx^3}\left(e^{2x}\sin x\right)\Big|_{x=0}$$

373 **uni-der-004**

R6 Boss+ • 210s • higher-derivative / taylor

$$\frac{d^4}{dx^4}\left(x^2 e^x\right)\Big|_{x=0}$$

374 **uni-der-005**

R6 Boss+ • 210s • higher-derivative / log

$$\frac{d^2}{dx^2}\log\left(x+\sqrt{1+x^2}\right)$$

375 **uni-der-006**

R6 Boss+ • 210s • fundamental-theorem / chain-rule

$$\frac{d}{dx}\left(\int_0^{\sin x} e^{t^2}\,dt\right)$$

376 **uni-der-007**

R6 Boss+ • 210s • fundamental-theorem / chain-rule

$$\frac{d}{dx}\left(\int_x^{2x} \log(1+t^2)\,dt\right)$$

377 **uni-der-008**

R6 Boss+ • 210s • inverse-trig / chain-rule

$$\frac{d}{dx}\arctan\left(\frac{2x}{1-x^2}\right)$$

378 **uni-der-009**

R6 Boss+ • 210s • parametric / exam-style

$x=t+\frac{1}{t},\ y=t-\frac{1}{t}$. 求 $\frac{dy}{dx}\Big|_{t=2}$

379 **uni-der-010**

R6 Boss+ • 210s • polar-curve / parametric

$r=1+\cos\theta$. 求切線斜率於$\theta=\frac{\pi}{2}$

380 **uni-der-011**

R6 Boss+ • 210s • implicit-differentiation / exam-style

$x^2+xy+y^2=3$. 求$\frac{dy}{dx}$ 於$(1,1)$

381 **uni-der-012**

R6 Boss+ • 210s • multivariable / hessian

求$f(x,y)=x^4+y^4-4xy$ 在$(1,1)$ 的 Hessian determinant

382 **uni-der-013**

R6 Boss+ • 210s • lagrange-multiplier / exam-style

在$x^2+4y^2=8$ 上求xy 的最大值

383 **uni-der-014**

R6 Boss+ • 210s • multivariable / directional-derivative

求$f(x,y)=e^{xy}$ 在$(0,2)$ 沿$(3/5,4/5)$ 的方向導數

384 **uni-der-015**

R6 Boss+ • 210s • multivariable / nabla

$\nabla\cdot(xe^y,\ ye^z,\ ze^x)$

385 **uni-der-016**

R6 Boss+ • 210s • multivariable / nabla

求$\nabla\times(xy,\ yz,\ zx)$ 的z-component

386 **uni-der-017**

R6 Boss+ • 210s • chain-rule / log

$$\frac{d}{dx}\log(\log(\log x))$$

387 **uni-der-018**

R6 Boss+ • 210s • chain-rule / log

$$\frac{d}{dx}\exp\left((\log x)^3\right)$$

388 **uni-der-019**

R6 Boss+ • 210s • higher-derivative / trig

$$\frac{d^5}{dx^5}\sin(2x)\Big|_{x=0}$$

389 **uni-der-020**

R6 Boss+ • 210s • log-differentiation / exam-style

$$\frac{d}{dx}\left((1+x^2)^x\right)$$

390 **depth-der-001**

R5 Boss • 220s • log-differentiation / power-exponential

$$\frac{d}{dx}\left(x^{x^2}\right)$$

391 **depth-der-002**

R6 Boss+ • 270s • log-differentiation / trig

$$\frac{d}{dx}\left((\sin x)^{\cos x}\right)$$

392 **depth-der-003**

R5 Boss • 220s • higher-derivative / product-rule

$$\frac{d^2}{dx^2}\left(xe^x\right)$$

393 **depth-der-004**

R5 Boss • 220s • higher-derivative / log

$$\frac{d^3}{dx^3}\left(x^2\log x\right)$$

394 **depth-der-005**

R6 Boss+ • 270s • fundamental-theorem / chain-rule

$$\frac{d}{dx}\left(\int_{x^2}^{x^3}\cos(t^2)\,dt\right)$$

395 **depth-der-006**

R5 Boss • 220s • chain-rule / log

$$\frac{d}{dx}\log\left(x+\sqrt{x^2+4}\right)$$

396 **depth-der-007**

R5 Boss • 220s • implicit-differentiation / exam-style

$$x^3+y^3=6xy,\quad \frac{dy}{dx}\Big|_{(3,3)}$$

397 **depth-der-008**

R5 Boss • 220s • parametric / exam-style

$$x=e^t,\ y=te^t.\quad \frac{dy}{dx}\Big|_{t=0}$$

398 **depth-der-009**

R6 Boss+ • 270s • parametric / higher-derivative

$$x=t^2,\ y=t^3.\quad \frac{d^2y}{dx^2}\Big|_{t=1}$$

399 **depth-der-010**

R6 Boss+ • 270s • inverse-function / higher-derivative

$f(x)=e^x+x^2,\ f(0)=1.$ find $(f^{-1})''(1)$

400 **depth-der-011**

R5 Boss • 220s • multivariable / hessian

Hessian determinant of $f(x,y)=x^2y+y^3$ at $(1,1)$

401 **depth-der-012**

R6 Boss+ • 270s • multivariable / nabla

$$\Delta\log(x^2+y^2)$$

402 **depth-der-013**
R5 Boss • 220s • multivariable / nabla
$\nabla \cdot (xyz,\ x^2z,\ xy^2)$

403 **depth-der-014**
R5 Boss • 220s • multivariable / nabla
z-component of $\nabla \times (x^2y,\ xy^2,\ z)$

404 **depth-der-015**
R6 Boss+ • 270s • inverse-trig / chain-rule
$$\frac{d}{dx} \arccos\left(\frac{1-x^2}{1+x^2}\right)$$

405 **depth-der-016**
R5 Boss • 220s • quotient-rule / log
$$\frac{d}{dx}\left(\frac{(\log x)^2}{x}\right)$$

406 **depth-der-017**
R5 Boss • 220s • higher-derivative / trig
$$\left.\frac{d^4}{dx^4}\cos(3x)\right|_{x=0}$$

407 **depth-der-018**
R5 Boss • 220s • chain-rule / radical
$$\frac{d}{dx}\sqrt{x+\sqrt{x}}$$

408 **burst-der-001**
R6 Boss+ • 240s • higher-derivative / super-high-derivative
$$\left.\frac{d^{20}}{dx^{20}}e^{2x}\right|_{x=0}$$

409 **burst-der-002**
R6 Boss+ • 240s • higher-derivative / super-high-derivative
$$\left.\frac{d^{15}}{dx^{15}}\sin(3x)\right|_{x=0}$$

410 **burst-der-003**
R6 Boss+ • 240s • higher-derivative / super-high-derivative
$$\left.\frac{d^{18}}{dx^{18}}\cosh(2x)\right|_{x=0}$$

411 **burst-der-004**
R6 Boss+ • 240s • higher-derivative / super-high-derivative
$$\left.\frac{d^{10}}{dx^{10}}(x^3e^x)\right|_{x=0}$$

412 **burst-der-005**
R6 Boss+ • 240s • higher-derivative / super-high-derivative
$$\left.\frac{d^{11}}{dx^{11}}(x^2\sin(2x))\right|_{x=0}$$

413 **burst-der-006**
R6 Boss+ • 240s • higher-derivative / super-high-derivative
$$\left.\frac{d^8}{dx^8}\log(1+x)\right|_{x=0}$$

414 **burst-der-007**
R6 Boss+ • 240s • higher-derivative / super-high-derivative
$$\left.\frac{d^{12}}{dx^{12}}\frac{1}{1-x}\right|_{x=0}$$

415 **burst-der-008**
R6 Boss+ • 240s • higher-derivative / super-high-derivative
$$\left.\frac{d^{12}}{dx^{12}}(e^x\cos x)\right|_{x=0}$$

416 **burst-der-009**
R6 Boss+ • 240s • higher-derivative / super-high-derivative
$$\left.\frac{d^{14}}{dx^{14}}(x^4e^{-x})\right|_{x=0}$$

417 **burst-der-010**

R6 Boss+ • 240s • higher-derivative / super-high-derivative

$$\frac{d^9}{dx^9}(e^{2x}\sin x)\Big|_{x=0}$$

418 **burst-boss-der-001**

R6 Boss+ • 300s • true-boss / boss-rank

$$\frac{d^{12}}{dx^{12}}(\log(1+x))^2\Big|_{x=0}$$

419 **burst-boss-der-002**

R6 Boss+ • 300s • true-boss / boss-rank

$$\frac{d^{11}}{dx^{11}}(\log(1+x))^2\Big|_{x=0}$$

420 **burst-boss-der-003**

R6 Boss+ • 300s • true-boss / boss-rank

$$\frac{d^{10}}{dx^{10}}(e^{x^2}\cos x)\Big|_{x=0}$$

421 **burst-boss-der-004**

R6 Boss+ • 300s • true-boss / boss-rank

$$\frac{d^{14}}{dx^{14}}(e^{x}\sin(2x))\Big|_{x=0}$$

422 **burst-boss-der-005**

R6 Boss+ • 300s • true-boss / boss-rank

$$\frac{d^{16}}{dx^{16}}(e^{x}\cos(2x))\Big|_{x=0}$$

423 **burst-boss-der-006**

R6 Boss+ • 300s • true-boss / boss-rank

$$\frac{d^{18}}{dx^{18}}(x^4e^{3x})\Big|_{x=0}$$

424 **burst-boss-der-007**

R6 Boss+ • 300s • true-boss / boss-rank

$$\frac{d^{16}}{dx^{16}}(x^3\sin(2x))\Big|_{x=0}$$

425 **burst-boss-der-008**

R6 Boss+ • 300s • true-boss / boss-rank

$$\frac{d^{18}}{dx^{18}}(x^2\cos(3x))\Big|_{x=0}$$

426 **burst-boss-der-009**

R6 Boss+ • 300s • true-boss / boss-rank

$$\frac{d^{10}}{dx^{10}}\frac{x^2}{(1-x)^4}\Big|_{x=0}$$

427 **burst-boss-der-010**

R6 Boss+ • 300s • true-boss / boss-rank

$$\frac{d^{15}}{dx^{15}}\frac{x^3}{1+x^2}\Big|_{x=0}$$

428 **burst-boss2-der-001**

R6 Boss+ • 360s • true-boss / boss-rank

$$\frac{d^{21}}{dx^{21}}(e^{x}\sin x)\Big|_{x=0}$$

429 **burst-boss2-der-002**

R6 Boss+ • 360s • true-boss / boss-rank

$$\frac{d^{20}}{dx^{20}}(e^{x}\cos x)\Big|_{x=0}$$

430 **burst-boss2-der-003**

R6 Boss+ • 360s • true-boss / boss-rank

$$\frac{d^{17}}{dx^{17}}(x^5e^{2x})\Big|_{x=0}$$

431 **burst-boss2-der-004**

R6 Boss+ • 360s • true-boss / boss-rank

$$\frac{d^{18}}{dx^{18}}(x^6\cos(2x))\Big|_{x=0}$$

432 **burst-boss2-der-005**

R6 Boss+ • 360s • true-boss / boss-rank

$$\frac{d^{19}}{dx^{19}}(x^4\sin(3x))\Big|_{x=0}$$

433 **burst-boss2-der-006**

R6 Boss+ • 360s • true-boss / boss-rank

$$\frac{d^{10}}{dx^{10}}\frac{1}{(1-x)^3}\Big|_{x=0}$$

434 **burst-boss2-der-007**

R6 Boss+ • 360s • true-boss / boss-rank

$$\frac{d^{10}}{dx^{10}}\frac{x^2}{(1-x)^5}\Big|_{x=0}$$

435 **burst-boss2-der-008**

R6 Boss+ • 360s • true-boss / boss-rank

$$\frac{d^{16}}{dx^{16}}\frac{x^4}{(1+x)^2}\Big|_{x=0}$$

436 **burst-boss3-der-001**

R6 Boss+ • 420s • true-boss / boss-rank

$$\frac{d^{20}}{dx^{20}}(\log(1+x))^2\Big|_{x=0}$$

437 **burst-boss3-der-002**

R6 Boss+ • 420s • true-boss / boss-rank

$$\frac{d^{18}}{dx^{18}}\frac{x^4}{(1-x)^7}\Big|_{x=0}$$

438 **burst-boss3-der-003**

R6 Boss+ • 420s • true-boss / boss-rank

$$\frac{d^{20}}{dx^{20}}\frac{x^5}{(1-x)^6}\Big|_{x=0}$$

439 **burst-boss3-der-004**

R6 Boss+ • 420s • true-boss / boss-rank

$$\frac{d^{16}}{dx^{16}}\frac{x^3}{(1-x)^8}\Big|_{x=0}$$

440 **burst-boss3-der-005**

R6 Boss+ • 420s • true-boss / boss-rank

$$\frac{d^{31}}{dx^{31}}\left(x^7e^{-x^2}\right)\Big|_{x=0}$$

441 **burst-boss3-der-006**

R6 Boss+ • 420s • true-boss / boss-rank

$$\frac{d^{30}}{dx^{30}}\left(x^6\cosh(x^2)\right)\Big|_{x=0}$$

442 **burst-boss3-der-007**

R6 Boss+ • 420s • true-boss / boss-rank

$$\frac{d^{27}}{dx^{27}}\left(x^5\sin(x^2)\right)\Big|_{x=0}$$

443 **burst-boss3-der-008**

R6 Boss+ • 420s • true-boss / boss-rank

$$\frac{d^{29}}{dx^{29}}\left(x^5\cos(x^2)\right)\Big|_{x=0}$$

積分 Integrals

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444 **int-001**

R1 Warm-up • 30s • rank-1 / beginner-friendly

$$\int 6x^2\,dx$$

445 **int-002**

R1 Warm-up • 25s • rank-1 / beginner-friendly

$$\int \cos x\,dx$$

446 **int-003**

R1 Warm-up • 30s • rank-1 / beginner-friendly

$$\int \frac{1}{x}\,dx$$

447 **int-004**

R1 Warm-up • 35s • rank-1 / beginner-friendly

$$\int e^{3x}\,dx$$

448 **int-005**

R2 Basic • 45s • rank-2 / beginner-friendly

$$\int xe^{x^2}\,dx$$

449 **int-006**

R2 Basic • 40s • rank-2 / beginner-friendly

$$\int \frac{2x}{1+x^2}\,dx$$

450 **int-007**

R2 Basic • 45s • rank-2 / beginner-friendly

$$\int x\cos(x^2)\,dx$$

451 **int-008**

R2 Basic • 35s • rank-2 / beginner-friendly

$$\int_0^1 3x^2\,dx$$

452 **int-009**

R3 Standard • 65s • rank-3

$$\int x\ln x\,dx$$

453 **int-010**

R3 Standard • 70s • rank-3

$$\int \sin^3 x\,dx$$

454 **int-011**

R3 Standard • 60s • rank-3

$$\int \frac{1}{x^2+4}\,dx$$

455 **int-012**

R2 Basic • 70s • rank-2 / beginner-friendly

$$\int_0^{\pi/2} \sin^2 x\,dx$$

456 **int-013**

R3 Standard • 100s • rank-3

$$\int (\ln x)^2\,dx$$

457 **int-014**

R3 Standard • 85s • rank-3

$$\int \frac{x^3}{x^2+1}\,dx$$

458 **int-015**

R4 Advanced • 95s • rank-4

$$\int e^x \sin x\,dx$$

459 **int-016**

R3 Standard • 85s • rank-3

$$\int_0^1 \ln(1+x)\,dx$$

460 **int-017**

R3 Standard • 80s • rank-3

$$\int_0^{\pi/2} x\sin x\,dx$$

461 **int-018**

R3 Standard • 70s • rank-3

$$\int \frac{\sin x}{1+\cos x}\,dx$$

462 **int-019**

R4 Advanced • 100s • rank-4

$$\int \frac{1}{1+\sqrt{x}}\,dx$$

463 **int-020**

R3 Standard • 70s • rank-3

$$\int_0^1 \frac{x^2}{1+x^3}\,dx$$

464 **td-int-001**

R6 Boss+ • 95s • rank-6 / boss-rank

$$\int_0^\infty \frac{\cos(3x)}{1+x^2}\,dx$$

465 **td-int-002**

R6 Boss+ • 100s • rank-6 / boss-rank

$$\int_0^\pi \log(5+3\cos x)\,dx$$

466 **td-int-003**

R4 Advanced • 85s • rank-4

$$\int_0^\infty \frac{e^{-5x}-e^{-2x}}{x}\,dx$$

467 **td-int-004**
R4 Advanced • 100s • rank-4

$$\int_0^{\infty} \frac{\sin(2x)\sin(5x)}{x}\,dx$$

468 **td-int-005**
R6 Boss+ • 80s • rank-6 / boss-rank

Area of $3x^2 + 2xy + 2y^2 \leq 1$

469 **td-int-006**
R6 Boss+ • 90s • rank-6 / boss-rank

$$\int_{-\infty}^{\infty} x^4 e^{-3x^2}\,dx$$

470 **int-021**
R4 Advanced • 60s • multivariable / double-integral

$$\int_0^1 \int_0^1 (x+y)\,dy\,dx$$

471 **int-022**
R4 Advanced • 55s • multivariable / area

$$\iint_D 1\,dA, \quad D = \{x \geq 0, y \geq 0, x+y \leq 2\}$$

472 **int-023**
R4 Advanced • 65s • multivariable / double-integral

$$\int_0^1 \int_0^x xy\,dy\,dx$$

473 **int-024**
R4 Advanced • 70s • multivariable / double-integral

$$\int_0^1 \int_0^{1-x} (x+y)\,dy\,dx$$

474 **int-025**
R4 Advanced • 75s • multivariable / polar-coordinates

$$\iint_{x^2+y^2 \leq 1} (x^2+y^2)\,dA$$

475 **int-026**
R4 Advanced • 55s • multivariable / double-integral

$$\int_0^2 \int_0^3 xy\,dy\,dx$$

476 **int-027**
R2 Basic • 55s • substitution / log

$$\int \frac{x}{1+x^2}\,dx$$

477 **int-028**
R3 Standard • 60s • integration-by-parts / log

$$\int \log x\,dx$$

478 **int-029**
R2 Basic • 55s • inverse-trig / standard-integral

$$\int \frac{1}{x^2+9}\,dx$$

479 **int-030**
R2 Basic • 55s • substitution / definite-integral

$$\int_0^{\pi/2} \sin^3 x \cos x\,dx$$

480 **int-031**
R4 Advanced • 70s • multivariable / double-integral

$$\int_0^1 \int_y^1 x^2\,dx\,dy$$

481 **int-032**
R4 Advanced • 60s • multivariable / polar-coordinates

$$\iint_{1 \leq x^2+y^2 \leq 4} 1\,dA$$

482 **int-033**
R4 Advanced • 85s • multivariable / polar-coordinates

$$\iint_{x^2+y^2 \leq 4} x^2\,dA$$

483 **int-034**
R4 Advanced • 60s • multivariable / area

$$\int_0^1 \int_0^{\sqrt{1-x^2}} 1\,dy\,dx$$

484 **int-035**
R4 Advanced • 60s • multivariable / polar-coordinates

$$\int_0^{2\pi} \int_0^1 r^2 \cdot r\,dr\,d\theta$$

485 **int-036**
R4 Advanced • 65s • multivariable / double-integral

$$\int_0^1 \int_0^{1-y} x\,dx\,dy$$

486 **int-037**
R3 Standard • 50s • substitution / rank-3

$$\int x e^{x^2}\,dx$$

487 **int-038**
R3 Standard • 65s • integration-by-parts / rank-3

$$\int x \cos x \, dx$$

488 **int-039**
R3 Standard • 50s • inverse-trig / definite-integral

$$\int_0^1 \frac{1}{1+x^2} \, dx$$

489 **int-040**
R5 Boss • 70s • improper-integral / gamma

$$\int_0^\infty x e^{-x} \, dx$$

490 **int-041**
R3 Standard • 70s • substitution / u-sub

$$\int \frac{x^3}{1+x^4} \, dx$$

491 **int-042**
R3 Standard • 75s • substitution / u-sub

$$\int x^2 \sqrt{1+x^3} \, dx$$

492 **int-043**
R4 Advanced • 95s • substitution / u-sub

$$\int \frac{x}{\sqrt{1+x^2}\,\left(1+\sqrt{1+x^2}\right)} \, dx$$

493 **int-044**
R3 Standard • 85s • substitution / u-sub

$$\int \frac{x e^{x^2}}{1+e^{x^2}} \, dx$$

494 **int-045**
R3 Standard • 70s • substitution / u-sub

$$\int \frac{(\log x)^2}{x} \, dx$$

495 **int-046**
R4 Advanced • 80s • substitution / u-sub

$$\int_0^1 \frac{2x}{(1+x^2)^3} \, dx$$

496 **int-047**
R3 Standard • 75s • trig-substitution / definite-integral

$$\int_0^2 \sqrt{4-x^2} \, dx$$

497 **int-048**
R4 Advanced • 90s • trig-substitution / definite-integral

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}} \, dx$$

498 **int-049**
R4 Advanced • 95s • trig-substitution / definite-integral

$$\int_0^2 \frac{x^2}{\sqrt{4-x^2}} \, dx$$

499 **int-050**
R4 Advanced • 95s • trig-substitution / inverse-trig

$$\int_1^2 \frac{1}{x\sqrt{x^2-1}} \, dx$$

500 **int-051**
R4 Advanced • 95s • trig-substitution / radical

$$\int \frac{1}{(x^2+4)^{3/2}} \, dx$$

501 **int-052**
R3 Standard • 90s • integration-by-parts / ibp

$$\int x^2 e^x \, dx$$

502 **int-053**
R4 Advanced • 100s • integration-by-parts / ibp

$$\int x^2 \sin x \, dx$$

503 **int-054**
R4 Advanced • 95s • integration-by-parts / ibp

$$\int e^x \sin x \, dx$$

504 **int-055**
R4 Advanced • 100s • integration-by-parts / ibp

$$\int (\log x)^2 \, dx$$

505 **int-056**
R4 Advanced • 100s • integration-by-parts / ibp

$$\int_0^1 x \arctan x \, dx$$

506 **int-057**
R5 Boss • 110s • ode-style / convolution

Find $F(x) = \displaystyle\int_0^x e^{x-t} \sin t \, dt$

507 **int-058**

R5 Boss • 105s • ode-style / convolution

Find $F(x) = \int_0^x e^{2(x-t)} t \, dt$

508 **int-059**

R5 Boss • 110s • ode-style / convolution

Find $F(x) = \int_0^x e^{-(x-t)} t^2 \, dt$

509 **int-060**

R5 Boss • 110s • ode-style / convolution

Find $F(x) = \int_0^x \sin(x-t) \cos t \, dt$

510 **int-061**

R5 Boss • 85s • kings-property / symmetry

$$\int_0^{\pi/2} \frac{\sin^3 x}{\sin^3 x + \cos^3 x} \, dx$$

511 **int-062**

R5 Boss • 80s • kings-property / symmetry

$$\int_0^1 \frac{e^x}{e^x + e^{1-x}} \, dx$$

512 **int-063**

R5 Boss • 90s • kings-property / symmetry

$$\int_0^{\pi/2} \frac{1 + \sin x}{2 + sinx + cosx} \, dx$$

513 **int-064**

R5 Boss • 110s • kings-property / symmetry

$$\int_0^{\pi} \frac{x \sin x}{1 + cos^2 x} \, dx$$

514 **int-065**

R5 Boss • 95s • frullani / improper-integral

$$\int_0^{\infty} \frac{e^{-3x} - e^{-7x}}{x} \, dx$$

515 **int-066**

R5 Boss • 90s • frullani / improper-integral

$$\int_0^{\infty} \frac{e^{-x} - e^{-4x}}{x} \, dx$$

516 **int-067**

R5 Boss • 90s • frullani / improper-integral

$$\int_0^{\infty} \frac{e^{-2x} - e^{-8x}}{x} \, dx$$

517 **int-068**

R5 Boss • 100s • frullani / cosine-integral

$$\int_0^{\infty} \frac{\cos(2x) - \cos(5x)}{x} \, dx$$

518 **int-069**

R5 Boss • 95s • frullani / cosine-integral

$$\int_0^{\infty} \frac{\cos x - \cos(4x)}{x} \, dx$$

519 **int-070**

R5 Boss • 95s • frullani / cosine-integral

$$\int_0^{\infty} \frac{\cos(3x) - \cos(9x)}{x} \, dx$$

520 **int-071**

R5 Boss • 110s • frullani / trig

$$\int_0^{\infty} \frac{\sin(3x)\sin(8x)}{x} \, dx$$

521 **int-072**

R5 Boss • 105s • frullani / trig

$$\int_0^{\infty} \frac{\sin(2x)\sin(6x)}{x} \, dx$$

522 **int-073**
R5 Boss • 105s • frullani / trig

$$\int_0^\infty \frac{\sin x \sin(4x)}{x}\,dx$$

523 **int-074**
R5 Boss • 115s • frullani / inverse-trig

$$\int_0^\infty \frac{\arctan(5x) - \arctan(2x)}{x}\,dx$$

524 **int-075**
R5 Boss • 110s • frullani / rational

$$\int_0^\infty \frac{1}{x}\left(\frac{1}{1+2x} - \frac{1}{1+5x}\right)dx$$

525 **int-076**
R5 Boss • 105s • frullani / rational

$$\int_0^\infty \frac{1}{x}\left(\frac{1}{1+x} - \frac{1}{1+6x}\right)dx$$

526 **gap-int-pf-001**
R3 Standard • 85s • partial-fraction / rational-integral

$$\int \frac{1}{x^2-1}\,dx$$

527 **gap-int-pf-002**
R3 Standard • 60s • partial-fraction / rational-integral

$$\int \frac{x+1}{x^2+x}\,dx$$

528 **gap-int-pf-003**
R3 Standard • 75s • partial-fraction / rational-integral

$$\int \frac{1}{x(x+2)}\,dx$$

529 **gap-int-pf-004**
R3 Standard • 75s • partial-fraction / rational-integral

$$\int \frac{2x+3}{x^2+3x+2}\,dx$$

530 **gap-int-pf-005**
R3 Standard • 55s • partial-fraction / repeated-factor

$$\int \frac{1}{(x+1)^2}\,dx$$

531 **gap-int-pf-006**
R4 Advanced • 95s • partial-fraction / rational-integral

$$\int \frac{3x+5}{x^2+x-2}\,dx$$

532 **gap-int-pf-007**
R4 Advanced • 90s • partial-fraction / rational-integral

$$\int \frac{x+4}{x^2+5x+6}\,dx$$

533 **gap-int-polar-001**
R2 Basic • 60s • polar-curve / polar-area

Area enclosed by $r = 2$ from $0 \le \theta \le \pi/2$

534 **gap-int-polar-002**
R3 Standard • 80s • polar-curve / polar-area

Area enclosed by $r = \sin\theta$ for $0 \le \theta \le \pi$

535 **gap-int-triple-001**
R6 Boss+ • 75s • triple-integral / iterated-integral

$$\int_0^1\int_0^1\int_0^1 xyz\,dz\,dy\,dx$$

536 **gap-int-triple-002**
R6 Boss+ • 75s • triple-integral / iterated-integral

$$\int_0^1\int_0^1\int_0^1 (x+y+z)\,dz\,dy\,dx$$

537 **gap-int-triple-003**
R6 Boss+ • 80s • triple-integral / volume

$$\iiint_{x+y+z\le 1,\ x,y,z\ge 0} 1\,dV$$

538 **gap-int-triple-004**
R6 Boss+ • 80s • triple-integral / iterated-integral

$$\int_0^1\int_0^x\int_0^y 1\,dz\,dy\,dx$$

539 **gap-int-triple-005**
R6 Boss+ • 70s • triple-integral / cylindrical-coordinates

Volume of the cylinder $x^2 + y^2 \le 4,\ 0 \le z \le 3$

540 **gap-int-triple-006**

R6 Boss+ • 70s • triple-integral / spherical-coordinates

Volume of the ball $x^2+y^2+z^2 \le 4$

541 **gap-int-cov-001**

R6 Boss+ • 75s • change-of-variables / jacobian

$u=x+y,\ v=x-y.\ \left|\frac{\partial(x,y)}{\partial(u,v)}\right| = ?$

542 **gap-int-cov-002**

R6 Boss+ • 80s • change-of-variables / jacobian

Area of $0 \le x+y \le 1,\ 0 \le x-y \le 2$

543 **gap-tech-003**

R2 Basic • 45s • technique-recognition / substitution

Technique for $\int xe^{x^2}\,dx$

544 **gap-tech-004**

R3 Standard • 45s • technique-recognition / integration-by-parts

Technique for $\int x\log x\,dx$

545 **gap-tech-006**

R3 Standard • 45s • technique-recognition / partial-fraction

Technique for $\int \frac{1}{x^2-1}\,dx$

546 **mob-tech-003**

R2 Basic • 45s • technique-sprint / technique-recognition

$\int xe^{x^2}\,dx$ Use which technique?

547 **mob-tech-004**

R2 Basic • 45s • technique-sprint / technique-recognition

$\int x\log x\,dx$ Use which technique?

548 **mob-tech-005**

R2 Basic • 45s • technique-sprint / technique-recognition

$\int \frac{dx}{x^2-1}$ Use which technique?

549 **mob-tech-006**

R2 Basic • 45s • technique-sprint / technique-recognition

$\int \frac{dx}{\sqrt{a^2-x^2}}$ Use which technique?

550 **mob-tech-010**

R2 Basic • 45s • technique-sprint / technique-recognition

$\int_0^1 f(x)\,dx$ with $f(x)+f(1-x)$ Use which technique?

551 **mob-tech-011**

R2 Basic • 45s • technique-sprint / technique-recognition

$\int_0^\infty \frac{e^{-ax}-e^{-bx}}{x}\,dx$ Use which technique?

552 **mob-tech-013**

R2 Basic • 45s • technique-sprint / technique-recognition

$\int_C \nabla\phi\cdot d\mathbf{r}$ Use which technique?

553 **mob-tech-015**

R2 Basic • 45s • technique-sprint / technique-recognition

$\int_0^1 x^{a-1}(1-x)^{b-1}\,dx$ Use which technique?

554 **mob-tech-016**

R2 Basic • 45s • technique-sprint / technique-recognition

$\int_0^\infty x^{s-1}e^{-x}\,dx$ Use which technique?

555 **mob-tech-017**

R2 Basic • 45s • technique-sprint / technique-recognition

$\int_0^{\pi/2} \sin^n x\,dx$ Use which technique?

556 **mob-trap-002**

R2 Basic • 45s • trap-drill / rank-2

$\int \cos(3x)\,dx$ Main trap?

557 **mob-trap-003**

R2 Basic • 45s • trap-drill / rank-2

$\int \frac{1}{x^2+1}\,dx$ Main trap?

558 **mob-trap-006**

R2 Basic • 45s • trap-drill / rank-2

$\int_0^\infty e^{-ax}\,dx$ Main trap?

559 **mob-trap-008**
R2 Basic • 45s • trap-drill / rank-2

$\iint_R f(x,y)\,dA$ under change of variables Main trap?

560 **mob-trap-010**
R2 Basic • 45s • trap-drill / rank-2

$\int \sec x \tan x\,dx$ Main trap?

561 **mob-trap-012**
R2 Basic • 45s • trap-drill / rank-2

$B(a,b) = \dfrac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$ Main trap?

562 **mob-beta-001**
R5 Boss • 70s • beta-function / special-functions

$B(1,1)$

563 **mob-beta-002**
R5 Boss • 70s • beta-function / special-functions

$B(2,1)$

564 **mob-beta-003**
R5 Boss • 70s • beta-function / special-functions

$B(2,2)$

565 **mob-beta-004**
R5 Boss • 70s • beta-function / special-functions

$B(3,2)$

566 **mob-beta-005**
R5 Boss • 70s • beta-function / special-functions

$B(1/2,1/2)$

567 **mob-beta-006**
R5 Boss • 70s • beta-function / special-functions

$B(1/2,1)$

568 **mob-beta-007**
R5 Boss • 70s • beta-function / special-functions

$B(3/2,1/2)$

569 **mob-beta-008**
R5 Boss • 70s • beta-function / special-functions

$\int_0^1 \sqrt{x(1-x)}\,dx$

570 **mob-beta-009**
R5 Boss • 70s • beta-function / special-functions

$B(3,3)$

571 **mob-beta-010**
R5 Boss • 70s • beta-function / special-functions

$B(4,1)$

572 **mob-gamma-001**
R5 Boss • 70s • gamma-function / special-functions

$\Gamma(1)$

573 **mob-gamma-002**
R5 Boss • 70s • gamma-function / special-functions

$\Gamma(1/2)$

574 **mob-gamma-003**
R5 Boss • 70s • gamma-function / special-functions

$\Gamma(3/2)$

575 **mob-gamma-004**
R5 Boss • 70s • gamma-function / special-functions

$\Gamma(5/2)$

576 **mob-gamma-005**

R5 Boss • 70s • gamma-function / special-functions

$\Gamma(6)$

577 **mob-gamma-006**

R5 Boss • 70s • gamma-function / special-functions

$$\int_0^\infty x^2 e^{-x}\, dx$$

578 **mob-gamma-007**

R5 Boss • 70s • gamma-function / special-functions

$$\int_0^\infty x^{1/2} e^{-x}\, dx$$

579 **mob-gamma-008**

R5 Boss • 70s • gamma-function / special-functions

$\Gamma(4)$

580 **mob-gamma-009**

R5 Boss • 70s • gamma-function / special-functions

$\Gamma(7/2)$

581 **mob-gamma-010**

R5 Boss • 70s • gamma-function / special-functions

$\Gamma(0+1)$

582 **mob-wallis-001**

R5 Boss • 70s • wallis / special-functions

$$\int_0^{\pi/2} \sin^2 x\, dx$$

583 **mob-wallis-002**

R5 Boss • 70s • wallis / special-functions

$$\int_0^{\pi/2} \sin^3 x\, dx$$

584 **mob-wallis-003**

R5 Boss • 70s • wallis / special-functions

$$\int_0^{\pi/2} \sin^4 x\, dx$$

585 **mob-wallis-004**

R5 Boss • 70s • wallis / special-functions

$$\int_0^{\pi/2} \sin^5 x\, dx$$

586 **mob-wallis-005**

R5 Boss • 70s • wallis / special-functions

$$\int_0^{\pi/2} \cos^2 x\, dx$$

587 **mob-wallis-006**

R5 Boss • 70s • wallis / special-functions

$$\int_0^{\pi/2} \cos^4 x\, dx$$

588 **mob-wallis-007**

R5 Boss • 70s • wallis / special-functions

$$\int_0^{\pi/2} \sin^2 x \cos^2 x\, dx$$

589 **mob-wallis-008**

R5 Boss • 70s • wallis / special-functions

$$\int_0^{\pi/2} \sin x\, dx$$

590 **mob-wallis-009**

R5 Boss • 70s • wallis / special-functions

$$\int_0^{\pi/2} \cos x\, dx$$

591 **mob-wallis-010**
R5 Boss • 70s • wallis / special-functions

$$\int_0^{\pi/2} \sin x \cos x \, dx$$

592 **rel-basic-013**
R1 Warm-up • 45s • basic-integral / beginner-foundation

$$\int_0^1 x^2 \, dx$$

593 **rel-basic-014**
R1 Warm-up • 45s • basic-integral / beginner-foundation

$$\int_0^\pi \sin x \, dx$$

594 **rel-basic-015**
R1 Warm-up • 45s • basic-integral / beginner-foundation

$$\int_0^1 e^{2x} \, dx$$

595 **rel-basic-016**
R1 Warm-up • 45s • basic-integral / log

$$\int_0^1 \frac{1}{1+x} \, dx$$

596 **rel-basic-017**
R1 Warm-up • 45s • basic-integral / beginner-foundation

$$\int_0^{\pi/2} \cos x \, dx$$

597 **rel-basic-018**
R1 Warm-up • 45s • basic-integral / beginner-foundation

$$\int_0^2 3x \, dx$$

598 **rel-adv-011**
R5 Boss • 75s • improper-integral / gamma-function

$$\int_0^\infty xe^{-3x} \, dx$$

599 **rel-adv-012**
R4 Advanced • 75s • double-integral / iterated-integral

$$\int_0^1 \int_0^x (x+y) \, dy \, dx$$

600 **rel-adv-013**
R4 Advanced • 75s • double-integral / polar

$$\iint_{x^2+y^2\le 1} (x^2+y^2) \, dA$$

601 **rel-adv-014**
R4 Advanced • 75s • double-integral / iterated-integral

$$\int_0^1 \int_0^1 (x+y)^2 \, dy \, dx$$

602 **rel-adv-009**
R3 Standard • 80s • substitution / u-sub

$$\int x^3 e^{x^2} \, dx$$

603 **rel-adv-010**
R3 Standard • 80s • substitution / log

$$\int x \log(1+x^2) \, dx$$

604 **rel-boss-001**
R5 Boss • 110s • improper-integral / dirichlet-integral

$$\int_0^\infty \frac{\sin(3x)}{x} \, dx$$

605 **rel-boss-002**
R5 Boss • 110s • frullani / improper-integral

$$\int_0^\infty \frac{e^{-2x} - e^{-5x}}{x} \, dx$$

606 **rel-boss-003**
R5 Boss • 110s • frullani / cosine-integral

$$\int_0^\infty \frac{\cos(2x) - \cos(7x)}{x} \, dx$$

607 **rel-boss-004**
R5 Boss • 110s • gamma-function / special-functions

$$\int_0^\infty x^{5/2} e^{-x} \, dx$$

608 **rel-boss-005**
R5 Boss • 110s • beta-function / special-functions

$$\int_0^1 x^{-1/2}(1-x)^{3/2} \, dx$$

609 **rel-boss-006**
R5 Boss • 110s • wallis / beta-function

$$\int_0^{\pi/2} \sin^6 x \cos^4 x \, dx$$

610 **rel-boss-008**
R6 Boss+ • 110s • complex / residue

$$\operatorname{Res}_{z=0} \frac{\sin z}{z^5}$$

611 **rel-boss-009**
R6 Boss+ • 110s • complex / residue

$$\operatorname{Res}_{z=0} \frac{e^z}{z^4}$$

612 **rel-boss-010**
R6 Boss+ • 110s • change-of-variables / jacobian

$$\iint_T \frac{xy}{x+y} \, dA, \quad T = \{x, y > 0,\ x + y < 1\}$$

613 **rel-boss-011**
R6 Boss+ • 110s • triple-integral / simplex

$$\iiint_{x,y,z \geq 0,\ x+y+z \leq 1} xyz \, dV$$

614 **rel-boss-012**
R6 Boss+ • 110s • triple-integral / spherical

$$\iiint_{x^2+y^2+z^2 \leq 4} (x^2 + y^2 + z^2) \, dV$$

615 **rel-boss-014**
R6 Boss+ • 110s • gamma-function / special-functions

$$\int_0^{\infty} \frac{x}{e^x - 1} \, dx$$

616 **rel-boss-015**
R6 Boss+ • 110s • gamma-function / special-functions

$$\int_0^{\infty} \frac{x^3}{e^x - 1} \, dx$$

617 **rel-boss-016**
R5 Boss • 110s • improper-integral / kings-property

$$\int_0^{\pi/2} \log(\sin x) \, dx$$

618 **rel-boss-017**
R5 Boss • 110s • improper-integral / gamma-function

$$\int_0^1 (\log x)^2 \, dx$$

619 **rel-boss-020**
R6 Boss+ • 110s • complex / residue

$$\operatorname{Res}_{z=0} \frac{1 - \cos z}{z^3}$$

620 **rel-hard-int-001**
R5 Boss • 125s • gamma-function / improper-integral

$$\int_0^{\infty} x^5 e^{-2x} \, dx$$

621 **rel-hard-int-002**
R5 Boss • 125s • gamma-function / improper-integral

$$\int_0^{\infty} x^{3/2} e^{-4x} \, dx$$

622 **rel-hard-int-003**
R5 Boss • 125s • parameter-integral / improper-integral

$$\int_0^1 x^2 \log x \, dx$$

623 **rel-hard-int-004**
R5 Boss • 125s • parameter-integral / gamma-function

$$\int_0^1 x^3 (\log x)^2 \, dx$$

624 **rel-hard-int-005**

R5 Boss • 125s • wallis / special-functions

$$\int_0^{\pi/2} \sin^8 x \, dx$$

625 **rel-hard-int-006**

R5 Boss • 125s • wallis / special-functions

$$\int_0^{\pi/2} \sin^7 x \, dx$$

626 **rel-hard-int-007**

R5 Boss • 125s • beta-function / wallis

$$\int_0^{\pi/2} \sin^4 x \cos^2 x \, dx$$

627 **rel-hard-int-008**

R5 Boss • 125s • frullani / improper-integral

$$\int_0^{\infty} \frac{e^{-3x} - e^{-11x}}{x} \, dx$$

628 **rel-hard-int-009**

R5 Boss • 125s • frullani / cosine-integral

$$\int_0^{\infty} \frac{\cos(4x) - \cos(10x)}{x} \, dx$$

629 **rel-hard-int-010**

R5 Boss • 125s • dirichlet-integral / improper-integral

$$\int_0^{\infty} \left(\frac{\sin x}{x}\right)^2 dx$$

630 **rel-hard-int-011**

R5 Boss • 125s • improper-integral / substitution

$$\int_0^{\infty} \frac{x}{x^4 + 1} \, dx$$

631 **rel-hard-int-012**

R5 Boss • 125s • improper-integral / residue

$$\int_0^{\infty} \frac{dx}{x^4 + 1}$$

632 **rel-hard-int-013**

R5 Boss • 125s • improper-integral / symmetry

$$\int_0^{\infty} \frac{\log x}{1 + x^2} \, dx$$

633 **rel-hard-int-014**

R5 Boss • 125s • parameter-integral / series-boss

$$\int_0^{1} \frac{\log x}{1 + x} \, dx$$

634 **rel-hard-int-015**

R5 Boss • 125s • parameter-integral / improper-integral

$$\int_0^{1} x^{-1/2} \log x \, dx$$

635 **rel-hard-int-016**

R5 Boss • 125s • substitution / log

$$\int_0^{1} \frac{x^2}{1 + x^3} \, dx$$

636 **rel-hard-int-017**

R5 Boss • 125s • kings-property / improper-integral

$$\int_0^{\pi/2} (\log(\sin x) + \log(\cos x)) \, dx$$

637 **rel-hard-int-018**

R5 Boss • 125s • improper-integral / trig-substitution

$$\int_0^{\infty} \frac{dx}{(x^2 + 4)^2}$$

638 **rel-hard-cpx-001**

R6 Boss+ • 115s • complex / residue

$$\operatorname{Res}_{z=0} \frac{\sin z}{z^4}$$

639 rel-hard-cpx-002

R6 Boss+ • 115s • complex / residue

$$\operatorname{Res}_{z=0} \frac{\cos z}{z^3}$$

640 rel-hard-cpx-003

R6 Boss+ • 115s • complex / residue

$$\operatorname{Res}_{z=0} \frac{e^z - 1 - z}{z^4}$$

641 rel-hard-cpx-004

R6 Boss+ • 115s • complex / residue

$$\operatorname{Res}_{z=0} \frac{\log(1+z)}{z^3}$$

642 rel-hard-cpx-005

R6 Boss+ • 115s • complex / residue

$$\operatorname{Res}_{z=0} \frac{1}{z^2(1-z)}$$

643 rel-hard-cpx-006

R6 Boss+ • 115s • complex / residue

$$\operatorname{Res}_{z=1} \frac{e^z}{(z-1)^2}$$

644 rel-hard-cpx-007

R6 Boss+ • 115s • complex / residue

$$\operatorname{Res}_{z=0} \frac{z}{\sin^2 z}$$

645 rel-hard-cpx-008

R6 Boss+ • 115s • complex / residue

$$\operatorname{Res}_{z=0} \cot z$$

646 rel-hard-cpx-009

R6 Boss+ • 115s • complex / residue

$$\operatorname{Res}_{z=0} \frac{e^{2z} - 1}{z^3}$$

647 rel-hard-cpx-010

R6 Boss+ • 115s • complex / residue

$$\operatorname{Res}_{z=0} \frac{1 - \cos z - z^2/2}{z^5}$$

648 rel-hard-mv-001

R5 Boss • 120s • double-integral / simplex

$$\iint_{x,y\geq 0,\ x+y\leq 1} x^2 y \, dA$$

649 rel-hard-mv-002

R5 Boss • 120s • double-integral / polar

$$\iint_{x^2+y^2\leq 1} x^2 y^2 \, dA$$

650 rel-hard-mv-003

R6 Boss+ • 120s • triple-integral / spherical

$$\iiint_{x^2+y^2+z^2\leq 1} (x^2+y^2+z^2) \, dV$$

651 rel-hard-mv-004

R6 Boss+ • 120s • triple-integral / spherical

$$\iiint_{x^2+y^2+z^2\leq 9} 1 \, dV$$

652 rel-hard-mv-008

R6 Boss+ • 120s • change-of-variables / jacobian

$$\iint_R (x+y) \, dA, \quad 0 \leq x+y \leq 1,\ 0 \leq x-y \leq 2$$

653 rel-hard-mv-009

R5 Boss • 120s • double-integral / polar

$$\iint_{x^2+y^2\leq 4} (x^2+y^2)^2 \, dA$$

654 rel-hard-mv-010

R6 Boss+ • 120s • triple-integral / simplex

$$\iiint_{x,y,z\geq 0,\ x+y+z\leq 1} x^2yz\,dV$$

655 hc-usub-001

R6 Boss+ • 160s • substitution / u-sub

$$\int \frac{x^3}{\sqrt{1+x^2}}\,dx$$

656 hc-usub-002

R6 Boss+ • 160s • substitution / u-sub

$$\int x^5 e^{x^2}\,dx$$

657 hc-usub-003

R6 Boss+ • 160s • substitution / u-sub

$$\int x\left(\log(1+x^2)\right)^2\,dx$$

658 hc-usub-004

R6 Boss+ • 160s • substitution / u-sub

$$\int \frac{x^2}{1+x^6}\,dx$$

659 hc-usub-005

R6 Boss+ • 160s • substitution / u-sub

$$\int \frac{x^3}{\sqrt{1+x^4}\left(1+\sqrt{1+x^4}\right)}\,dx$$

660 hc-usub-006

R6 Boss+ • 160s • substitution / u-sub

$$\int e^x\sqrt{1+e^x}\,dx$$

661 hc-usub-007

R6 Boss+ • 150s • substitution / u-sub

$$\int_0^1 \frac{x^2}{\sqrt{1-x^6}}\,dx$$

662 hc-usub-008

R6 Boss+ • 150s • substitution / u-sub

$$\int_0^1 \frac{x^3}{(1+x^2)^3}\,dx$$

663 hc-usub-009

R6 Boss+ • 150s • substitution / u-sub

$$\int_0^{\pi/2} \frac{\sin x\cos x}{(1+\sin^2 x)^2}\,dx$$

664 hc-usub-010

R6 Boss+ • 150s • substitution / u-sub

$$\int_0^1 \frac{x^2 e^{x^3}}{1+e^{x^3}}\,dx$$

665 hc-ibp-001

R6 Boss+ • 160s • integration-by-parts / ibp

$$\int x^3 e^{2x}\,dx$$

666 hc-ibp-002

R6 Boss+ • 160s • integration-by-parts / ibp

$$\int x^2\cos(3x)\,dx$$

667 hc-ibp-003

R6 Boss+ • 160s • integration-by-parts / ibp

$$\int x^2 e^x \sin x\,dx$$

668 hc-ibp-004

R6 Boss+ • 160s • integration-by-parts / ibp

$$\int (\log x)^3\,dx$$

669 hc-ibp-005

R6 Boss+ • 160s • integration-by-parts / ibp

$$\int x(\arctan x)^2\,dx$$

670 hc-ibp-006

R6 Boss+ • 160s • integration-by-parts / ibp

$$\int e^{2x}\sin(3x)\,dx$$

671 hc-ibp-007

R6 Boss+ • 160s • integration-by-parts / ibp

$$\int x^3\log x\,dx$$

672 hc-ibp-008

R6 Boss+ • 150s • integration-by-parts / ibp

$$\int_0^1 x^4\log(1+x)\,dx$$

673 hc-ibp-009

R6 Boss+ • 150s • integration-by-parts / ibp

$$\int_0^{\pi/2} x^2\cos x\,dx$$

674 hc-ibp-010

R6 Boss+ • 150s • integration-by-parts / ibp

$$\int_0^1 x^3\arctan x\,dx$$

675 hc-rad-001

R6 Boss+ • 160s • trig-substitution / radical

$$\int \frac{dx}{x^2\sqrt{x^2-1}}$$

676 hc-rad-002

R6 Boss+ • 160s • complete-square / inverse-trig

$$\int \frac{dx}{x^2+2x+5}$$

677 hc-rad-003

R6 Boss+ • 160s • partial-fraction / rational

$$\int \frac{x^2+1}{x^3-x}\,dx$$

678 hc-rad-004

R6 Boss+ • 160s • partial-fraction / rational

$$\int \frac{dx}{x^4-1}$$

679 hc-rad-005

R6 Boss+ • 150s • trig-substitution / radical

$$\int_2^4 \frac{\sqrt{x^2-4}}{x}\,dx$$

680 hc-rad-006

R6 Boss+ • 150s • trig / rationalize

$$\int_0^{\pi/2} \frac{dx}{1+sinx}$$

681 hc-rad-007

R6 Boss+ • 150s • trig / beta-function

$$\int_0^{\pi/2} \sin^5 x\cos^4 x\,dx$$

682 hc-rad-008

R6 Boss+ • 150s • improper-integral / rational

$$\int_0^{\infty} \frac{dx}{1+x^6}$$

683 hc-rad-009

R6 Boss+ • 150s • improper-integral / rational

$$\int_0^{\infty} \frac{x^2}{1+x^6}\,dx$$

684 hc-rad-010

R6 Boss+ • 150s • substitution / u-sub

$$\int_0^1 \frac{x}{\sqrt{1-x^4}}\,dx$$

685 **hc-spec-001**

R6 Boss+ • 160s • gamma-function / improper-integral

$$\int_0^\infty x^4 e^{-3x}\,dx$$

686 **hc-spec-002**

R6 Boss+ • 160s • beta-function / improper-integral

$$\int_0^\infty \frac{\sqrt{x}}{(1+x)^3}\,dx$$

687 **hc-spec-003**

R6 Boss+ • 160s • beta-function / special-functions

$$\int_0^1 x^3(1-x)^4\,dx$$

688 **hc-spec-004**

R6 Boss+ • 160s • wallis / trig

$$\int_0^{\pi/2} \sin^{10} x\,dx$$

689 **hc-spec-005**

R6 Boss+ • 160s • gamma-function / zeta

$$\int_0^\infty \frac{x^5}{e^x-1}\,dx$$

690 **hc-spec-006**

R6 Boss+ • 160s • gamma-function / gaussian-integral

$$\int_0^\infty x^4 e^{-2x^2}\,dx$$

691 **exam-int-001**

R5 Boss • 210s • substitution / rational

$$\int \frac{x}{(1+x^2)^2}\,dx$$

692 **exam-int-002**

R5 Boss • 210s • algebra / rational

$$\int \frac{x^3}{1+x^2}\,dx$$

693 **exam-int-003**

R5 Boss • 210s • integration-by-parts / exponential

$$\int x^2 e^x\,dx$$

694 **exam-int-004**

R5 Boss • 210s • integration-by-parts / log

$$\int x\log x\,dx$$

695 **exam-int-005**

R5 Boss • 210s • substitution / log

$$\int \frac{\log x}{x}\,dx$$

696 **exam-int-006**

R5 Boss • 210s • trig-integral / exam-style

$$\int \sin^3 x\,dx$$

697 **exam-int-007**

R5 Boss • 210s • trig-integral / substitution

$$\int \sec^2 x\tan x\,dx$$

698 **exam-int-008**

R5 Boss • 210s • inverse-trig / exam-style

$$\int \frac{1}{x^2+4}\,dx$$

699 **exam-int-009**

R5 Boss • 210s • partial-fraction / exam-style

$$\int \frac{1}{x^2-1}\,dx$$

700 **exam-int-010**

R5 Boss • 210s • substitution / radical

$$\int \frac{\sqrt{x}}{1+x^{3/2}}\,dx$$

701 **exam-int-011**

R5 Boss • 210s • substitution / radical

$$\int \frac{x}{\sqrt{4-x^2}}\,dx$$

702 **exam-int-012**

R5 Boss • 210s • integration-by-parts / exponential

$$\int e^{2x}\sin x\,dx$$

703 **exam-int-013**

R5 Boss • 210s • integration-by-parts / inverse-trig

$$\int \arctan x\,dx$$

704 **exam-int-014**

R5 Boss • 210s • algebra / inverse-trig

$$\int \frac{x^2}{x^2+1}\,dx$$

705 **exam-int-015**

R5 Boss • 210s • substitution / log

$$\int \frac{1}{x\log x}\,dx$$

706 **exam-int-016**

R5 Boss • 210s • substitution / trig

$$\int \frac{\cos(\log x)}{x}\,dx$$

707 **exam-int-017**

R5 Boss • 210s • substitution / rational

$$\int \frac{2x+3}{x^2+3x+5}\,dx$$

708 **exam-int-018**

R5 Boss • 210s • substitution / inverse-trig

$$\int \frac{x}{x^4+1}\,dx$$

709 **exam-int-019**

R5 Boss • 210s • substitution / radical

$$\int \frac{1}{\sqrt{x}(1+x)}\,dx$$

710 **exam-int-020**

R5 Boss • 210s • substitution / radical

$$\int \frac{x^2}{\sqrt{1+x^3}}\,dx$$

711 **exam-int-021**

R5 Boss • 180s • definite-integral / substitution

$$\int_0^1 \frac{x}{1+x^2}\,dx$$

712 **exam-int-022**

R5 Boss • 180s • definite-integral / substitution

$$\int_0^1 \frac{x^2}{1+x^3}\,dx$$

713 **exam-int-023**

R5 Boss • 180s • definite-integral / trig

$$\int_0^{\pi/2} \sin^2 x\,dx$$

714 **exam-int-024**

R5 Boss • 180s • definite-integral / substitution

$$\int_0^{\pi/2} \frac{\sin x}{1+\cos x}\,dx$$

715 **exam-int-025**

R5 Boss • 180s • definite-integral / improper-integral

$$\int_0^1 \log x\,dx$$

716 **exam-int-026**

R5 Boss • 180s • definite-integral / improper-integral

$$\int_0^\infty e^{-2x}\,dx$$

717 **exam-int-027**

R5 Boss • 180s • definite-integral / inverse-trig

$$\int_0^1 \frac{1}{1+x^2}\,dx$$

718 **exam-int-028**

R5 Boss • 180s • definite-integral / substitution

$$\int_0^1 x\log(1+x^2)\,dx$$

719 **exam-int-029**

R5 Boss • 180s • definite-integral / integration-by-parts

$$\int_0^\pi x\sin x\,dx$$

720 **exam-int-030**

R5 Boss • 180s • definite-integral / exam-style

$$\int_0^1 (1-x)^5\,dx$$

721 **exam-multi-006**

R5 Boss • 180s • multivariable / double-integral

$$\int_0^1\int_0^x (x+y)\,dy\,dx$$

722 **exam-multi-007**

R5 Boss • 180s • multivariable / polar-coordinates

求$r = 2\cos\theta$ 所圍面積

723 **uni-int-001**

R6 Boss+ • 240s • algebra / rational

$$\int \frac{x^5}{1+x^2}\,dx$$

724 **uni-int-002**

R6 Boss+ • 240s • substitution / exponential

$$\int x^3 e^{x^2}\,dx$$

725 **uni-int-003**

R6 Boss+ • 240s • integration-by-parts / log

$$\int \frac{\log x}{x^2}\,dx$$

726 **uni-int-004**

R6 Boss+ • 240s • integration-by-parts / inverse-trig

$$\int \frac{\arctan x}{x^2}\,dx$$

727 **uni-int-005**

R6 Boss+ • 240s • substitution / radical

$$\int x\sqrt{1+x^2}\,dx$$

728 **uni-int-006**

R6 Boss+ • 240s • substitution / trig-integral

$$\int \sin^5 x\cos x\,dx$$

729 **uni-int-007**

R6 Boss+ • 240s • trig-integral / exam-style

$$\int \sec^4 x\,dx$$

730 **uni-int-008**

R6 Boss+ • 240s • inverse-trig / rational

$$\int \frac{1}{x^2+2x+5}\,dx$$

731 **uni-int-009**

R6 Boss+ • 240s • substitution / rational

$$\int \frac{2x+1}{x^2+x+1}\,dx$$

732 **uni-int-010**

R6 Boss+ • 240s • integration-by-parts / exponential

$$\int e^x \cos x\,dx$$

733 **uni-int-011**

R6 Boss+ • 240s • integration-by-parts / log

$$\int x^2 \log x\,dx$$

734 **uni-int-012**

R6 Boss+ • 240s • integration-by-parts / inverse-trig

$$\int x \arctan x\,dx$$

735 **uni-int-013**

R6 Boss+ • 240s • definite-integral / algebra

$$\int_0^1 \frac{x^3}{1+x^2}\,dx$$

736 **uni-int-014**

R6 Boss+ • 240s • improper-integral / inverse-trig

$$\int_0^\infty \frac{1}{(1+x^2)^2}\,dx$$

737 **uni-int-015**

R6 Boss+ • 240s • definite-integral / integration-by-parts

$$\int_0^1 x \log(1+x)\,dx$$

738 **uni-int-016**

R6 Boss+ • 240s • definite-integral / beta-function

$$\int_0^{\pi/2} \sin^3 x \cos^2 x\,dx$$

739 **uni-int-017**

R6 Boss+ • 240s • definite-integral / trig-substitution

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx$$

740 **uni-int-018**

R6 Boss+ • 240s • improper-integral / substitution

$$\int_0^\infty xe^{-x^2}\,dx$$

741 **uni-int-019**

R6 Boss+ • 240s • improper-integral / gamma-function

$$\int_0^1 (\log x)^2\,dx$$

742 **uni-int-020**

R6 Boss+ • 240s • definite-integral / inverse-trig

$$\int_0^1 \frac{1}{1+x+x^2}\,dx$$

743 **uni-int-021**

R6 Boss+ • 240s • double-integral / exam-style

$$\int_0^1 \int_0^x xy\,dy\,dx$$

744 **uni-int-022**

R6 Boss+ • 240s • double-integral / polar-coordinates

$$\iint_{x^2+y^2\le 1} (x^2+y^2)\,dA$$

745 uni-int-023

R6 Boss+ • 240s • parametric / area

求參數曲線$x = 2\cos t,\ y = 3\sin t$ 所圍面積

746 uni-int-024

R6 Boss+ • 240s • triple-integral / exam-style

$$\iiint_{x+y+z\le 1,\ x,y,z\ge 0} 1\,dV$$

747 uni-int-025

R6 Boss+ • 240s • improper-integral / inverse-trig

$$\int_0^\infty \frac{1}{x^2+4}\,dx$$

748 depth-int-001

R5 Boss • 220s • substitution / rational

$$\int \frac{x^3}{1+x^4}\,dx$$

749 depth-int-002

R6 Boss+ • 270s • algebra / rational

$$\int \frac{x^5}{1+x^3}\,dx$$

750 depth-int-003

R5 Boss • 220s • substitution / radical

$$\int \frac{x}{(1+x^2)^{3/2}}\,dx$$

751 depth-int-004

R5 Boss • 220s • substitution / log

$$\int \frac{(\log x)^2}{x}\,dx$$

752 depth-int-005

R6 Boss+ • 270s • integration-by-parts / exponential

$$\int xe^x \sin x\,dx$$

753 depth-int-006

R5 Boss • 220s • integration-by-parts / exponential

$$\int e^{2x}\cos(3x)\,dx$$

754 depth-int-007

R6 Boss+ • 270s • substitution / integration-by-parts

$$\int \frac{\arctan\sqrt{x}}{\sqrt{x}}\,dx$$

755 depth-int-008

R5 Boss • 220s • substitution / log

$$\int \frac{1}{x\sqrt{\log x}}\,dx$$

756 depth-int-009

R6 Boss+ • 270s • substitution / exponential

$$\int \sin(\log x)\,dx$$

757 depth-int-010

R5 Boss • 220s • substitution / radical

$$\int \frac{1}{\sqrt{x}(1+\sqrt{x})}\,dx$$

758 depth-int-011

R6 Boss+ • 270s • rational / inverse-trig

$$\int \frac{x^2}{(1+x^2)^2}\,dx$$

759 depth-int-012

R5 Boss • 220s • substitution / radical

$$\int x^2\sqrt{1+x^3}\,dx$$

760 **depth-int-013**
R5 Boss • 220s • substitution / trig-integral

$$\int \frac{\sec^2 x}{1+\tan x}\,dx$$

761 **depth-int-014**
R6 Boss+ • 270s • integration-by-parts / log

$$\int x\log(1+x)\,dx$$

762 **depth-int-015**
R6 Boss+ • 260s • definite-integral / rational

$$\int_0^1 \frac{x}{1+x+x^2}\,dx$$

763 **depth-int-016**
R5 Boss • 260s • definite-integral / integration-by-parts

$$\int_0^{\pi/2} x\cos x\,dx$$

764 **depth-int-017**
R6 Boss+ • 260s • definite-integral / taylor

$$\int_0^1 \frac{\log(1+x)}{x}\,dx$$

765 **depth-int-018**
R6 Boss+ • 260s • definite-integral / substitution

$$\int_0^1 \frac{x^3}{(1+x^2)^2}\,dx$$

766 **depth-int-019**
R5 Boss • 260s • improper-integral / substitution

$$\int_0^{\infty} \frac{x}{(1+x^2)^2}\,dx$$

767 **depth-int-020**
R6 Boss+ • 260s • improper-integral / partial-fraction

$$\int_0^{\infty} \frac{1}{(x^2+1)(x^2+4)}\,dx$$

768 **depth-int-021**
R5 Boss • 260s • definite-integral / trig-integral

$$\int_0^{\pi} \sin^4 x\,dx$$

769 **depth-int-022**
R5 Boss • 260s • double-integral / region

$$\int_0^1\int_y^1 x\,dx\,dy$$

770 **depth-int-023**
R6 Boss+ • 260s • polar-coordinates / polar-curve

area enclosed by $r = 1+\cos\theta$

771 **depth-int-024**
R6 Boss+ • 260s • double-integral / polar-coordinates

$$\iint_{x^2+y^2\le 4} x^2\,dA$$

772 **depth-int-025**
R5 Boss • 260s • double-integral / region

$$\int_0^1\int_0^{1-x} (x+y)\,dy\,dx$$

773 **depth-int-026**
R5 Boss • 260s • improper-integral / gamma-function

$$\int_0^{\infty} x^3 e^{-x^2}\,dx$$

774 **depth-int-027**
R5 Boss • 260s • definite-integral / trig-substitution

$$\int_0^1 \frac{1}{\sqrt{1-x^2}}\,dx$$

775 **depth-int-028**

R5 Boss • 260s • improper-integral / integration-by-parts

$$\int_0^1 x^2 \log x \, dx$$

776 **burst-int-001**

R6 Boss+ • 330s • multi-ibp / gamma-function

$$\int_0^\infty x^4 e^{-3x} \, dx$$

777 **burst-int-002**

R6 Boss+ • 330s • multi-ibp / exponential

$$\int_0^\infty x^2 e^{-x} \sin x \, dx$$

778 **burst-int-003**

R6 Boss+ • 330s • multi-ibp / exponential

$$\int_0^\infty x^2 e^{-2x} \cos x \, dx$$

779 **burst-int-004**

R5 Boss • 330s • integration-by-parts / multi-ibp

$$\int_0^1 x^5 \log x \, dx$$

780 **burst-int-005**

R6 Boss+ • 330s • multi-ibp / parameter-integral

$$\int_0^1 x^4 (\log x)^2 \, dx$$

781 **burst-int-006**

R6 Boss+ • 330s • multi-ibp / parameter-integral

$$\int_0^1 x^2 (\log x)^3 \, dx$$

782 **burst-int-007**

R5 Boss • 330s • beta-function / special-function

$$\int_0^1 x^3 (1-x)^4 \, dx$$

783 **burst-int-008**

R6 Boss+ • 330s • beta-function / wallis

$$\int_0^1 \sqrt{x(1-x)} \, dx$$

784 **burst-int-009**

R6 Boss+ • 330s • gamma-function / special-function

$$\int_0^\infty x^4 e^{-x^2} \, dx$$

785 **burst-int-010**

R5 Boss • 330s • gamma-function / special-function

$$\int_0^\infty x^5 e^{-x^2} \, dx$$

786 **burst-int-011**

R6 Boss+ • 330s • wallis / special-function

$$\int_0^{\pi/2} \log(\sin x) \, dx$$

787 **burst-int-012**

R6 Boss+ • 330s • wallis / special-function

$$\int_0^{\pi/2} (\log(\sin x))^2 \, dx$$

788 **burst-int-013**

R6 Boss+ • 330s • wallis / trig-power

$$\int_0^{\pi} x \sin^8 x \, dx$$

789 **burst-int-014**

R5 Boss • 330s • wallis / trig-power

$$\int_0^{\pi/2} \sin^{10} x \, dx$$

790 **burst-int-015**

R5 Boss • 330s • wallis / trig-power

$$\int_0^{\pi/2} \cos^{11} x \, dx$$

791 **burst-int-016**

R6 Boss+ • 330s • beta-function / trig-power

$$\int_0^{\pi/2} \sin^6 x \cos^4 x \, dx$$

792 **burst-int-017**

R5 Boss • 330s • beta-function / trig-power

$$\int_0^{\pi/2} \sin^4 x \cos^4 x \, dx$$

793 **burst-int-018**

R5 Boss • 330s • beta-function / trig-power

$$\int_0^{\pi/2} \sin^5 x \cos^3 x \, dx$$

794 **burst-int-019**

R6 Boss+ • 330s • beta-function / trig-power

$$\int_0^{\pi/2} \sin^8 x \cos^2 x \, dx$$

795 **burst-int-020**

R5 Boss • 330s • wallis / trig-power

$$\int_0^{\pi} \sin^8 x \, dx$$

796 **burst-int-021**

R5 Boss • 330s • wallis / trig-power

$$\int_0^{\pi/2} \sin^7 x \, dx$$

797 **burst-int-022**

R5 Boss • 330s • wallis / trig-power

$$\int_0^{\pi/2} \cos^8 x \, dx$$

798 **burst-int-023**

R6 Boss+ • 330s • beta-function / trig-power

$$\int_0^{\pi/2} \sin^3 x \cos^6 x \, dx$$

799 **burst-int-024**

R6 Boss+ • 330s • beta-function / trig-power

$$\int_0^{\pi/2} \sin^2 x \cos^6 x \, dx$$

800 **burst-int-025**

R6 Boss+ • 330s • special-function / improper-integral

$$\int_0^{\infty} \frac{x^3}{e^x - 1} \, dx$$

801 **burst-int-026**

R6 Boss+ • 330s • special-function / improper-integral

$$\int_0^{\infty} \frac{x}{e^x - 1} \, dx$$

802 **burst-int-027**

R6 Boss+ • 330s • improper-integral / special-function

$$\int_0^{\infty} \frac{1}{1 + x^4} \, dx$$

803 **burst-int-028**

R6 Boss+ • 330s • improper-integral / kings-property

$$\int_0^{\infty} \frac{\log x}{1 + x^2} \, dx$$

804 **burst-int-029**

R6 Boss+ • 330s • beta-function / special-function

$$\int_0^{\infty} \frac{\sqrt{x}}{1 + x^2} \, dx$$

805 **burst-int-030**

R6 Boss+ • 330s • series-integral / special-function

$$\int_0^1 \frac{\log x}{1+x}\,dx$$

806 **burst-int-031**

R5 Boss • 330s • trig-power / wallis

$$\int_0^{\pi/2} \sin^4(2x)\,dx$$

807 **burst-int-032**

R6 Boss+ • 330s • beta-function / trig-power

$$\int_0^{\pi/2} \sin^6 x \cos^6 x\,dx$$

808 **burst-int-033**

R6 Boss+ • 330s • wallis / trig-power

$$\int_0^{\pi/2} \sin^{12} x\,dx$$

809 **burst-int-034**

R6 Boss+ • 330s • wallis / trig-power

$$\int_0^{\pi/2} \cos^{13} x\,dx$$

810 **burst-int-035**

R6 Boss+ • 330s • beta-function / trig-power

$$\int_0^{\pi/2} \sin^9 x \cos^5 x\,dx$$

811 **burst-int-036**

R6 Boss+ • 330s • multi-ibp / integration-by-parts

$$\int x^3 e^{2x}\,dx$$

812 **burst-int-037**

R6 Boss+ • 330s • multi-ibp / integration-by-parts

$$\int x^2 \sin(3x)\,dx$$

813 **burst-int-038**

R6 Boss+ • 330s • multi-ibp / integration-by-parts

$$\int (\log x)^3\,dx$$

814 **burst-int-039**

R5 Boss • 330s • integration-by-parts / log

$$\int x^4 \log x\,dx$$

815 **burst-int-040**

R6 Boss+ • 330s • integration-by-parts / inverse-trig

$$\int x^3 \arctan x\,dx$$

816 **burst-int-041**

R6 Boss+ • 330s • multi-ibp / integration-by-parts

$$\int x^3 e^{-x}\,dx$$

817 **burst-int-042**

R6 Boss+ • 330s • integration-by-parts / log

$$\int x^2 \log(1+x)\,dx$$

818 **burst-rec-001**

R6 Boss+ • 270s • recurrence-formula / parameter-integral

$I_n = \int_0^1 x^n \log x\,dx$. Find I_n

819 **burst-rec-002**

R6 Boss+ • 270s • recurrence-formula / parameter-integral

$I_n = \int_0^1 x^n (\log x)^2\,dx$. Find I_n

820 **burst-rec-003**

R6 Boss+ • 270s • recurrence-formula / beta-function

$I_n = \int_0^1 x^n (1-x)^2\, dx$. Find I_n

821 **burst-rec-004**

R6 Boss+ • 270s • recurrence-formula / wallis

$W_n = \int_0^{\pi/2} \sin^n x\, dx$. Find W_n / W_{n-2}

822 **burst-rec-005**

R6 Boss+ • 270s • recurrence-formula / wallis

$A_n = \int_0^{\pi/2} \sin^{2n} x\, dx$. Find A_n / A_{n-1}

823 **burst-rec-006**

R6 Boss+ • 270s • recurrence-formula / wallis

$B_n = \int_0^{\pi/2} \sin^{2n+1} x\, dx$. Find B_n / B_{n-1}

824 **burst-rec-007**

R6 Boss+ • 270s • recurrence-formula / gamma-function

$G_n = \int_0^{\infty} x^n e^{-x}\, dx$. Find G_n / G_{n-1}

825 **burst-rec-008**

R6 Boss+ • 270s • recurrence-formula / integration-by-parts

$J_n = \int_0^1 x^n e^x\, dx,\ J_n = e - n\, J_{n-1}$. Find the coefficient of J_{n-1}

826 **burst-rec-009**

R5 Boss • 270s • recurrence-formula / definite-integral

$K_n = \int_0^1 x^n (1+x)\, dx$. Find K_n

827 **burst-rec-010**

R5 Boss • 270s • recurrence-formula / definite-integral

$L_n = \int_0^1 x^n (1-x)\, dx$. Find L_n

828 **burst-boss-int-001**

R6 Boss+ • 420s • true-boss / boss-rank

$$\int_0^{\infty} \frac{(\log x)^2}{1+x^2}\, dx$$

829 **burst-boss-int-002**

R6 Boss+ • 420s • true-boss / boss-rank

$$\int_0^{\infty} \frac{\log x}{1+x^4}\, dx$$

830 **burst-boss-int-003**

R6 Boss+ • 420s • true-boss / boss-rank

$$\int_0^{\infty} \frac{\sqrt{x}\log x}{1+x^2}\, dx$$

831 **burst-boss-int-004**

R6 Boss+ • 420s • true-boss / boss-rank

$$\int_0^{\infty} \frac{x^4}{1+x^6}\, dx$$

832 **burst-boss-int-005**

R5 Boss • 330s • true-boss / boss-rank

$$\int_0^{\infty} \frac{x^2}{1+x^6}\, dx$$

833 **burst-boss-int-006**

R5 Boss • 330s • true-boss / boss-rank

$$\int_0^{\infty} \frac{1}{1+x^6}\, dx$$

834 **burst-boss-int-007**

R5 Boss • 330s • true-boss / boss-rank

$$\int_0^{\infty} \frac{x}{1+x^4}\, dx$$

835 **burst-boss-int-008**

R6 Boss+ • 420s • true-boss / boss-rank

$$\int_0^\infty \frac{x^2}{1+x^4}\,dx$$

836 **burst-boss-int-009**

R6 Boss+ • 420s • true-boss / boss-rank

$$\int_0^1 \log x \log(1-x)\,dx$$

837 **burst-boss-int-010**

R6 Boss+ • 420s • true-boss / boss-rank

$$\int_0^1 \frac{\log x}{1-x}\,dx$$

838 **burst-boss-int-011**

R5 Boss • 330s • true-boss / boss-rank

$$\int_0^1 (\log x)^4\,dx$$

839 **burst-boss-int-012**

R6 Boss+ • 420s • true-boss / boss-rank

$$\int_0^1 x^3(\log x)^4\,dx$$

840 **burst-boss-int-013**

R6 Boss+ • 420s • true-boss / boss-rank

$$\int_0^1 x^2(\log x)^5\,dx$$

841 **burst-boss-int-014**

R6 Boss+ • 420s • true-boss / boss-rank

$$\int_0^{\pi/2} \log(\sin x)\log(\cos x)\,dx$$

842 **burst-boss-int-015**

R6 Boss+ • 420s • true-boss / boss-rank

$$\int_0^{\pi/2} (\log(\tan x))^2\,dx$$

843 **burst-boss-int-016**

R6 Boss+ • 420s • true-boss / boss-rank

$$\int_0^\pi x\sin^{10} x\,dx$$

844 **burst-boss-int-017**

R6 Boss+ • 420s • true-boss / boss-rank

$$\int_0^{\pi/2} \sin^8 x\cos^8 x\,dx$$

845 **burst-boss-int-018**

R6 Boss+ • 420s • true-boss / boss-rank

$$\int_0^{\pi/2} \sin^{14} x\,dx$$

846 **burst-boss-int-019**

R6 Boss+ • 420s • true-boss / boss-rank

$$\int_0^{\pi/2} \sin^{15} x\,dx$$

847 **burst-boss-int-020**

R6 Boss+ • 420s • true-boss / boss-rank

$$\int_0^\infty x^5e^{-2x}\sin x\,dx$$

848 **burst-boss-anti-001**

R6 Boss+ • 420s • true-boss / boss-rank

$$\int x^4e^x\,dx$$

849 **burst-boss-anti-002**

R6 Boss+ • 420s • true-boss / boss-rank

$$\int x^4e^{-2x}\,dx$$

850 **burst-boss-anti-003**
R6 Boss+ • 420s • true-boss / boss-rank

$$\int x^3 \sin(2x)\,dx$$

851 **burst-boss-anti-004**
R6 Boss+ • 420s • true-boss / boss-rank

$$\int x^3 \cos(2x)\,dx$$

852 **burst-boss-anti-005**
R6 Boss+ • 420s • true-boss / boss-rank

$$\int x^4 (\log x)^2\,dx$$

853 **burst-boss-anti-006**
R6 Boss+ • 420s • true-boss / boss-rank

$$\int (\log x)^4\,dx$$

854 **burst-boss-anti-007**
R5 Boss • 330s • true-boss / boss-rank

$$\int x^2 \arctan x\,dx$$

855 **burst-boss-anti-008**
R6 Boss+ • 420s • true-boss / boss-rank

$$\int x^4 \arctan x\,dx$$

856 **burst-boss-anti-009**
R6 Boss+ • 420s • true-boss / boss-rank

$$\int x^2 \log(1+x^2)\,dx$$

857 **burst-boss-anti-010**
R6 Boss+ • 420s • true-boss / boss-rank

$$\int x^3 \log(1+x^2)\,dx$$

858 **burst-boss-param-001**
R6 Boss+ • 360s • true-boss / boss-rank

$$I(a) = \int_0^\infty e^{-ax} \sin x\,dx. \text{ Find } I(a)$$

859 **burst-boss-param-002**
R5 Boss • 300s • true-boss / boss-rank

$$I(a) = \int_0^\infty x e^{-ax}\,dx. \text{ Find } I(a)$$

860 **burst-boss-param-003**
R5 Boss • 300s • true-boss / boss-rank

$$I(a) = \int_0^\infty \frac{1}{1+ax^2}\,dx. \text{ Find } I(a)$$

861 **burst-boss-param-004**
R6 Boss+ • 360s • true-boss / boss-rank

$$I(a) = \int_0^{\pi/2} \log(1 + a\sin^2 x)\,dx. \text{ Find } I(a)$$

862 **burst-boss-param-005**
R6 Boss+ • 360s • true-boss / boss-rank

$$J_n = \int_0^\infty \frac{x^n}{1+x^2}\,dx \text{ for } -1 < n < 1. \text{ Find } J_n$$

863 **burst-boss-param-006**
R6 Boss+ • 360s • true-boss / boss-rank

$$K_n = \int_0^1 x^n (\log x)^3\,dx. \text{ Find } K_n$$

864 **burst-boss2-int-001**
R6 Boss+ • 480s • true-boss / boss-rank

$$\int_0^\infty \frac{\log x}{1+x^3}\,dx$$

865 **burst-boss2-int-002**

R6 Boss+ • 480s • true-boss / boss-rank

$$\int_0^\infty \frac{(\log x)^2}{1+x^3}\,dx$$

866 **burst-boss2-int-003**

R6 Boss+ • 480s • true-boss / boss-rank

$$\int_0^\infty \frac{\sqrt{x}}{1+x^3}\,dx$$

867 **burst-boss2-int-004**

R6 Boss+ • 480s • true-boss / boss-rank

$$\int_0^\infty \frac{x^{1/3}}{1+x^2}\,dx$$

868 **burst-boss2-int-005**

R6 Boss+ • 480s • true-boss / boss-rank

$$\int_0^\infty \frac{\arctan x}{x(1+x^2)}\,dx$$

869 **burst-boss2-int-006**

R6 Boss+ • 480s • true-boss / boss-rank

$$\int_0^\infty \left(\frac{\sin x}{x}\right)^2\,dx$$

870 **burst-boss2-int-007**

R6 Boss+ • 480s • true-boss / boss-rank

$$\int_0^\infty \frac{1-\cos x}{x^2}\,dx$$

871 **burst-boss2-int-008**

R6 Boss+ • 480s • true-boss / boss-rank

$$\int_0^1 \frac{(\log x)^3}{1-x}\,dx$$

872 **burst-boss2-int-009**

R6 Boss+ • 480s • true-boss / boss-rank

$$\int_0^1 \frac{(\log x)^5}{1-x}\,dx$$

873 **burst-boss2-int-010**

R6 Boss+ • 480s • true-boss / boss-rank

$$\int_0^1 \frac{(\log x)^3}{1+x}\,dx$$

874 **burst-boss2-int-011**

R6 Boss+ • 480s • true-boss / boss-rank

$$\int_0^1 \frac{(\log x)^5}{1+x}\,dx$$

875 **burst-boss2-int-012**

R6 Boss+ • 480s • true-boss / boss-rank

$$\int_0^{\pi/2} x\sin^2 x\,dx$$

876 **burst-boss2-int-013**

R6 Boss+ • 480s • true-boss / boss-rank

$$\int_0^{\pi/2} x\sin x\cos x\,dx$$

877 **burst-boss2-int-014**

R6 Boss+ • 480s • true-boss / boss-rank

$$\int_0^{\pi} x^2\sin x\,dx$$

878 **burst-boss2-param-001**

R6 Boss+ • 420s • true-boss / boss-rank

$F(a,b)=\int_0^\infty e^{-ax}\sin(bx)\,dx$. Find $F(a,b)$

879 **burst-boss2-param-002**

R6 Boss+ • 420s • true-boss / boss-rank

$F(a,b)=\int_0^\infty e^{-ax}\cos(bx)\,dx$. Find $F(a,b)$

880 **burst-boss2-param-003**

R6 Boss+ • 420s • true-boss / boss-rank

$F(a)=\int_0^\infty xe^{-ax}\sin x\,dx$. Find $F(a)$

881 burst-boss2-param-004

R6 Boss+ • 420s • true-boss / boss-rank

$F(a) = \int_0^\infty x e^{-ax} \cos x \, dx$. Find $F(a)$

882 burst-boss2-param-005

R6 Boss+ • 420s • true-boss / boss-rank

$F(a) = \int_0^1 \frac{x^a - 1}{\log x} \, dx$. Find $F(a)$

883 burst-boss2-param-006

R6 Boss+ • 420s • true-boss / boss-rank

$F(a) = \int_0^1 x^a (\log x)^5 \, dx$. Find $F(a)$

884 burst-boss2-anti-001

R6 Boss+ • 420s • true-boss / boss-rank

$\int x^5 e^{2x} \, dx$

885 burst-boss2-anti-002

R6 Boss+ • 420s • true-boss / boss-rank

$\int x^4 \sin x \, dx$

886 burst-boss2-anti-003

R6 Boss+ • 420s • true-boss / boss-rank

$\int x^4 \cos x \, dx$

887 burst-boss2-anti-004

R6 Boss+ • 420s • true-boss / boss-rank

$\int (\log x)^5 \, dx$

888 burst-boss2-anti-005

R6 Boss+ • 420s • true-boss / boss-rank

$\int x^5 \arctan x \, dx$

889 burst-boss2-anti-006

R6 Boss+ • 420s • true-boss / boss-rank

$\int \frac{x^5}{1 + x^2} \, dx$

890 burst-boss3-int-001

R6 Boss+ • 540s • true-boss / boss-rank

$\int_0^\infty \int_0^\infty x^3 y^4 e^{-2x-3y} \, dx \, dy$

891 burst-boss3-int-002

R6 Boss+ • 540s • true-boss / boss-rank

$\int_0^1 \int_0^1 xy^2 (\log x)^2 (\log y)^3 \, dx \, dy$

892 burst-boss3-int-003

R6 Boss+ • 540s • true-boss / boss-rank

$\int_0^1 (\log(x(1-x)))^2 \, dx$

893 burst-boss3-int-004

R6 Boss+ • 540s • true-boss / boss-rank

$\int_0^1 \left(\log \frac{x}{1-x}\right)^2 \, dx$

894 burst-boss3-int-005

R6 Boss+ • 540s • true-boss / boss-rank

$\int_0^\pi x \log(\sin x) \, dx$

895 burst-boss3-int-006

R6 Boss+ • 540s • true-boss / boss-rank

$\int_0^\pi x^3 \sin x \, dx$

896 burst-boss3-int-007

R6 Boss+ • 540s • true-boss / boss-rank

$$\int_0^{\pi} x^4 \sin x \, dx$$

897 burst-boss3-int-008

R6 Boss+ • 540s • true-boss / boss-rank

$$\int_0^{\infty} x^3 e^{-2x} \cos(3x) \, dx$$

898 burst-boss3-int-009

R6 Boss+ • 540s • true-boss / boss-rank

$$\int_0^{\infty} x^3 e^{-2x} \sin(3x) \, dx$$

899 burst-boss3-int-010

R6 Boss+ • 540s • true-boss / boss-rank

$$\int_0^{\infty} x^4 e^{-x} \cos(2x) \, dx$$

900 burst-boss3-int-011

R6 Boss+ • 540s • true-boss / boss-rank

$$\int_0^{\infty} x^4 e^{-x} \sin(2x) \, dx$$

901 burst-boss3-int-012

R6 Boss+ • 540s • true-boss / boss-rank

$$\int_0^{\infty} \frac{x^5}{e^x + 1} \, dx$$

902 burst-boss3-int-013

R6 Boss+ • 540s • true-boss / boss-rank

$$\int_0^{\infty} \frac{x^3}{e^x + 1} \, dx$$

903 burst-boss3-param-001

R6 Boss+ • 480s • true-boss / boss-rank

$F(a) = \int_0^{\infty} e^{-ax} \sin(3x) \, dx$. Find $F''(a)$

904 burst-boss3-param-002

R6 Boss+ • 480s • true-boss / boss-rank

$G(a) = \int_0^{\infty} x^2 e^{-ax} \cos x \, dx$. Find $G(a)$

905 burst-boss3-param-003

R6 Boss+ • 480s • true-boss / boss-rank

$H(a) = \int_0^{\infty} x^3 e^{-ax} \sin x \, dx$. Find $H(a)$

906 burst-boss3-param-004

R6 Boss+ • 480s • true-boss / boss-rank

$B(a) = \int_0^{1} x^a (\log x)^4 \, dx$. Find $B(a)$

907 burst-boss3-param-005

R6 Boss+ • 480s • true-boss / boss-rank

$C(a) = \int_0^{1} x^a (\log x)^6 \, dx$. Find $C(a)$

908 burst-boss3-param-006

R6 Boss+ • 480s • true-boss / boss-rank

$Q(a) = \frac{d^3}{da^3} \int_0^{1} \frac{x^a - 1}{\log x} \, dx$. Find $Q(a)$

909 burst-boss3-anti-001

R6 Boss+ • 540s • true-boss / boss-rank

$$\int x^6 e^{-3x} \, dx$$

910 burst-boss3-anti-002

R6 Boss+ • 540s • true-boss / boss-rank

$$\int x^5 \sin(2x) \, dx$$

911 burst-boss3-anti-003

R6 Boss+ • 540s • true-boss / boss-rank

$$\int x^5 \cos(2x)\,dx$$

912 **burst-boss3-anti-004**

R6 Boss+ • 540s • true-boss / boss-rank

$$\int x^6 \arctan x\,dx$$

913 **burst-boss3-anti-005**

R6 Boss+ • 540s • true-boss / boss-rank

$$\int x^6 \log(1+x)\,dx$$

914 **burst-boss3-anti-006**

R6 Boss+ • 540s • true-boss / boss-rank

$$\int \frac{x^7}{1+x^2}\,dx$$

級數 Series

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915 **ser-001**

R1 Warm-up • 25s • rank-1 / beginner-friendly

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

916 **ser-002**

R1 Warm-up • 25s • rank-1 / beginner-friendly

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

917 **ser-003**

R1 Warm-up • 35s • rank-1 / beginner-friendly

$$\sum_{n=0}^{\infty} \left(\frac{1}{3}\right)^n$$

918 **ser-004**

R1 Warm-up • 35s • rank-1 / beginner-friendly

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$$

919 **ser-005**

R2 Basic • 45s • rank-2 / beginner-friendly

$$\sum_{n=1}^{\infty} \frac{n}{n^2+1}$$

920 **ser-006**

R2 Basic • 50s • rank-2 / beginner-friendly

$$\sum_{n=1}^{\infty} \frac{n!}{n^n}$$

921 **ser-007**

R2 Basic • 55s • rank-2 / beginner-friendly

$$\sum_{n=1}^{\infty} \frac{3^n}{n!}$$

922 **ser-008**

R2 Basic • 45s • rank-2 / beginner-friendly

Radius of convergence of $\sum_{n=0}^{\infty} n!x^n$

923 **ser-009**

R3 Standard • 45s • rank-3

Radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n}$

924 **ser-010**

R3 Standard • 45s • rank-3

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$$

925 **ser-011**

R3 Standard • 70s • rank-3

$$\sum_{n=1}^{\infty} \frac{2^n+n}{3^n}$$

926 **ser-012**

R2 Basic • 60s • rank-2 / beginner-friendly

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$$

927 **ser-013**

R2 Basic • 55s • rank-2 / beginner-friendly

$$\sum_{n=1}^{\infty} (-1)^n \frac{n}{n+1}$$

928 **ser-014**

R3 Standard • 55s • rank-3

$$\sum_{n=1}^{\infty} \frac{n}{2^n}$$

929 **ser-015**

R3 Standard • 70s • rank-3

$$\sum_{n=1}^{\infty} \frac{n^2}{2^n}$$

930 **ser-016**

R4 Advanced • 70s • rank-4

$$\sum_{n=1}^{\infty} \frac{(n!)^2}{(2n)!}$$

931 **ser-017**

R3 Standard • 60s • rank-3

Radius of convergence of $\sum_{n=1}^{\infty} \frac{n^2 x^n}{3^n}$

932 **ser-018**

R2 Basic • 45s • rank-2 / beginner-friendly

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$$

933 **ser-019**

R2 Basic • 75s • rank-2 / beginner-friendly

$$\sum_{n=1}^{\infty} \frac{1}{n(n+2)}$$

934 **ser-020**

R3 Standard • 60s • rank-3

$$\sum_{n=2}^{\infty} \frac{\ln n}{n}$$

935 **td-ser-001**

R6 Boss+ • 100s • rank-6 / boss-rank

Coefficient of x^6 in $(1+x)^x$

936 **td-ser-002**

R6 Boss+ • 110s • rank-6 / boss-rank

Coefficient of x^5 in $e^{2\arcsin x}$

937 **td-ser-003**

R6 Boss+ • 105s • rank-6 / boss-rank

$$\sum_{n=1}^{\infty} \frac{(2n-1)!!}{(2n)!!} \cdot \frac{1}{n}$$

938 **td-ser-004**

R4 Advanced • 60s • rank-4

$$\sum_{n=1}^{\infty} \frac{n^2}{3^n}$$

939 **ser-021**

R2 Basic • 55s • telescoping / rank-2

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$$

940 **ser-022**

R2 Basic • 55s • power-series / geometric-derivative

$$\sum_{n=1}^{\infty} \frac{n}{2^n}$$

941 **ser-023**

R2 Basic • 55s • alternating-series / conditional-convergence

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$$

942 **ser-024**

R2 Basic • 60s • integral-test / log

$$\sum_{n=2}^{\infty} \frac{1}{n \log n}$$

943 **ser-025**

R3 Standard • 70s • ratio-test / rank-3

$$\sum_{n=1}^{\infty} \frac{n!}{n^n}$$

944 **ser-026**

R2 Basic • 60s • power-series / radius

Radius of convergence of $\sum_{n=1}^{\infty} \frac{n^2 x^n}{3^n}$

945 **ser-027**

R2 Basic • 50s • taylor / coefficient

Coefficient of x^4 in e^{2x}

946 **ser-028**

R1 Warm-up • 45s • p-series / rank-1

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

947 **ser-029**
R2 Basic • 55s • exponential-series / rank-2

$$\sum_{n=1}^{\infty} \frac{3^n}{n!}$$

948 **ser-030**
R2 Basic • 55s • power-series / radius

Radius of convergence of $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n5^n}$

949 **ser-031**
R3 Standard • 45s • p-series / rank-3

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$$

950 **ser-032**
R3 Standard • 65s • term-test / rank-3

$$\sum_{n=1}^{\infty} \left(\frac{n}{n+1}\right)^n$$

951 **ser-033**
R3 Standard • 65s • alternating-series / conditional-convergence

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$$

952 **ser-034**
R3 Standard • 60s • power-series / radius

Radius of convergence of $\sum_{n=1}^{\infty} n!x^n$

953 **ser-035**
R3 Standard • 60s • power-series / radius

Radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n!}$

954 **ser-036**
R3 Standard • 70s • comparison / geometric-series

$$\sum_{n=1}^{\infty} \frac{2^n}{3^n+n}$$

955 **ser-037**
R3 Standard • 70s • integral-test / log

$$\sum_{n=2}^{\infty} \frac{1}{n(\log n)^2}$$

956 **ser-038**
R2 Basic • 50s • taylor / coefficient

Coefficient of x^3 in $\log(1+x)$

957 **ser-039**
R3 Standard • 55s • power-series / radius

Radius of convergence of $\sum_{n=1}^{\infty} \frac{(2x)^n}{n}$

958 **ser-040**
R3 Standard • 60s • power-series / geometric-derivative

$$\sum_{n=1}^{\infty} \frac{n^2}{2^n}$$

959 **gap-ser-root-001**
R3 Standard • 75s • root-test / power-series

Radius of convergence of $\sum_{n=1}^{\infty} \frac{3^n}{n^2}x^n$

960 **gap-ser-root-002**
R3 Standard • 85s • root-test / power-series

Radius of convergence of $\sum_{n=1}^{\infty} \left(\frac{n}{n+1}\right)^n x^n$

961 **gap-ser-lc-001**
R3 Standard • 75s • limit-comparison / harmonic

$\sum_{n=1}^{\infty} \frac{3n^2+1}{n^3+2}$ converges or diverges?

962 **gap-ser-lc-002**
R3 Standard • 75s • limit-comparison / p-series

$\sum_{n=1}^{\infty} \frac{n+1}{n^3+1}$ converges or diverges?

963 **gap-ser-cond-001**
R3 Standard • 75s • alternating-series / absolute-conditional

$\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ absolute, conditional, or divergent?

964 **gap-ser-end-001**
R4 Advanced • 85s • endpoint-analysis / power-series

$\sum_{n=1}^{\infty} \frac{x^n}{n}$ at $x=-1$ is absolute, conditional, or divergent?

965 **gap-ser-end-002**
R4 Advanced • 80s • endpoint-analysis / power-series

$\sum_{n=1}^{\infty} \frac{x^n}{n}$ at $x=1$ converges or diverges?

966 **gap-ser-taylor-001**
R3 Standard • 70s • taylor / coefficient

Coefficient of x^5 in the Maclaurin series for $\sin x$

967 **gap-ser-taylor-002**
R3 Standard • 80s • taylor / full-expansion
Degree 4 Maclaurin polynomial for $\cos x$

968 **gap-tech-005**
R3 Standard • 45s • technique-recognition / root-test
Technique for $\sum \left(\frac{n}{n+1}\right)^n x^n$

969 **mob-tech-007**
R2 Basic • 45s • technique-sprint / technique-recognition
$\sum_{n=1}^{\infty} \frac{n!}{3^n}$ Use which technique?

970 **mob-tech-008**
R2 Basic • 45s • technique-sprint / technique-recognition
$\sum_{n=1}^{\infty} \left(\frac{n}{2n+1}\right)^n$ Use which technique?

971 **mob-tech-019**
R2 Basic • 45s • technique-sprint / technique-recognition
Find interval of convergence after radius is known Use which technique?

972 **mob-trap-004**
R2 Basic • 45s • trap-drill / rank-2
$\sum (-1)^n/n$ is not absolutely convergent Main trap?

973 **mob-trap-005**
R2 Basic • 45s • trap-drill / rank-2
$\sum x^n/n$ after finding $R=1$ Main trap?

974 **mob-conv-001**
R3 Standard • 45s • convergence-test / series-test
$\sum_{n=1}^{\infty} \frac{1}{n}$

975 **mob-conv-002**
R3 Standard • 45s • convergence-test / series-test
$\sum_{n=1}^{\infty} \frac{1}{n^2}$

976 **mob-conv-003**
R3 Standard • 45s • convergence-test / series-test
$\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$

977 **mob-conv-004**
R3 Standard • 45s • convergence-test / series-test
$\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$

978 **mob-conv-005**
R3 Standard • 45s • convergence-test / series-test
$\sum_{n=1}^{\infty} \frac{n!}{n^n}$

979 **mob-conv-006**
R3 Standard • 45s • convergence-test / series-test
$\sum_{n=1}^{\infty} \frac{2^n}{n!}$

980 **mob-conv-007**
R3 Standard • 45s • convergence-test / series-test
$\sum_{n=1}^{\infty} \frac{n}{n^2+1}$

981 **mob-conv-008**
R3 Standard • 45s • convergence-test / series-test
$\sum_{n=2}^{\infty} \frac{1}{n\log n}$

982 **mob-conv-009**
R3 Standard • 45s • convergence-test / series-test
$\sum_{n=2}^{\infty} \frac{1}{n(\log n)^2}$

983 **mob-conv-010**
R3 Standard • 45s • convergence-test / series-test
$\sum_{n=1}^{\infty} \frac{n+1}{3n+2}$

984 **rel-basic-019**
R1 Warm-up • 45s • geometric-series / beginner-foundation
$\sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n$

985 **rel-basic-022**
R1 Warm-up • 45s • power-series / beginner-foundation
Radius of convergence of $\sum_{n=0}^{\infty} 3^n x^n$

986 **rel-basic-024**
R1 Warm-up • 45s • geometric-series / beginner-foundation
$\sum_{n=1}^{\infty} \left(\frac{1}{3}\right)^n$

987 **rel-basic-020**

R1 Warm-up • 40s • p-series / beginner-foundation

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

988 **rel-basic-021**

R1 Warm-up • 40s • p-series / beginner-foundation

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

989 **rel-basic-023**

R2 Basic • 50s • alternating / series-test

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$$

990 **rel-adv-017**

R3 Standard • 75s • taylor / power-series

coefficient of x^6 in e^{x^2}

991 **rel-adv-019**

R3 Standard • 75s • power-series / radius-of-convergence

Radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n^2}$

992 **rel-adv-015**

R3 Standard • 55s • root-test / series-test

$$\sum_{n=1}^{\infty} \left(\frac{n}{3n+1}\right)^n$$

993 **rel-adv-016**

R3 Standard • 55s • alternating / series-test

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$$

994 **rel-adv-018**

R3 Standard • 55s • limit-comparison / series-test

$$\sum_{n=1}^{\infty} \frac{3n^2+1}{n^3+n}$$

995 **rel-adv-020**

R3 Standard • 55s • endpoint-analysis / power-series

$$\sum_{n=1}^{\infty} \frac{x^n}{n} \text{ at } x=-1$$

996 **rel-boss-013**

R5 Boss • 110s • telescoping / series-boss

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)}$$

997 **rel-boss-018**

R5 Boss • 110s • ratio-test / power-series

Radius of convergence of $\sum_{n=1}^{\infty} n!x^n$

998 **rel-hard-ser-001**

R5 Boss • 100s • power-series / ratio-test

Radius of convergence of $\sum_{n=0}^{\infty} \frac{(2n)!}{(n!)^2}x^n$

999 **rel-hard-ser-002**

R5 Boss • 100s • power-series / ratio-test

Radius of convergence of $\sum_{n=0}^{\infty} \frac{(n!)^2}{(2n)!}x^n$

1000 **rel-hard-ser-003**

R5 Boss • 100s • taylor / coefficient

coefficient of x^8 in $\cos(x^2)$

1001 **rel-hard-ser-004**

R5 Boss • 100s • taylor / coefficient

coefficient of x^{10} in $\sin(x^2)$

1002 **rel-hard-ser-005**

R5 Boss • 100s • telescoping / series-boss

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)(n+3)}$$

1003 **rel-hard-ser-006**

R5 Boss • 100s • generating-function / series-boss

$$\sum_{n=1}^{\infty} \frac{n^2}{2^n}$$

1004 **rel-hard-ser-007**

R5 Boss • 100s • alternating / series-boss

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$$

1005 **rel-hard-ser-008**

R5 Boss • 100s • telescoping / partial-fraction

$$\sum_{n=1}^{\infty} \frac{1}{4n^2 - 1}$$

1006 **rel-hard-ser-012**

R5 Boss • 100s • power-series / ratio-test

Radius of convergence of $\sum_{n=1}^{\infty} \frac{n^n}{n!} x^n$

1007 **rel-hard-ser-009**

R5 Boss • 80s • root-test / series-boss

$$\sum_{n=1}^{\infty} \frac{n!}{n^n}$$

1008 **rel-hard-ser-010**

R5 Boss • 80s • term-test / series-boss

$$\sum_{n=1}^{\infty} \frac{n^n}{n!}$$

1009 **rel-hard-ser-011**

R5 Boss • 80s • endpoint-analysis / alternating

$\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$ at the endpoint

1010 **hc-spec-007**

R6 Boss+ • 160s • series / telescoping

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)(n+3)(n+4)}$$

1011 **hc-spec-008**

R6 Boss+ • 160s • series / generating-function

$$\sum_{n=1}^{\infty} \frac{n^3}{2^n}$$

1012 **hc-spec-009**

R6 Boss+ • 160s • series / power-series

Radius of convergence of $\sum_{n=0}^{\infty} \frac{(3n)!}{(n!)^3} x^n$

1013 **hc-spec-010**

R6 Boss+ • 160s • series / bessel

coefficient of x^8 in $J_0(x)$

1014 **exam-ser-001**

R5 Boss • 180s • power-series / radius

收斂半徑：$\sum_{n=1}^{\infty} \frac{nx^n}{3^n}$

1015 **exam-ser-002**

R5 Boss • 180s • power-series / radius

收斂半徑：$\sum_{n=1}^{\infty} \frac{n!}{n^n} x^n$

1016 **exam-ser-003**

R5 Boss • 180s • power-series / radius

收斂半徑：$\sum_{n=1}^{\infty} \frac{(x-2)^n}{n}$

1017 **exam-ser-004**

R5 Boss • 180s • telescoping-series / exam-style

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$$

1018 **exam-ser-005**

R5 Boss • 180s • power-series / sum-series

$$\sum_{n=1}^{\infty} \frac{n}{2^n}$$

1019 **exam-ser-006**

R5 Boss • 180s • power-series / sum-series

$$\sum_{n=1}^{\infty} \frac{n^2}{2^n}$$

1020 **exam-ser-007**

R5 Boss • 180s • taylor / sum-series

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!}$$

1021 **exam-ser-008**

R5 Boss • 180s • taylor / sum-series

$$\sum_{n=0}^{\infty} \frac{1}{(2n+1)!}$$

1022 **exam-ser-009**

R5 Boss • 180s • taylor / coefficient

求e^{2x} 的x^5 係數

1023 **exam-ser-010**

R5 Boss • 180s • taylor / coefficient

求 $\sin(x^2)$ 的x^6 係數

1024 **exam-ser-011**

R5 Boss • 180s • taylor / coefficient

求 $\log(1+x)$ 的x^4 係數

1025 **exam-ser-012**

R5 Boss • 180s • geometric-series / exam-style

$$\sum_{n=0}^{\infty} \left(\frac{1}{3}\right)^n$$

1026 **exam-ser-013**

R5 Boss • 180s • power-series / sum-series

$$\sum_{n=1}^{\infty} \frac{n(n+1)}{2^n}$$

1027 **exam-ser-014**

R5 Boss • 180s • power-series / radius

收斂半徑：$\displaystyle\sum_{n=1}^{\infty} \frac{4^n x^n}{\sqrt{n+1}}$

1028 **exam-ser-015**

R5 Boss • 180s • alternating-series / sum-series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$$

1029 **exam-ser-016**

R5 Boss • 180s • p-series / sum-series

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

1030 **exam-ser-017**

R5 Boss • 180s • taylor / sum-series

$$\sum_{n=0}^{\infty} \frac{2^n}{n!}$$

1031 **exam-ser-018**

R5 Boss • 180s • binomial / coefficient

求$(1+x)^7$ 的x^3 係數

1032 **exam-ser-019**

R5 Boss • 180s • geometric-series / coefficient

求$\dfrac{1}{1-2x}$ 的x^4 係數

1033 **exam-ser-020**

R5 Boss • 180s • power-series / sum-series

$$\sum_{n=1}^{\infty} \frac{n}{3^n}$$

1034 **uni-ser-001**

R6 Boss+ • 210s • sum-series / power-series

$$\sum_{n=1}^{\infty} \frac{n}{4^n}$$

1035 **uni-ser-002**

R6 Boss+ • 210s • sum-series / power-series

$$\sum_{n=1}^{\infty} \frac{n^2}{3^n}$$

1036 **uni-ser-003**

R6 Boss+ • 210s • telescoping-series / exam-style

$$\sum_{n=1}^{\infty} \frac{1}{n(n+2)}$$

1037 **uni-ser-004**

R6 Boss+ • 210s • taylor / coefficient

求x^8 在 $\sin(x^2)$ 的係數

1038 **uni-ser-005**

R6 Boss+ • 210s • taylor / coefficient

求x^6 在$e^x \cos x$ 的係數

1039 **uni-ser-006**

R6 Boss+ • 210s • taylor / coefficient

求x^5 在$x^2 \log(1+x)$ 的係數

1040 **uni-ser-007**

R6 Boss+ • 210s • power-series / radius

收斂半徑：$\sum_{n=1}^{\infty} \frac{n^2}{5^n} x^n$

1041 **uni-ser-008**

R6 Boss+ • 210s • power-series / radius

收斂半徑：$\sum_{n=1}^{\infty} \frac{(n!)^2}{(2n)!} x^n$

1042 **uni-ser-009**

R6 Boss+ • 210s • power-series / radius

收斂半徑：$\sum_{n=1}^{\infty} \frac{n^n}{n!} x^n$

1043 **uni-ser-010**

R6 Boss+ • 210s • taylor / alternating-series

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1}$$

1044 **uni-ser-011**

R6 Boss+ • 210s • taylor / sum-series

$$\sum_{n=0}^{\infty} \frac{1}{(2n)!}$$

1045 **uni-ser-012**

R6 Boss+ • 210s • sum-series / power-series

$$\sum_{n=0}^{\infty} \frac{n(n-1)}{2^n}$$

1046 **uni-ser-013**

R6 Boss+ • 210s • coefficient / binomial

求x^4 在$\frac{1}{(1-x)^3}$ 的係數

1047 **uni-ser-014**

R6 Boss+ • 210s • power-series / radius

收斂半徑：$\sum_{n=1}^{\infty} \frac{3^n}{n^2} (x-1)^n$

1048 **uni-ser-015**

R6 Boss+ • 210s • sum-series / alternating-series

$$\sum_{n=1}^{\infty}(-1)^{n+1}\frac{n}{2^n}$$

1049 uni-ser-016

R6 Boss+ • 210s • taylor / coefficient

求x^7 在 $\sin x\cos x$ 的係數

1050 uni-ser-017

R6 Boss+ • 210s • sum-series / power-series

$$\sum_{n=1}^{\infty}\frac{n^3}{2^n}$$

1051 uni-ser-018

R6 Boss+ • 210s • coefficient / binomial

求x^6 在$(1+x)^{10}$ 的係數

1052 uni-ser-019

R6 Boss+ • 210s • power-series / radius

收斂半徑：$\displaystyle\sum_{n=1}^{\infty}\frac{x^n}{\sqrt{n}}$

1053 uni-ser-020

R6 Boss+ • 210s • p-series / sum-series

$$\sum_{n=1}^{\infty}\frac{1}{n^2}$$

1054 depth-ser-001

R5 Boss • 230s • sum-series / power-series

$$\sum_{n=1}^{\infty}\frac{n(n+1)}{3^n}$$

1055 depth-ser-002

R5 Boss • 230s • sum-series / power-series

$$\sum_{n=1}^{\infty}\frac{n^2}{4^n}$$

1056 depth-ser-003

R5 Boss • 230s • telescoping-series / exam-style

$$\sum_{n=1}^{\infty}\frac{1}{(2n-1)(2n+1)}$$

1057 depth-ser-004

R5 Boss • 230s • taylor / coefficient

coefficient of x^7 in e^{3x}

1058 depth-ser-005

R5 Boss • 230s • taylor / coefficient

coefficient of x^8 in $\cos(x^2)$

1059 depth-ser-006

R5 Boss • 230s • taylor / coefficient

coefficient of x^6 in $\dfrac{\sin x}{x}$

1060 depth-ser-007

R5 Boss • 230s • taylor / sum-series

$$\sum_{n=0}^{\infty}\frac{1}{(2n)!}$$

1061 depth-ser-008

R5 Boss • 230s • sum-series / power-series

$$\sum_{n=1}^{\infty}\frac{n}{3^{n+1}}$$

1062 depth-ser-009

R6 Boss+ • 230s • power-series / radius

radius of convergence of $\displaystyle\sum_{n=0}^{\infty}\binom{2n}{n}x^n$

1063 **depth-ser-010**

R6 Boss+ • 230s • power-series / radius

radius of convergence of $\sum_{n=1}^{\infty} \frac{(n!)^2}{(2n)!} 3^n x^n$

1064 **depth-ser-011**

R5 Boss • 230s • coefficient / geometric-series

coefficient of x^5 in $\frac{1}{(1-2x)^2}$

1065 **depth-ser-012**

R5 Boss • 230s • sum-series / power-series

$\sum_{n=1}^{\infty} \frac{n^2}{2^{n+1}}$

1066 **depth-ser-013**

R5 Boss • 230s • taylor / sum-series

$\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!}$

1067 **depth-ser-014**

R5 Boss • 230s • telescoping-series / exam-style

$\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)}$

1068 **depth-ser-015**

R6 Boss+ • 230s • taylor / coefficient

coefficient of x^3 in $e^x \log(1+x)$

1069 **depth-ser-016**

R6 Boss+ • 230s • sum-series / power-series

$\sum_{n=1}^{\infty} \frac{n^3}{3^n}$

1070 **depth-ser-017**

R5 Boss • 230s • power-series / radius

radius of convergence of $\sum_{n=1}^{\infty} \frac{n^3+1}{5^n} x^n$

1071 **depth-ser-018**

R5 Boss • 230s • taylor / sum-series

$\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!}$

1072 **depth-ser-019**

R6 Boss+ • 230s • taylor / coefficient

coefficient of x^4 in $(\log(1+x))^2$

1073 **depth-ser-020**

R5 Boss • 230s • power-series / radius

radius of convergence of $\sum_{n=1}^{\infty} \frac{2^n}{n^2}(x+1)^n$

1074 **burst-boss-ser-001**

R5 Boss • 240s • true-boss / boss-rank

$\sum_{n=1}^{\infty} \frac{n^3}{2^n}$

1075 **burst-boss-ser-002**

R5 Boss • 240s • true-boss / boss-rank

$\sum_{n=1}^{\infty} \frac{n^2}{3^n}$

1076 **burst-boss-ser-003**

R6 Boss+ • 300s • true-boss / boss-rank

$H_n = \sum_{k=1}^{n} \frac{1}{k}$. Find $\sum_{n=1}^{\infty} \frac{H_n}{2^n}$

1077 **burst-boss-ser-004**

R6 Boss+ • 300s • true-boss / boss-rank

$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$

1078 burst-boss-ser-005

R6 Boss+ • 300s • true-boss / boss-rank

Coefficient of x^6 in $(\log(1+x))^2$

1079 burst-boss-ser-006

R6 Boss+ • 300s • true-boss / boss-rank

Coefficient of x^8 in $\frac{x^2 e^{3x}}{1-x}$

1080 burst-boss2-ser-001

R6 Boss+ • 360s • true-boss / boss-rank

$$\sum_{n=1}^{\infty} \frac{1}{n^4}$$

1081 burst-boss2-ser-002

R6 Boss+ • 360s • true-boss / boss-rank

$$\sum_{n=1}^{\infty} \frac{1}{(2n-1)^4}$$

1082 burst-boss2-ser-003

R6 Boss+ • 360s • true-boss / boss-rank

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^4}$$

1083 burst-boss2-ser-004

R6 Boss+ • 360s • true-boss / boss-rank

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^6}$$

1084 burst-boss2-ser-005

R6 Boss+ • 360s • true-boss / boss-rank

$$\sum_{n=1}^{\infty} \frac{n^4}{3^n}$$

1085 burst-boss2-ser-006

R6 Boss+ • 360s • true-boss / boss-rank

$$\sum_{n=1}^{\infty} \frac{n^5}{3^n}$$

1086 burst-boss2-ser-007

R6 Boss+ • 360s • true-boss / boss-rank

$H_n = \sum_{k=1}^{n} \frac{1}{k}$. Find $\sum_{n=1}^{\infty} \frac{H_n}{3^n}$

1087 burst-boss2-ser-008

R6 Boss+ • 360s • true-boss / boss-rank

$H_n = \sum_{k=1}^{n} \frac{1}{k}$. Find $\sum_{n=1}^{\infty} \frac{H_n}{4^n}$

1088 burst-boss3-ser-001

R6 Boss+ • 420s • true-boss / boss-rank

$$\sum_{n=1}^{\infty} \frac{n^6}{2^n}$$

1089 burst-boss3-ser-002

R6 Boss+ • 420s • true-boss / boss-rank

$$\sum_{n=1}^{\infty} \frac{n^7}{2^n}$$

1090 burst-boss3-ser-003

R6 Boss+ • 420s • true-boss / boss-rank

$$\sum_{n=1}^{\infty} \frac{n^5}{4^n}$$

1091 burst-boss3-ser-004

R6 Boss+ • 420s • true-boss / boss-rank

$$\sum_{n=1}^{\infty} \frac{n^6}{3^n}$$

1092 burst-boss3-ser-005

R6 Boss+ • 420s • true-boss / boss-rank

$H_n = \sum_{k=1}^{n} \frac{1}{k}$. Find $\sum_{n=1}^{\infty} \frac{H_n}{n2^n}$

1093 **burst-boss3-ser-006**

R6 Boss+ • 420s • true-boss / boss-rank

Coefficient of x^{20} in $(\log(1+x))^2$

1094 **burst-boss3-ser-007**

R6 Boss+ • 420s • true-boss / boss-rank

Coefficient of x^{18} in $\frac{e^{2x}}{(1-x)^3}$

1095 **burst-boss3-ser-008**

R6 Boss+ • 420s • true-boss / boss-rank

Coefficient of x^{16} in $\frac{e^{2x}}{(1-x)^4}$

答案速查

0001. lim-001 `1`

0002. lim-002 `1/2`

0003. lim-003 `4`

0004. lim-004 `3/2`

0005. lim-005 `1`

0006. lim-006 `3`

0007. lim-007 `2`

0008. lim-008 `1/2`

0009. lim-009 `-1/6`

0010. lim-010 `2`

0011. lim-011 `0`

0012. lim-012 `1/3`

0013. lim-013 `1/2`

0014. lim-014 `-1/2`

0015. lim-015 `1/2`

0016. lim-016 `-1/8`

0017. lim-017 `1/3`

0018. lim-018 `1/3`

0019. lim-019 `-e/2`

0020. lim-020 `-1/3`

0021. td-lim-001 `exp(1/2)`

0022. td-lim-002 `sqrt(e)/3`

0023. td-lim-003 `11*e/24`

0024. td-lim-004 `-1/2`

0025. lim-021 `1/3`

0026. lim-022 `3/2`

0027. lim-023 `2`

0028. lim-024 `-1/2`

0029. lim-025 `1`

0030. lim-026 `0`

0031. lim-027 `dne`

0032. lim-028 `1/2`

0033. lim-029 `0`

0034. lim-030 `9/2`

0035. lim-031 `0`

0036. lim-032 `dne`

0037. lim-033 `1`

0038. lim-034 `0`

0039. lim-035 `5/2`

0040. lim-036 `1/2`

0041. lim-037 `2`

0042. lim-038 `1/3`

0043. lim-039 `dne`

0044. lim-040 `2`

0045. gap-tech-001 `rationalize`

0046. gap-tech-002 `taylor`

0047. gap-tech-008 `pathtest`

0048. mob-tech-001 `rationalize`

0049. mob-tech-002 `Taylor`

0050. mob-tech-009 `pathtest`

0051. mob-trap-009 `indeterminate`

0052. mob-limtrap-001 `-1/6`

0053. mob-limtrap-002 `1/3`

0054. mob-limtrap-003 `-1/24`

0055. mob-limtrap-004 `1/3`

0056. mob-limtrap-005 `1/6`

0057. mob-limtrap-006 `-1/8`

0058. mob-limtrap-007 `0`

0059. mob-limtrap-008 `0`

0060. mob-limtrap-009 `DNE`

0061. mob-limtrap-010 `DNE`

0062. rel-basic-001
5

0063. rel-basic-002
9/2

0064. rel-basic-003
4

0065. rel-basic-004
2

0066. rel-basic-005
1/2

0067. rel-basic-006
4

0068. rel-adv-001
1/120

0069. rel-adv-002
1/24

0070. rel-adv-003
-1/4

0071. rel-adv-004
2/15

0072. rel-boss-019
1

0073. rel-hard-lim-001
-1/5040

0074. rel-hard-lim-002
17/315

0075. rel-hard-lim-003
1/5

0076. rel-hard-lim-004
1/360

0077. rel-hard-lim-005
-e/2

0078. rel-hard-lim-007
1/2

0079. rel-hard-lim-008
1

0080. rel-hard-lim-009
1/5

0081. rel-hard-lim-010
11*e/24

0082. rel-hard-lim-006
DNE

0083. exam-lim-001
81/40

0084. exam-lim-002
8/3

0085. exam-lim-003
1

0086. exam-lim-004
1/3

0087. exam-lim-005
2

0088. exam-lim-006
-2

0089. exam-lim-007
-9/8

0090. exam-lim-008
1/3

0091. exam-lim-009
1/3

0092. exam-lim-010
-8/3

0093. exam-lim-011
1

0094. exam-lim-012
exp(6)

0095. exam-lim-013
exp(-1/6)

0096. exam-lim-014
-1/2

0097. exam-lim-015
1

0098. exam-lim-016
1/2

0099. exam-lim-017
-1/12

0100. exam-lim-018
1/2

0101. exam-lim-019
1

0102. exam-lim-020
1/6

0103. uni-lim-001
1/120

0104. uni-lim-002
-1/4

0105. uni-lim-003
3/2

0106. uni-lim-004
1/2

0107. uni-lim-005
1/8

0108. uni-lim-006
1/2

0109. uni-lim-007
-1

0110. uni-lim-008
-9*exp(3)/2

0111. uni-lim-009
`1`

0112. uni-lim-010
`-1/2`

0113. uni-lim-011
`-1/2`

0114. uni-lim-012
`1`

0115. uni-lim-013
`exp(1/6)`

0116. uni-lim-014
`1/3`

0117. uni-lim-015
`1/2`

0118. depth-lim-001
`-1/5040`

0119. depth-lim-002
`1/5`

0120. depth-lim-003
`1/2`

0121. depth-lim-004
`1/6`

0122. depth-lim-005
`2/3`

0123. depth-lim-006
`1/2`

0124. depth-lim-007
`-exp(1)/2`

0125. depth-lim-008
`1/2`

0126. depth-lim-009
`1/3`

0127. depth-lim-010
`1/2`

0128. depth-lim-011
`-1/2`

0129. depth-lim-012
`exp(1)/2`

0130. burst-boss-lim-001
`2/15`

0131. burst-boss-lim-002
`1/120`

0132. burst-boss-lim-003
`-1/4`

0133. burst-boss-lim-004
`-1/8`

0134. burst-boss-lim-005
`1/6`

0135. burst-boss-lim-006
`-1/7`

0136. burst-boss-lim-007
`-e/2`

0137. burst-boss-lim-008
`-e/2`

0138. burst-boss2-lim-001
`exp(-1/6)`

0139. burst-boss2-lim-002
`-1/45`

0140. burst-boss2-lim-003
`7/360`

0141. burst-boss2-lim-004
`-1/12`

0142. burst-boss2-lim-005
`1/10`

0143. burst-boss2-lim-006
`1/21`

0144. burst-boss2-lim-007
`-1/4`

0145. burst-boss2-lim-008
`-1/4`

0146. burst-boss2-lim-009
`(1/sqrt(2)-1)/2`

0147. burst-boss2-lim-010
`(2-e)/2`

0148. burst-boss3-lim-001
`1/exp(1)`

0149. burst-boss3-lim-002
`4/exp(2)`

0150. burst-boss3-lim-003
`1/2`

0151. burst-boss3-lim-004
`(e-1)/12`

0152. burst-boss3-lim-005
`-1/24`

0153. burst-boss3-lim-006
`-1/24`

0154. burst-boss3-lim-007
`11/72`

0155. burst-boss3-lim-008
`-1/42`

0156. burst-boss3-lim-009
`-1/180`

0157. burst-boss3-lim-010
`-1/12`

0158. burst-boss3-lim-011
`1/16`

0159. burst-boss3-lim-012
`exp(1/6)`

0160. der-001 `5*x^4-6*x`

0161. der-002 `cos(x)-sin(x)`

0162. der-003 `exp(x)*log(x)+exp(x)/x`

0163. der-004 `1/(2*sqrt(x))`

0164. der-005 `2*x/(1+x^2)`

0165. der-006 `2*x*sin(x)+x^2*cos(x)`

0166. der-007 `1/(1+x^2)`

0167. der-008 `1/(x+1)^2`

0168. der-009 `x^x*(log(x)+1)`

0169. der-010 `exp(-x)*(2*x*cos(x^2)-sin(x^2))`

0170. der-011 `cos(x)/sin(x)`

0171. der-012 `((x^2+1)/x-2*x*log(x))/(x^2+1)^2`

0172. der-013 `x^sin(x)*(cos(x)*log(x)+sin(x)/x)`

0173. der-014 `1/(1+x^2)`

0174. der-015 `1/sqrt(1+x^2)`

0175. der-016 `1/(1+cos(x))`

0176. der-017 `exp(x^2)*(2*x*cos(3*x)-3*sin(3*x))`

0177. der-018 `1/sqrt(x^2-1)`

0178. der-019 `10*x*(x^2+1)^4`

0179. der-020 `1/(2*sqrt(x)*(1+x))`

0180. td-der-001
`2*(1-3*x^2)/(1+x^2)^3`

0181. td-der-002
`(35*x^3-15*x)/2`

0182. td-der-003
`log(1+x)+x/(1+x)`

0183. td-der-004
`-3*sin(x)/(5+3*cos(x))`

0184. der-021 `2*x*y+y*cos(x*y)`

0185. der-022 `x^2*exp(x*y)+3*y^2`

0186. der-023 `exp(x*y)*(1+x*y)`

0187. der-024 `22/5`

0188. der-025 `2*x/(x^2+y^2)`

0189. der-026 `2*x*sin(3*x)+3*x^2*cos(3*x)`

0190. der-027 `x/(1+x^2)`

0191. der-028 `exp(x)/(1+exp(2*x))`

0192. der-029 `exp(x)*(x+2)`

0193. der-030 `-(2*x+y)/(x+2*y)`

0194. der-031 `2*x*exp(x^2+y^2)`

0195. der-032 `x/(x^2+y^2)`

0196. der-033 `2*y-sin(x)`

0197. der-034 `x-y`

0198. der-035 `exp(t)*(4*t^3+t^4)`

0199. der-036 `4`

0200. der-037 `27`

0201. der-038 `(x^2-2*x-1)/(x-1)^2`

0202. der-039 `2*sin(x)*cos(x)`

0203. der-040 `2*x^log(x)*log(x)/x`

0204. der-041 `-0.04`

0205. der-042 `-0.028`

0206. der-043 `0`

0207. der-044 `y*exp(x*y)*dx+x*exp(x*y)*dy`

0208. der-045 `-y*dx/(x^2+y^2)+x*dy/(x^2+y^2)`

0209. der-046 `4.92`

0210. der-047 `-9/2`

0211. der-048 `-9`

0212. der-049 `-1`

0213. der-050 `5`

0214. der-051 `3`

0215. der-052 `144`

0216. der-053 `-1`

0217. der-054 `128`

0218. der-055 `localminimum`

0219. der-056 `saddlepoint`

0220. der-057 `0`

0221. der-058 `-4`

0222. der-059 `-1`

0223. der-060 `1`

0224. der-061 `exp(3*x)`

0225. der-062 `x^2`

0226. der-063 `2`

0227. der-064 `exp(2*x)`

0228. der-065 `1/x`

0229. der-066 `x^4`

0230. der-067 `4*x`

0231. der-068 `exp(sin(x))*(2*x+x^2*cos(x))`

0232. der-069 `4*(x^2+y^2)`

0233. der-070 `-4*y`

0234. der-071 `r`

0235. der-072 `-2*x/y`

0236. der-073 `20*((2*x+y)^2+(x+3*y)^2)`

0237. der-074 `-4`

0238. der-075 `x^3-3*x*y^2`

0239. der-076 `3*x^2*y-y^3`

0240. der-077 `sqrt(2)`

0241. der-078 `-4`

0242. der-079 `holomorphic`

0243. der-080 `notholomorphic`

0244. der-081 `1/2`

0245. der-082 `0`

0246. der-083 `1/2`

0247. der-084 `1`

0248. der-085 `-8`

0249. der-086 `4*(x^2+y^2)`

0250. der-087 `0`

0251. der-088 `harmonic`

0252. gap-der-param-001
`3*t/2`

0253. gap-der-param-002
`-cos(t)/sin(t)`

0254. gap-der-param-003
`2`

0255. gap-der-param-004
`3`

0256. gap-der-polar-001
`1`

0257. gap-der-param-005
`5`

0258. gap-der-app-001
`72*pi`

0259. gap-der-app-002
`2*pi`

0260. gap-der-app-003
`-3/2`

0261. gap-der-app-004
`-1/4`

0262. gap-der-app-005
`2.01`

0263. gap-der-app-006
`3/2`

0264. gap-der-app-007
`2`

0265. gap-der-ode-001
`2*exp(-x)`

0266. gap-der-ode-002
`3*exp(2*x)`

0267. gap-der-ode-003
`(exp(x)-exp(-x))/2`

0268. gap-der-ode-004
`sin(x)`

0269. gap-der-ode-005
`1/(1+exp(-x))`

0270. gap-tech-007
`parametricdifferentiation`

0271. mob-tech-012
`Lagrangemultiplier`

0272. mob-tech-014
`nabla`

0273. mob-tech-018
`Besselequation`

0274. mob-tech-020
`logarithmicdifferentiation`

0275. mob-trap-001
`missingchainfactor`

0276. mob-trap-007
`treatyconstant`

0277. mob-trap-011
`zero`

0278. mob-lm-001
`25`

0279. mob-lm-002
`18`

0280. mob-lm-003
`64`

0281. mob-lm-004
`16`

0282. mob-lm-005
`16`

0283. mob-lm-006
`3`

0284. mob-lm-007
`25`

0285. mob-lm-008
`1/3`

0286. mob-lm-009
`25`

0287. mob-lm-010
`2`

0288. mob-bessel-001
`1`

0289. mob-bessel-002
`0`

0290. mob-bessel-003
`-1/4`

0291. mob-bessel-004
1/2

0292. mob-bessel-005
-J_1(x)

0293. mob-bessel-006
Besselequation

0294. mob-bessel-007
regularsingular

0295. mob-bessel-008
J_2(x)

0296. mob-bessel-009
-J_1(x)

0297. mob-bessel-010
second-orderODE

0298. mob-nabla-001
3

0299. mob-nabla-002
-2

0300. mob-nabla-003
6

0301. mob-nabla-004
2*sqrt(2)

0302. mob-nabla-005
0

0303. mob-nabla-006
0

0304. mob-nabla-007
2

0305. mob-nabla-008
Laplacian

0306. mob-nabla-009
conservativefield

0307. mob-nabla-010
curl

0308. rel-basic-007
12

0309. rel-basic-008
1

0310. rel-basic-009
2

0311. rel-basic-010
1/4

0312. rel-basic-011
1/e

0313. rel-basic-012
1

0314. rel-adv-005
20

0315. rel-adv-006
8/3

0316. rel-adv-007
128

0317. rel-adv-008
9

0318. rel-boss-007
1/64

0319. rel-hard-mv-005
4

0320. rel-hard-mv-006
1/27

0321. rel-hard-mv-007
2

0322. rel-hard-der-001
1

0323. rel-hard-der-002
-1

0324. rel-hard-der-003
-4

0325. rel-hard-der-004
1/(2*sqrt(2))

0326. rel-hard-der-005
5/3

0327. rel-hard-der-006
3/8

0328. rel-hard-der-007
5/3

0329. rel-hard-der-008
6

0330. rel-hard-der-009
0

0331. rel-hard-der-010
2

0332. hc-der-001
x^sin(x)*(cos(x)*log(x)+sin(x)/x)

0333. hc-der-002
sin(x)^cos(x)*(-sin(x)*log(sin(x))+cos(x)*cot(x))

0334. hc-der-003
exp(exp(2*x)+x^2)*(2*exp(2*x)+2*x)

0335. hc-der-004
360

0336. hc-der-005
42

0337. hc-der-006
1680

0338. hc-der-007
-1/11

0339. hc-der-008
`16`

0340. hc-der-009
`4096`

0341. hc-der-010
`16`

0342. exam-der-001
`x^x*(log(x)+1)`

0343. exam-der-002
`x^sin(x)*(cos(x)*log(x)+sin(x)/x)`

0344. exam-der-003
`(1-log(x))/x^2`

0345. exam-der-004
`1/(1+x^2)`

0346. exam-der-005
`exp(-x ) *(2*x*sin(x ) -x^2*sin(x ) +x^2*cos(x))`

0347. exam-der-006
`1/sqrt(1+x^2)`

0348. exam-der-007
`(2+4*x^2)*exp(x^2)`

0349. exam-der-008
`1/(1+cos(x))`

0350. exam-der-009
`-1/(1+sin(x))`

0351. exam-der-010
`log(1+x^2)+2*x^2/(1+x^2)`

0352. exam-der-011
`(x^2+1)^2*(6*x*(x-1)-(x^2+1))/(x-1)^2`

0353. exam-der-012
`2/x+1`

0354. exam-der-013
`11/4`

0355. exam-der-014
`2*x*exp(x^4)`

0356. exam-der-015
`2*x*log(1+x^4)-log(1+x^2)`

0357. exam-der-016
`1/2`

0358. exam-der-017
`1/(2*sqrt(x)*(1+x))`

0359. exam-der-018
`sec(x)`

0360. exam-der-019
`1/(1+x^2)^(3/2)`

0361. exam-der-020
`exp(x)/(1+exp(x))^2`

0362. exam-multi-001
`2*x*y+y*cos(x*y)`

0363. exam-multi-002
`6*x*y^2+(1+x*y)*exp(x*y)`

0364. exam-multi-003
`20`

0365. exam-multi-004
`27`

0366. exam-multi-005
`16/5`

0367. exam-multi-008
`4*(x^2+y^2)`

0368. exam-multi-009
`2*x*y+2*y*z+2*z*x`

0369. exam-multi-010
`-2*y`

0370. uni-der-001
`x^cos(x)*(-sin(x)*log(x)+cos(x)/x)`

0371. uni-der-002
`sin(x)^x*(log(sin(x))+x*cot(x))`

0372. uni-der-003
`11`

0373. uni-der-004
`12`

0374. uni-der-005
`-x/(1+x^2)^(3/2)`

0375. uni-der-006
`cos(x)*exp(sin(x)^2)`

0376. uni-der-007
`2*log(1+4*x^2)-log(1+x^2)`

0377. uni-der-008
`2/(1+x^2)`

0378. uni-der-009
`5/3`

0379. uni-der-010
`1`

0380. uni-der-011
`-1`

0381. uni-der-012
`128`

0382. uni-der-013
`2`

0383. uni-der-014
`6/5`

0384. uni-der-015
`exp(y)+exp(z)+exp(x)`

0385. uni-der-016
`-x`

0386. uni-der-017
`1/(x*log(x)*log(log(x)))`

0387. uni-der-018
`3*log(x)^2*exp(log(x)^3)/x`

0388. uni-der-019
`32`

0389. uni-der-020
`(1+x^2)^x*(log(1+x^2)+2*x^2/(1+x^2))`

0390. depth-der-001
`x^(x^2)*(2*x*log(x)+x)`

0391. depth-der-002
`sin(x)^cos(x)*(-sin(x)*log(sin(x))+cos(x)*cot(x))`

0392. depth-der-003
`exp(x)*(x+2)`

0393. depth-der-004
`2/x`

0394. depth-der-005
`3*x^2*cos(x^6)-2*x*cos(x^4)`

0395. depth-der-006
`1/sqrt(x^2+4)`

0396. depth-der-007
`-1`

0397. depth-der-008
`1`

0398. depth-der-009
`3/4`

0399. depth-der-010
`-3`

0400. depth-der-011
`8`

0401. depth-der-012
`0`

0402. depth-der-013
`y*z`

0403. depth-der-014
`y^2-x^2`

0404. depth-der-015
`2/(1+x^2)`

0405. depth-der-016
`(2*log(x)-log(x)^2)/x^2`

0406. depth-der-017
`81`

0407. depth-der-018
`(1+1/(2*sqrt(x)))/(2*sqrt(x+sqrt(x)))`

0408. burst-der-001
`1048576`

0409. burst-der-002
`-14348907`

0410. burst-der-003
`262144`

0411. burst-der-004
`720`

0412. burst-der-005
`56320`

0413. burst-der-006
`-5040`

0414. burst-der-007
`479001600`

0415. burst-der-008
`-64`

0416. burst-der-009
`24024`

0417. burst-der-010
`-1199`

0418. burst-boss-der-001
`241087680`

0419. burst-boss-der-002
`-21257280`

0420. burst-boss-der-003
`-22591`

0421. burst-boss-der-004
`16124`

0422. burst-boss-der-005
`164833`

0423. burst-boss-der-006
`351261243360`

0424. burst-boss-der-007
`27525120`

0425. burst-boss-der-008
`13172296626`

0426. burst-boss-der-009
`598752000`

0427. burst-boss-der-010
`1307674368000`

0428. burst-boss2-der-001
`-1024`

0429. burst-boss2-der-002
`-1024`

0430. burst-boss2-der-003
`3041525760`

0431. burst-boss2-der-004
`54747463680`

0432. burst-boss2-der-005
`-1334792724768`

0433. burst-boss2-der-006
`239500800`

0434. burst-boss2-der-007
`1796256000`

0435. burst-boss2-der-008
`271996268544000`

0436. burst-boss3-der-001
`863130293635276800`

0437. burst-boss3-der-002
`2481560048340172800000`

0438. burst-boss3-der-003
`3771971273477062656 0000`

0439. burst-boss3-der-004
`1621934672117760000`

0440. burst-boss3-der-005
`171666204333720864768 00000`

0441. burst-boss3-der-006
`5537619494636156928 00000`

0442. burst-boss3-der-007
`-27278913766680576 0000`

0443. burst-boss3-der-008
`18458731648787189760000`

0444. int-001 `2*x^3`

0445. int-002 `sin(x)`

0446. int-003 `log(abs(x))`

0447. int-004 `exp(3*x)/3`

0448. int-005 `exp(x^2)/2`

0449. int-006 `log(1+x^2)`

0450. int-007 `sin(x^2)/2`

0451. int-008 `1`

0452. int-009 `x^2*log(x)/2-x^2/4`

0453. int-010 `-cos(x)+cos(x)^3/3`

0454. int-011 `atan(x/2)/2`

0455. int-012 `pi/4`

0456. int-013 `x*(log(x)^2-2*log(x)+2)`

0457. int-014 `x^2/2-log(x^2+1)/2`

0458. int-015 `exp(x)*(sin(x)-cos(x))/2`

0459. int-016 `2*log(2)-1`

0460. int-017 `1`

0461. int-018 `-log(1+cos(x))`

0462. int-019 `2*sqrt(x)-2*log(1+sqrt(x))`

0463. int-020 `log(2)/3`

0464. td-int-001
`pi*exp(-3)/2`

0465. td-int-002
`pi*log(9/2)`

0466. td-int-003
`log(2/5)`

0467. td-int-004
`log(7/3)/2`

0468. td-int-005
`pi/sqrt(5)`

0469. td-int-006
`sqrt(pi)/(12*sqrt(3))`

0470. int-021 `1`

0471. int-022 `2`

0472. int-023 `1/8`

0473. int-024 `1/3`

0474. int-025 `pi/2`

0475. int-026 `9`

0476. int-027 `log(1+x^2)/2`

0477. int-028 `x*log(x)-x`

0478. int-029 `atan(x/3)/3`

0479. int-030 `1/4`

0480. int-031 `1/4`

0481. int-032 `3*pi`

0482. int-033 `4*pi`

0483. int-034 `pi/4`

0484. int-035 `pi/2`

0485. int-036 `1/6`

0486. int-037 `exp(x^2)/2`

0487. int-038 `x*sin(x)+cos(x)`

0488. int-039 `pi/4`

0489. int-040 `1`

0490. int-041 `log(1+x^4)/4`

0491. int-042 `(2/9)*(1+x^3)^(3/2)`

0492. int-043 `log(1+sqrt(1+x^2))`

0493. int-044 `log(1+exp(x^2))/2`

0494. int-045 `log(x)^3/3`

0495. int-046 `3/8`

0496. int-047 `pi`

0497. int-048 `pi/4`

0498. int-049 `pi`

0499. int-050 `pi/3`

0500. int-051 `x/(4*sqrt(x^2+4))`

0501. int-052 `exp(x)*(x^2-2*x+2)`

0502. int-053 `-x^2*cos(x)+2*x*sin(x)+2*cos(x)`

0503. **int-054** `exp(x)*(sin(x)-cos(x))/2`

0504. **int-055** `x*(log(x)^2-2*log(x)+2)`

0505. **int-056** `pi/4-1/2`

0506. **int-057** `(exp(x)-sin(x)-cos(x))/2`

0507. **int-058** `(exp(2*x)-2*x-1)/4`

0508. **int-059** `x^2-2*x+2-2*exp(-x)`

0509. **int-060** `x*sin(x)/2`

0510. **int-061** `pi/4`

0511. **int-062** `1/2`

0512. **int-063** `pi/4`

0513. **int-064** `pi^2/4`

0514. **int-065** `log(7/3)`

0515. **int-066** `log(4)`

0516. **int-067** `log(4)`

0517. **int-068** `log(5/2)`

0518. **int-069** `log(4)`

0519. **int-070** `log(3)`

0520. **int-071** `log(11/5)/2`

0521. **int-072** `log(2)/2`

0522. **int-073** `log(5/3)/2`

0523. **int-074** `pi*log(5/2)/2`

0524. **int-075** `log(5/2)`

0525. **int-076** `log(6)`

0526. **gap-int-pf-001**
`(log(x-1)-log(x+1))/2`

0527. **gap-int-pf-002**
`log(x)`

0528. **gap-int-pf-003**
`(log(x)-log(x+2))/2`

0529. **gap-int-pf-004**
`log(x+1)+log(x+2)`

0530. **gap-int-pf-005**
`-1/(x+1)`

0531. **gap-int-pf-006**
`8*log(x-1)/3+log(x+2)/3`

0532. **gap-int-pf-007**
`2*log(x+2)-log(x+3)`

0533. **gap-int-polar-001**
`pi`

0534. **gap-int-polar-002**
`pi/4`

0535. **gap-int-triple-001**
`1/8`

0536. **gap-int-triple-002**
`3/2`

0537. **gap-int-triple-003**
`1/6`

0538. **gap-int-triple-004**
`1/6`

0539. **gap-int-triple-005**
`12*pi`

0540. **gap-int-triple-006**
`32*pi/3`

0541. **gap-int-cov-001**
`1/2`

0542. **gap-int-cov-002**
`1`

0543. **gap-tech-003**
`substitution`

0544. **gap-tech-004**
`integrationbyparts`

0545. **gap-tech-006**
`partialfraction`

0546. **mob-tech-003**
`u-substitution`

0547. **mob-tech-004**
`integrationbyparts`

0548. **mob-tech-005**
`partialfraction`

0549. **mob-tech-006**
`trigsubstitution`

0550. **mob-tech-010**
`King'sproperty`

0551. **mob-tech-011**
`Frullaniintegral`

0552. **mob-tech-013**
`gradienttheorem`

0553. **mob-tech-015**
`Betafunction`

0554. **mob-tech-016**
`Gammafunction`

0555. **mob-tech-017**
`Wallisreduction`

0556. **mob-trap-002**
`dividebyinnercoefficient`

0557. **mob-trap-003**
`arctannotlog`

0558. **mob-trap-006**
`requiresapositive`

0559. **mob-trap-008**
`multiplybyJacobian`

0560. mob-trap-010
`secx`

0561. mob-trap-012
`denominatorGamma(a+b)`

0562. mob-beta-001
`1`

0563. mob-beta-002
`1/2`

0564. mob-beta-003
`1/6`

0565. mob-beta-004
`1/12`

0566. mob-beta-005
`pi`

0567. mob-beta-006
`2`

0568. mob-beta-007
`pi/2`

0569. mob-beta-008
`pi/8`

0570. mob-beta-009
`1/30`

0571. mob-beta-010
`1/4`

0572. mob-gamma-001
`1`

0573. mob-gamma-002
`sqrt(pi)`

0574. mob-gamma-003
`sqrt(pi)/2`

0575. mob-gamma-004
`3*sqrt(pi)/4`

0576. mob-gamma-005
`120`

0577. mob-gamma-006
`2`

0578. mob-gamma-007
`sqrt(pi)/2`

0579. mob-gamma-008
`6`

0580. mob-gamma-009
`15*sqrt(pi)/8`

0581. mob-gamma-010
`1`

0582. mob-wallis-001
`pi/4`

0583. mob-wallis-002
`2/3`

0584. mob-wallis-003
`3*pi/16`

0585. mob-wallis-004
`8/15`

0586. mob-wallis-005
`pi/4`

0587. mob-wallis-006
`3*pi/16`

0588. mob-wallis-007
`pi/16`

0589. mob-wallis-008
`1`

0590. mob-wallis-009
`1`

0591. mob-wallis-010
`1/2`

0592. rel-basic-013
`1/3`

0593. rel-basic-014
`2`

0594. rel-basic-015
`(e^2-1)/2`

0595. rel-basic-016
`log(2)`

0596. rel-basic-017
`1`

0597. rel-basic-018
`6`

0598. rel-adv-011
`1/9`

0599. rel-adv-012
`1/2`

0600. rel-adv-013
`pi/2`

0601. rel-adv-014
`5/6`

0602. rel-adv-009
`exp(x^2)*(x^2-1)/2`

0603. rel-adv-010
`(1+x^2)*log(1+x^2)/2-(1+x^2)/2`

0604. rel-boss-001
`pi/2`

0605. rel-boss-002
`log(5/2)`

0606. rel-boss-003
`log(7/2)`

0607. rel-boss-004
`15*sqrt(pi)/8`

0608. rel-boss-005
`3*pi/8`

0609. rel-boss-006
`3*pi/512`

0610. rel-boss-008
`0`

0611. rel-boss-009
`1/6`

0612. rel-boss-010
`1/18`

0613. rel-boss-011
`1/720`

0614. rel-boss-012
`128*pi/5`

0615. rel-boss-014
`pi^2/6`

0616. rel-boss-015
`pi^4/15`

0617. rel-boss-016
`-pi*log(2)/2`

0618. rel-boss-017
`2`

0619. rel-boss-020
`1/2`

0620. rel-hard-int-001
`15/8`

0621. rel-hard-int-002
`3*sqrt(pi)/128`

0622. rel-hard-int-003
`-1/9`

0623. rel-hard-int-004
`1/32`

0624. rel-hard-int-005
`35*pi/256`

0625. rel-hard-int-006
`16/35`

0626. rel-hard-int-007
`pi/32`

0627. rel-hard-int-008
`log(11/3)`

0628. rel-hard-int-009
`log(5/2)`

0629. rel-hard-int-010
`pi/2`

0630. rel-hard-int-011
`pi/4`

0631. rel-hard-int-012
`pi/(2*sqrt(2))`

0632. rel-hard-int-013
`0`

0633. rel-hard-int-014
`-1*pi^2/12`

0634. rel-hard-int-015
`-4`

0635. rel-hard-int-016
`log(2)/3`

0636. rel-hard-int-017
`-pi*log(2)`

0637. rel-hard-int-018
`pi/32`

0638. rel-hard-cpx-001
`-1/6`

0639. rel-hard-cpx-002
`-1/2`

0640. rel-hard-cpx-003
`1/6`

0641. rel-hard-cpx-004
`-1/2`

0642. rel-hard-cpx-005
`1`

0643. rel-hard-cpx-006
`e`

0644. rel-hard-cpx-007
`1`

0645. rel-hard-cpx-008
`1`

0646. rel-hard-cpx-009
`2`

0647. rel-hard-cpx-010
`-1/24`

0648. rel-hard-mv-001
`1/60`

0649. rel-hard-mv-002
`pi/24`

0650. rel-hard-mv-003
`4*pi/5`

0651. rel-hard-mv-004
`36*pi`

0652. rel-hard-mv-008
`1/2`

0653. rel-hard-mv-009
`64*pi/3`

0654. rel-hard-mv-010
`1/2520`

0655. hc-usub-001
`(1+x^2)^(3/2)/3-sqrt(1+x^2)`

0656. hc-usub-002
`exp(x^2)*(x^4-2*x^2+2)/2`

0657. hc-usub-003
`(1+x^2)*(log(1+x^2)^2-2*log(1+x^2)+2)/2`

0658. hc-usub-004
`atan(x^3)/3`

0659. hc-usub-005
`log(1+sqrt(1+x^4))/2`

0660. hc-usub-006
`2*(1+exp(x))^(3/2)/3`

0661. hc-usub-007
`pi/6`

0662. hc-usub-008
`1/16`

0663. hc-usub-009
`1/4`

0664. hc-usub-010
`log((1+e)/2)/3`

0665. hc-ibp-001
`exp(2*x)*(4*x^3-6*x^2+6*x-3)/8`

0666. hc-ibp-002
`x^2*sin(3*x) /3 + 2*x*cos(3*x)/9-2*sin(3*x)/27`

0667. hc-ibp-003
`exp(x)*((x^2-1)*sin(x)+(2*x-1-x^2)*cos(x))/2`

0668. hc-ibp-004
`x*(log(x)^3-3*log(x)^2+6*log(x)-6)`

0669. hc-ibp-005
`(x^2 + 1 ) *atan(x ) ^2 / 2-x*atan(x )+log(1+x^2)/2`

0670. hc-ibp-006
`exp(2*x)*(2*sin(3*x)-3*cos(3*x))/13`

0671. hc-ibp-007
`x^4*log(x)/4-x^4/16`

0672. hc-ibp-008
`2*log(2)/5-47/300`

0673. hc-ibp-009
`pi^2/4-2`

0674. hc-ibp-010
`1/6`

0675. hc-rad-001
`sqrt(x^2-1)/x`

0676. hc-rad-002
`atan((x+1)/2)/2`

0677. hc-rad-003
`-log(abs(x ) ) + log(abs(x-1 ) ) + log(abs(x+1))`

0678. hc-rad-004
`(log(abs(x-1 ) )-log(abs(x + 1 ) ) )/4-atan(x)/2`

0679. hc-rad-005
`2*sqrt(3)-2*pi/3`

0680. hc-rad-006
`1`

0681. hc-rad-007
`8/315`

0682. hc-rad-008
`pi/3`

0683. hc-rad-009
`pi/6`

0684. hc-rad-010
`pi/4`

0685. hc-spec-001
`8/81`

0686. hc-spec-002
`pi/8`

0687. hc-spec-003
`1/280`

0688. hc-spec-004
`63*pi/512`

0689. hc-spec-005
`8*pi^6/63`

0690. hc-spec-006
`3*sqrt(pi)/(32*sqrt(2))`

0691. exam-int-001
`-1/(2*(1+x^2))`

0692. exam-int-002
`x^2/2-log(1+x^2)/2`

0693. exam-int-003
`exp(x)*(x^2-2*x+2)`

0694. exam-int-004
`x^2*log(x)/2-x^2/4`

0695. exam-int-005
`log(x)^2/2`

0696. exam-int-006
`-cos(x)+cos(x)^3/3`

0697. exam-int-007
`tan(x)^2/2`

0698. exam-int-008
`atan(x/2)/2`

0699. exam-int-009
`log(abs((x-1)/(x+1)))/2`

0700. exam-int-010
`2*log(1+x^(3/2))/3`

0701. exam-int-011
`-sqrt(4-x^2)`

0702. exam-int-012
`exp(2*x)*(2*sin(x)-cos(x))/5`

0703. exam-int-013
`x*atan(x)-log(1+x^2)/2`

0704. exam-int-014
`x-atan(x)`

0705. exam-int-015
`log(log(x))`

0706. exam-int-016
`sin(log(x))`

0707. exam-int-017
`log(x^2+3*x+5)`

0708. exam-int-018
`atan(x^2)/2`

0709. exam-int-019
`2*atan(sqrt(x))`

0710. exam-int-020
`2*sqrt(1+x^3)/3`

0711. exam-int-021
`log(2)/2`

0712. exam-int-022
`log(2)/3`

0713. exam-int-023
`pi/4`

0714. exam-int-024
`log(2)`

0715. exam-int-025
`-1`

0716. exam-int-026
`1/2`

0717. exam-int-027
`pi/4`

0718. exam-int-028
`log(2)-1/2`

0719. exam-int-029
`pi`

0720. exam-int-030
`1/6`

0721. exam-multi-006
`1/2`

0722. exam-multi-007
`pi`

0723. uni-int-001
`x^4/4-x^2/2+log(1+x^2)/2`

0724. uni-int-002
`exp(x^2)*(x^2-1)/2`

0725. uni-int-003
`-(log(x)+1)/x`

0726. uni-int-004
`-atan(x)/x+log(x)-log(1+x^2)/2`

0727. uni-int-005
`(1+x^2)^(3/2)/3`

0728. uni-int-006
`sin(x)^6/6`

0729. uni-int-007
`tan(x)+tan(x)^3/3`

0730. uni-int-008
`atan((x+1)/2)/2`

0731. uni-int-009
`log(x^2+x+1)`

0732. uni-int-010
`exp(x)*(sin(x)+cos(x))/2`

0733. uni-int-011
`x^3*log(x)/3-x^3/9`

0734. uni-int-012
`(x^2+1)*atan(x)/2-x/2`

0735. uni-int-013
`1/2-log(2)/2`

0736. uni-int-014
`pi/4`

0737. uni-int-015
`1/4`

0738. uni-int-016
`2/15`

0739. uni-int-017
`pi/4`

0740. uni-int-018
`1/2`

0741. uni-int-019
`2`

0742. uni-int-020
`pi/(3*sqrt(3))`

0743. uni-int-021
`1/8`

0744. uni-int-022
`pi/2`

0745. uni-int-023
`6*pi`

0746. uni-int-024
`1/6`

0747. uni-int-025
`pi/4`

0748. depth-int-001
`log(1+x^4)/4`

0749. depth-int-002
`x^3/3-log(1+x^3)/3`

0750. depth-int-003
`-1/sqrt(1+x^2)`

0751. depth-int-004
`log(x)^3/3`

0752. depth-int-005
`exp(x)*(x*(sin(x)-cos(x))+cos(x))/2`

0753. depth-int-006
`exp(2*x)*(2*cos(3*x)+3*sin(3*x))/13`

0754. depth-int-007
`2*sqrt(x)*atan(sqrt(x))-log(1+x)`

0755. depth-int-008
`2*sqrt(log(x))`

0756. depth-int-009
`x*(sin(log(x))-cos(log(x)))/2`

0757. depth-int-010
`2*log(1+sqrt(x))`

0758. depth-int-011
`atan(x)/2-x/(2*(1+x^2))`

0759. depth-int-012
`2*(1+x^3)^(3/2)/9`

0760. depth-int-013
`log(1+tan(x))`

0761. depth-int-014
`x^2*log(1+x)/2-x^2/4+x/2-log(1+x)/2`

0762. depth-int-015
`log(3)/2-pi/(3*sqrt(3))`

0763. depth-int-016
`pi/2-1`

0764. depth-int-017
`pi^2/12`

0765. depth-int-018
`log(2)/2-1/4`

0766. depth-int-019
`1/2`

0767. depth-int-020
`pi/12`

0768. depth-int-021
`3*pi/8`

0769. depth-int-022
`1/3`

0770. depth-int-023
`3*pi/2`

0771. depth-int-024
`4*pi`

0772. depth-int-025
`1/3`

0773. depth-int-026
`1/2`

0774. depth-int-027
`pi/2`

0775. depth-int-028
`-1/9`

0776. burst-int-001
`8/81`

0777. burst-int-002
`1/2`

0778. burst-int-003
`4/125`

0779. burst-int-004
`-1/36`

0780. burst-int-005
`2/125`

0781. burst-int-006
`-2/27`

0782. burst-int-007
`1/280`

0783. burst-int-008
`pi/8`

0784. burst-int-009
`3*sqrt(pi)/8`

0785. burst-int-010
`1`

0786. burst-int-011
`-pi*log(2)/2`

0787. burst-int-012
`pi*log(2)^2/2+pi^3/24`

0788. burst-int-013
`35*pi^2/256`

0789. burst-int-014
`63*pi/512`

0790. burst-int-015
`256/693`

0791. burst-int-016
`3*pi/512`

0792. burst-int-017
`3*pi/256`

0793. burst-int-018
`1/24`

0794. burst-int-019
`7*pi/512`

0795. burst-int-020
`35*pi/128`

0796. burst-int-021
`16/35`

0797. burst-int-022
`35*pi/256`

0798. burst-int-023
`2/63`

0799. burst-int-024
`5*pi/256`

0800. burst-int-025
`pi^4/15`

0801. burst-int-026
`pi^2/6`

0802. burst-int-027
`pi/(2*sqrt(2))`

0803. burst-int-028
`0`

0804. burst-int-029
`pi/sqrt(2)`

0805. burst-int-030
`-pi*pi/12`

0806. burst-int-031
`3*pi/16`

0807. burst-int-032
`5*pi/2048`

0808. burst-int-033
`231*pi/2048`

0809. burst-int-034
`1024/3003`

0810. burst-int-035
`1/210`

0811. burst-int-036
`exp(2*x)*(x^3/2-3*x^2/4+3*x/4-3/8)`

0812. burst-int-037
`-(x^2)*cos(3*x)/3+2*x*sin(3*x)/9+2*cos(3*x)/27`

0813. burst-int-038
`x*(log(x)^3-3*log(x)^2+6*log(x)-6)`

0814. burst-int-039
`x^5*log(x)/5-x^5/25`

0815. burst-int-040
`(x^4-1)*atan(x)/4-x^3/12+x/4`

0816. burst-int-041
`-exp(-x)*(x^3+3*x^2+6*x+6)`

0817. burst-int-042
`x^3*log(1+x)/3-x^3/9+x^2/6-x/3+log(1+x)/3`

0818. burst-rec-001
`-1/(n+1)^2`

0819. burst-rec-002
`2/(n+1)^3`

0820. burst-rec-003
`2/((n+1)*(n+2)*(n+3))`

0821. burst-rec-004
`(n-1)/n`

0822. burst-rec-005
`(2*n-1)/(2*n)`

0823. burst-rec-006
`(2*n)/(2*n+1)`

0824. burst-rec-007
`n`

0825. burst-rec-008
`-n`

0826. burst-rec-009
`1/(n+1)+1/(n+2)`

0827. burst-rec-010
`1/((n+1)*(n+2))`

0828. burst-boss-int-001
`pi^3/8`

0829. burst-boss-int-002
`-(pi^2)/(8*sqrt(2))`

0830. burst-boss-int-003
`pi^2/(2*sqrt(2))`

0831. burst-boss-int-004
`pi/3`

0832. burst-boss-int-005
`pi/6`

0833. burst-boss-int-006
`pi/3`

0834. burst-boss-int-007
`pi/4`

0835. burst-boss-int-008
`pi/(2*sqrt(2))`

0836. burst-boss-int-009
`2-pi^2/6`

0837. burst-boss-int-010
`-(pi^2)/6`

0838. burst-boss-int-011
`24`

0839. burst-boss-int-012
`3/128`

0840. burst-boss-int-013
`-40/243`

0841. burst-boss-int-014
`pi*log(2)^2/2-pi^3/48`

0842. burst-boss-int-015
`pi^3/8`

0843. burst-boss-int-016
`63*pi^2/512`

0844. burst-boss-int-017
`35*pi/65536`

0845. burst-boss-int-018
`429*pi/4096`

0846. burst-boss-int-019
`2048/6435`

0847. burst-boss-int-020
`5280/15625`

0848. burst-boss-anti-001
`exp(x)*(x^4-4*x^3+12*x^2-24*x+24)`

0849. burst-boss-anti-002
`-exp(-2*x)*(x^4/2+x^3+3*x^2/2+3*x/2+3/4)`

0850. burst-boss-anti-003
`-(x^3)*cos(2*x)/2+3*x^2*sin(2*x)/4+3*x*cos(2*x)/4-3*sin(2*x)/8`

0851. burst-boss-anti-004
`x^3*sin(2*x ) /2 + 3*x^2*cos(2*x )/4-3*x*sin(2*x)/4-3*cos(2*x)/8`

0852. burst-boss-anti-005
`x^5*(log(x)^2/5-2*log(x)/25+2/125)`

0853. burst-boss-anti-006
`x*(log(x ) ^4-4*log(x ) ^3 + 12*log(x ) ^2-24*log(x)+24)`

0854. burst-boss-anti-007
`x^3*atan(x)/3-x^2/6+log(1+x^2)/6`

0855. burst-boss-anti-008
`x^5*atan(x)/5-x^4/20+x^2/10-log(1+x^2)/10`

0856. burst-boss-anti-009
`x^3*log(1 + x^2 ) /3-2*x^3 / 9 + 2*x / 3-2*atan(x)/3`

0857. burst-boss-anti-010
`x^4*log(1+x^2)/4-x^4/8+x^2/4-log(1+x^2)/4`

0858. burst-boss-param-001
`1/(a^2+1)`

0859. burst-boss-param-002
`1/a^2`

0860. burst-boss-param-003
`pi/(2*sqrt(a))`

0861. burst-boss-param-004
`pi*log((1+sqrt(1+a))/2)`

0862. burst-boss-param-005
`pi/(2*sin(pi*(n+1)/2))`

0863. burst-boss-param-006
`-6/(n+1)^4`

0864. burst-boss2-int-001
`-(2*pi^2)/27`

0865. burst-boss2-int-002
`10*pi^3/(81*sqrt(3))`

0866. burst-boss2-int-003
`pi/3`

0867. burst-boss2-int-004
`pi/sqrt(3)`

0868. burst-boss2-int-005
`pi*log(2)/2`

0869. burst-boss2-int-006
`pi/2`

0870. burst-boss2-int-007
`pi/2`

0871. burst-boss2-int-008
`-pi^4/15`

0872. burst-boss2-int-009
`-8*pi^6/63`

0873. burst-boss2-int-010
`-7*pi^4/120`

0874. burst-boss2-int-011
`-31*pi^6/252`

0875. burst-boss2-int-012
`pi^2/16+1/4`

0876. burst-boss2-int-013
`pi/8`

0877. burst-boss2-int-014
`pi^2-4`

0878. burst-boss2-param-001
`b/(a^2+b^2)`

0879. burst-boss2-param-002
`a/(a^2+b^2)`

0880. burst-boss2-param-003
`2*a/(a^2+1)^2`

0881. burst-boss2-param-004
`(a^2-1)/(a^2+1)^2`

0882. burst-boss2-param-005
`log(a+1)`

0883. burst-boss2-param-006
`-120/(a+1)^6`

0884. burst-boss2-anti-001
`exp(2*x ) *(x^5 / 2-5*x^4 / 4 + 5*x^3 / 2-15*x^2/4+15*x/4-15/8)`

0885. burst-boss2-anti-002
`-(x^4 ) *cos(x ) +4*x^3*sin(x ) +12*x^2*cos(x)-24*x*sin(x)-24*cos(x)`

0886. burst-boss2-anti-003
`x^4*sin(x)+4*x^3*cos(x)-12*x^2*sin(x) -24*x*cos(x)+24*sin(x)`

0887. burst-boss2-anti-004
`x*(log(x ) ^5-5*log(x ) ^4 + 20*log(x ) ^3-60*log(x)^2+120*log(x)-120)`

0888. burst-boss2-anti-005
`x^6*atan(x ) /6-x^5 / 30 + x^3 / 18-x / 6 + atan(x)/6`

0889. burst-boss2-anti-006
`x^4/4-x^2/2+log(1+x^2)/2`

0890. burst-boss3-int-001
`1/27`

0891. burst-boss3-int-002
`-1/54`

0892. burst-boss3-int-003
`8-pi^2/3`

0893. burst-boss3-int-004
`pi^2/3`

0894. burst-boss3-int-005
`-pi^2*log(2)/2`

0895. burst-boss3-int-006
`pi^3-6*pi`

0896. burst-boss3-int-007
`pi^4-12*pi^2+48`

0897. burst-boss3-int-008
`-714/28561`

0898. burst-boss3-int-009
`-720/28561`

0899. burst-boss3-int-010
`984/3125`

0900. burst-boss3-int-011
`-912/3125`

0901. burst-boss3-int-012
`31*pi^6/252`

0902. burst-boss3-int-013
`7*pi^4/120`

0903. burst-boss3-param-001
`18*(a^2-3)/(a^2+9)^3`

0904. burst-boss3-param-002
`2*(a^3-3*a)/(a^2+1)^3`

0905. burst-boss3-param-003
`24*a*(a^2-1)/(a^2+1)^4`

0906. burst-boss3-param-004
`24/(a+1)^5`

0907. burst-boss3-param-005
`720/(a+1)^7`

0908. burst-boss3-param-006
`2/(a+1)^3`

0909. burst-boss3-anti-001
`-exp(-3*x)*(x^6/3+2*x^5/3+10*x^4/9+40*x^3/27+40*x^2/27+80*x/81+80/243)`

0910. burst-boss3-anti-002
`-x^5*cos(2*x)/2+5*x^4*sin(2*x)/4+5*x^3*cos(2*x)/2-15*x^2*sin(2*x)/4-15*x*cos(2*x)/4+15*sin(2*x)/8`

0911. burst-boss3-anti-003
`x^5*sin(2*x)/2 + 5*x^4*cos(2*x)/4-5*x^3*sin(2*x)/2-15*x^2*cos(2*x)/4+15*x*sin(2*x)/4+15*cos(2*x)/8`

0912. burst-boss3-anti-004
`x^7*atan(x)/7-x^6/42+x^4/28-x^2/14+log(1+x^2)/14`

0913. burst-boss3-anti-005
`x^7*log(1+x)/7-x^7/49+x^6/42-x^5/35+x^4/28-x^3/21+x^2/14-x/7+log(1+x)/7`

0914. burst-boss3-anti-006
`x^6/6-x^4/4+x^2/2-log(1+x^2)/2`

0915. ser-001 收斂

0916. ser-002 發散

0917. ser-003 `3/2`

0918. ser-004 條件收斂

0919. ser-005 發散

0920. ser-006 收斂

0921. ser-007 `exp(3)-1`

0922. ser-008 `0`

0923. ser-009 `1`

0924. ser-010 條件收斂

0925. ser-011 `9/4`

0926. ser-012 `1`

0927. ser-013 發散

0928. ser-014 `2`

0929. ser-015 `6`

0930. ser-016 收斂

0931. ser-017 `3`

0932. ser-018 絕對收斂

0933. ser-019 `3/4`

0934. ser-020 發散

0935. td-ser-001
`33/40`

0936. td-ser-002
`13/12`

0937. td-ser-003
`2*log(2)`

0938. td-ser-004
`3/2`

0939. ser-021 `1`

0940. ser-022 `2`

0941. ser-023 條件收斂

0942. ser-024 發散

0943. ser-025 收斂

0944. ser-026 `3`

0945. ser-027 `2/3`

0946. ser-028 收斂

0947. ser-029 `exp(3)-1`

0948. ser-030 `5`

0949. ser-031 發散

0950. ser-032 發散

0951. ser-033 條件收斂

0952. ser-034 `0`

0953. ser-035 `Infinity`

0954. ser-036 收斂

0955. ser-037 收斂

0956. ser-038 `1/3`

0957. ser-039 `1/2`

0958. ser-040 `6`

0959. gap-ser-root-001
`1/3`

0960. gap-ser-root-002
`1`

0961. gap-ser-lc-001
`divergent`

0962. gap-ser-lc-002
`convergent`

0963. gap-ser-cond-001
`conditional`

0964. gap-ser-end-001
`conditional`

0965. gap-ser-end-002
`divergent`

0966. gap-ser-taylor-001
`1/120`

0967. gap-ser-taylor-002
`1-x^2/2+x^4/24`

0968. gap-tech-005
`roottest`

0969. mob-tech-007
`ratiotest`

0970. mob-tech-008
`roottest`

0971. mob-tech-019
`endpointanalysis`

0972. mob-trap-004
`conditional`

0973. mob-trap-005
`checkendpoints`

0974. mob-conv-001
`divergent`

0975. mob-conv-002
`convergent`

0976. mob-conv-003
`conditional`

0977. mob-conv-004
`absolute`

0978. mob-conv-005
`convergent`

0979. mob-conv-006
`convergent`

0980. mob-conv-007
`divergent`

0981. mob-conv-008
`divergent`

0982. mob-conv-009
`convergent`

0983. mob-conv-010
`divergent`

0984. rel-basic-019
`2`

0985. rel-basic-022
`1/3`

0986. rel-basic-024
`1/2`

0987. rel-basic-020
`convergent`

0988. rel-basic-021
`divergent`

0989. rel-basic-023
`conditional`

0990. rel-adv-017
`1/6`

0991. rel-adv-019
`1`

0992. rel-adv-015
`convergent`

0993. rel-adv-016
`conditional`

0994. rel-adv-018
`divergent`

0995. rel-adv-020
`conditional`

0996. rel-boss-013
`1/4`

0997. rel-boss-018
`0`

0998. rel-hard-ser-001
`1/4`

0999. rel-hard-ser-002
`4`

1000. rel-hard-ser-003
`1/24`

1001. rel-hard-ser-004
`1/120`

1002. rel-hard-ser-005
`1/18`

1003. **rel-hard-ser-006**
6

1004. **rel-hard-ser-007**
`pi^2/12`

1005. **rel-hard-ser-008**
`1/2`

1006. **rel-hard-ser-012**
`1/e`

1007. **rel-hard-ser-009**
`convergent`

1008. **rel-hard-ser-010**
`divergent`

1009. **rel-hard-ser-011**
`conditional`

1010. **hc-spec-007**
`1/96`

1011. **hc-spec-008**
`26`

1012. **hc-spec-009**
`1/27`

1013. **hc-spec-010**
`1/147456`

1014. **exam-ser-001**
`3`

1015. **exam-ser-002**
`e`

1016. **exam-ser-003**
`1`

1017. **exam-ser-004**
`1`

1018. **exam-ser-005**
`2`

1019. **exam-ser-006**
`6`

1020. **exam-ser-007**
`cos(1)`

1021. **exam-ser-008**
`(exp(1)-exp(-1))/2`

1022. **exam-ser-009**
`4/15`

1023. **exam-ser-010**
`-1/6`

1024. **exam-ser-011**
`-1/4`

1025. **exam-ser-012**
`3/2`

1026. **exam-ser-013**
`8`

1027. **exam-ser-014**
`1/4`

1028. **exam-ser-015**
`log(2)`

1029. **exam-ser-016**
`pi^2/6`

1030. **exam-ser-017**
`exp(2)`

1031. **exam-ser-018**
`35`

1032. **exam-ser-019**
`16`

1033. **exam-ser-020**
`3/4`

1034. **uni-ser-001**
`4/9`

1035. **uni-ser-002**
`3/2`

1036. **uni-ser-003**
`3/4`

1037. **uni-ser-004**
`0`

1038. **uni-ser-005**
`0`

1039. **uni-ser-006**
`1/3`

1040. **uni-ser-007**
`5`

1041. **uni-ser-008**
`4`

1042. **uni-ser-009**
`1/e`

1043. **uni-ser-010**
`pi/4`

1044. **uni-ser-011**
`(exp(1)+exp(-1))/2`

1045. **uni-ser-012**
`4`

1046. **uni-ser-013**
`15`

1047. **uni-ser-014**
`1/3`

1048. **uni-ser-015**
`2/9`

1049. **uni-ser-016**
`-4/315`

1050. **uni-ser-017**
`26`

1051. **uni-ser-018**
`210`

1052. uni-ser-019
`1`

1053. uni-ser-020
`pi^2/6`

1054. depth-ser-001
`9/4`

1055. depth-ser-002
`20/27`

1056. depth-ser-003
`1/2`

1057. depth-ser-004
`243/560`

1058. depth-ser-005
`1/24`

1059. depth-ser-006
`-1/5040`

1060. depth-ser-007
`(exp(1)+exp(-1))/2`

1061. depth-ser-008
`1/4`

1062. depth-ser-009
`1/4`

1063. depth-ser-010
`4/3`

1064. depth-ser-011
`192`

1065. depth-ser-012
`3`

1066. depth-ser-013
`cos(1)`

1067. depth-ser-014
`1/4`

1068. depth-ser-015
`1/3`

1069. depth-ser-016
`33/8`

1070. depth-ser-017
`5`

1071. depth-ser-018
`sin(1)`

1072. depth-ser-019
`11/12`

1073. depth-ser-020
`1/2`

1074. burst-boss-ser-001
`26`

1075. burst-boss-ser-002
`3/2`

1076. burst-boss-ser-003
`2*log(2)`

1077. burst-boss-ser-004
`pi^2/12`

1078. burst-boss-ser-005
`137/180`

1079. burst-boss-ser-006
`1553/80`

1080. burst-boss2-ser-001
`pi^4/90`

1081. burst-boss2-ser-002
`pi^4/96`

1082. burst-boss2-ser-003
`7*pi^4/720`

1083. burst-boss2-ser-004
`31*pi^6/30240`

1084. burst-boss2-ser-005
`15`

1085. burst-boss2-ser-006
`273/4`

1086. burst-boss2-ser-007
`3*log(3/2)/2`

1087. burst-boss2-ser-008
`4*log(4/3)/3`

1088. burst-boss3-ser-001
`9366`

1089. burst-boss3-ser-002
`94586`

1090. burst-boss3-ser-003
`4108/243`

1091. burst-boss3-ser-004
`1491/4`

1092. burst-boss3-ser-005
`pi^2/12`

1093. burst-boss3-ser-006
`275295799/775975200`

1094. burst-boss3-ser-007
`111165691613818/97692469875`

1095. burst-boss3-ser-008
`1094053506107/212837625`

答案與解法提示

0001 **lim-001** 極限・R1
Answer. 1
Note. 標準極限：sin x 和 x 在 0 附近同階，所以答案是 1。

0002 **lim-002** 極限・R1
Answer. 1/2
Note. 用 1-cos x 約等於 x^2/2。

0003 **lim-003** 極限・R1
Answer. 4
Note. 因式分解為 (x-2)(x+2)，約掉後代入 x=2。

0004 **lim-004** 極限・R1
Answer. 3/2
Note. 最高次項係數比是 3/2。

0005 **lim-005** 極限・R2
Answer. 1
Note. e^x 在 0 的線性項是 x。

0006 **lim-006** 極限・R2
Answer. 3
Note. ln(1+u) 約等於 u，令 u=3x。

0007 **lim-007** 極限・R2
Answer. 2
Note. tan(4x) 約等於 4x，sin(2x) 約等於 2x。

0008 **lim-008** 極限・R2
Answer. 1/2
Note. 有理化後是 x/(sqrt(x^2+1)+x)，趨近 1/2。

0009 **lim-009** 極限・R3
Answer. −1/6
Note. sin x = x - x^3/6 + ...。

0010 **lim-010** 極限・R3
Answer. 2
Note. e^{2x}=1+2x+2x^2+...。

0011 **lim-011** 極限・R3
Answer. 0
Note. 令 x=1/t，變成 -(ln t)/t，趨近 0。

0012 **lim-012** 極限・R3
Answer. 1/3
Note. ln(1+x)=x-x^2/2+x^3/3-...。

0013 lim-013 極限・R3
Answer. 1/2
Note. tan x = x+x^3/3+...，sin x = x-x^3/6+...。

0014 lim-014 極限・R4
Answer. -1/2
Note. 二階項來自 ln(1+x) 的 -x^2/2。

0015 lim-015 極限・R3
Answer. 1/2
Note. sec x = 1+x^2/2+...。

0016 lim-016 極限・R4
Answer. -1/8
Note. 展開 sqrt(1+1/x)，下一項是 -1/(8x^2)。

0017 lim-017 極限・R3
Answer. 1/3
Note. e^x-e^{-x}=2x+x^3/3+...。

0018 lim-018 極限・R3
Answer. 1/3
Note. sin x 與 x cos x 的三階項相減。

0019 lim-019 極限・R4
Answer. -e/2
Note. (1+x)^{1/x}=e(1-x/2+...)。

0020 lim-020 極限・R3
Answer. -1/3
Note. arctan x = x-x^3/3+...。

0021 td-lim-001 極限・R6
Answer. exp(1/2)
Note. 取 log 後，sum k/n^2 趨近 1/2，其餘高階項趨近 0。

0022 td-lim-002 極限・R6
Answer. sqrt(e)/3
Note. 把 log 展開到 1/n 項：log P_n=1/2+1/(3n)+O(1/n^2)。

0023 td-lim-003 極限・R6
Answer. 11*e/24
Note. 展開 n log(1+1/n)=1-1/(2n)+1/(3n^2)+...。

0024 td-lim-004 極限・R4
Answer. -1/2
Note. (1+x)^x=exp(x log(1+x))=1+x^2-x^3/2+...。

0025 lim-021 極限・R2
Answer. 1/3

Note. tan x = x + x^3/3 + O(x^5)，所以分子約為 x^3/3，極限為 1/3。

0026 lim-022 極限・R2
Answer. 3/2
Note. 有理化後為 3x/(sqrt(x^2+3x)+x)，上下同除以 x 得 3/(sqrt(1+3/x)+1)，極限為 3/2。

0027 lim-023 極限・R2
Answer. 2
Note. e^{2x}=1+2x+2x^2+O(x^3)，分子為 2x^2+O(x^3)，所以極限為 2。

0028 lim-024 極限・R2
Answer. -1/2
Note. log(1+x)=x-x^2/2+O(x^3)，分子為 -x^2/2+O(x^3)，極限為 -1/2。

0029 lim-025 極限・R4
Answer. 1
Note. 令 u=xy，則 u->0，標準極限 sin u/u = 1。

0030 lim-026 極限・R4
Answer. 0
Note. 令 x=r cos theta, y=r sin theta，原式 = r cos^2 theta sin theta，當 r->0 時趨近 0。

0031 lim-027 極限・R4
Answer. dne
Note. 沿 y=x 得 1/2；沿 y=-x 得 -1/2。路徑極限不同，所以 DNE。

0032 lim-028 極限・R4
Answer. 1/2
Note. 設 r^2=x^2+y^2。有理化後為 1/(sqrt(1+r^2)+1)，令 r->0 得 1/2。

0033 lim-029 極限・R4
Answer. 0
Note. 因 x^4+y^4 <= (x^2+y^2)^2，所以原式 <= x^2+y^2，極限為 0。

0034 lim-030 極限・R2
Answer. 9/2
Note. 1-cos(3x) ~ (3x)^2/2，所以極限為 9/2。

0035 lim-031 極限・R4
Answer. 0
Note. With x=r cos t, y=r sin t, the expression is r^2 cos^2 t sin^2 t, so the limit is 0.

0036 lim-032 極限・R4
Answer. dne
Note. Along y=0 the value is 1; along x=0 the value is -1. The limit does not exist.

0037 lim-033 極限・R4
Answer. 1
Note. Set u=x+y. The expression becomes (e^u-1)/u, so the limit is 1.

0038 lim-034 極限・R4

Answer. `0`

Note. Let r=sqrt(x^2+y^2). Since |x|<=r, the absolute value is <= r^3/r^2=r, hence the limit is 0.

0039 lim-035 極限・R2

Answer. `5/2`

Note. As x->0, sin(5x)~5x and sin(2x)~2x, so the limit is 5/2.

0040 lim-036 極限・R3

Answer. `1/2`

Note. log(1+x+x^2)=x+x^2-(x^2/2)+O(x^3)=x+x^2/2+O(x^3). The limit is 1/2.

0041 lim-037 極限・R2

Answer. `2`

Note. After rationalizing, the expression is 4/(sqrt(1+4x)+1), whose limit is 2.

0042 lim-038 極限・R3

Answer. `1/3`

Note. sin x-x cos x=(x-x^3/6)-(x-x^3/2)+O(x^5)=x^3/3+O(x^5).

0043 lim-039 極限・R4

Answer. `dne`

Note. Along y=kx^2 the expression is k/(1+k^2). Since this depends on k, the limit does not exist.

0044 lim-040 極限・R2

Answer. `2`

Note. Since log(1+2/x)~2/x, the limit is 2.

0045 gap-tech-001 極限・R2

Answer. `rationalize`

Note. The right reflex is to rationalize with the conjugate.

0046 gap-tech-002 極限・R2

Answer. `taylor`

Note. Use Taylor expansion of e^{2x} around 0.

0047 gap-tech-008 極限・R2

Answer. `pathtest`

Note. Use a path test such as y=mx^2.

0048 mob-tech-001 極限・R2

Answer. `rationalize`

Note. Use rationalize.

0049 mob-tech-002 極限・R2

Answer. `Taylor`

Note. Use Taylor.

0050 **mob-tech-009** 極限・R2
Answer. `pathtest`
Note. Use path test.

0051 **mob-trap-009** 極限・R2
Answer. `indeterminate`
Note. indeterminate

0052 **mob-limtrap-001** 極限・R3
Answer. `-1/6`
Note. The limit is -1/6.

0053 **mob-limtrap-002** 極限・R3
Answer. `1/3`
Note. The limit is 1/3.

0054 **mob-limtrap-003** 極限・R3
Answer. `-1/24`
Note. The limit is -1/24.

0055 **mob-limtrap-004** 極限・R3
Answer. `1/3`
Note. The limit is 1/3.

0056 **mob-limtrap-005** 極限・R3
Answer. `1/6`
Note. The limit is 1/6.

0057 **mob-limtrap-006** 極限・R3
Answer. `-1/8`
Note. The limit is -1/8.

0058 **mob-limtrap-007** 極限・R3
Answer. `0`
Note. The limit is 0.

0059 **mob-limtrap-008** 極限・R3
Answer. `0`
Note. The limit is 0.

0060 **mob-limtrap-009** 極限・R3
Answer. `DNE`
Note. Different paths give different values.

0061 **mob-limtrap-010** 極限・R3
Answer. `DNE`
Note. The axes give 1 and -1.

0062 **rel-basic-001** 極限・R1
Answer. `5`

Note. sin(5x)/(5x) tends to 1.

0063 **rel-basic-002** 極限・R1
Answer. 9/2
Note. Use 1-cos u ~ u^2/2 with u=3x.

0064 **rel-basic-003** 極限・R1
Answer. 4
Note. Use e^u-1 ~ u.

0065 **rel-basic-004** 極限・R1
Answer. 2
Note. Use log(1+u) ~ u.

0066 **rel-basic-005** 極限・R1
Answer. 1/2
Note. Rationalize the numerator.

0067 **rel-basic-006** 極限・R1
Answer. 4
Note. Factor x^2-4=(x-2)(x+2).

0068 **rel-adv-001** 極限・R3
Answer. 1/120
Note. sin x=x-x^3/6+x^5/120+...

0069 **rel-adv-002** 極限・R3
Answer. 1/24
Note. The next Taylor coefficient is 1/24.

0070 **rel-adv-003** 極限・R3
Answer. -1/4
Note. The x^4 coefficient in log(1+x) is -1/4.

0071 **rel-adv-004** 極限・R3
Answer. 2/15
Note. tan x=x+x^3/3+2x^5/15+...

0072 **rel-boss-019** 極限・R5
Answer. 1
Note. Rewrite as n log(1+1/n).

0073 **rel-hard-lim-001** 極限・R5
Answer. -1/5040
Note. Use the x^7 term of sin x.

0074 **rel-hard-lim-002** 極限・R5
Answer. 17/315
Note. Use the x^7 term of tan x.

0075 rel-hard-lim-003 极限・R5

Answer. `1/5`
Note. Use the x^5 term of log(1+x).

0076 rel-hard-lim-004 極限・R5

Answer. `1/360`
Note. Use the even expansion of e^x+e^{-x}.

0077 rel-hard-lim-005 極限・R5

Answer. `-e/2`
Note. Expand exp(log(1+x)/x).

0078 rel-hard-lim-007 極限・R5

Answer. `1/2`
Note. Use 1-cos u ~ u^2/2.

0079 rel-hard-lim-008 極限・R5

Answer. `1`
Note. Use e^u-1 ~ u.

0080 rel-hard-lim-009 極限・R5

Answer. `1/5`
Note. Use arctan x=x-x^3/3+x^5/5-...

0081 rel-hard-lim-010 極限・R5

Answer. `11*e/24`
Note. The second correction term is 11e/(24n^2).

0082 rel-hard-lim-006 極限・R5

Answer. `DNE`
Note. Along y=0 the limit is 0; along y=x it is 1/4.

0083 exam-lim-001 極限・R5

Answer. `81/40`
Note. Expand sin(3x) through the x^5 term.

0084 exam-lim-002 極限・R5

Answer. `8/3`
Note. Use log(1+u)=u-u^2/2+u^3/3+... with u=2x.

0085 exam-lim-003 極限・R5

Answer. `1`
Note. The constant and linear terms cancel; the x^2 coefficient is 1.

0086 exam-lim-004 極限・R5

Answer. `1/3`
Note. tan x=x+x^3/3+...

0087 **exam-lim-005** 極限・R5

Answer. 2

Note. 1-cos u is asymptotic to u^2/2.

0088 **exam-lim-006** 極限・R5

Answer. -2

Note. sqrt(1+4x)=1+2x-2x^2+...

0089 **exam-lim-007** 極限・R5

Answer. -9/8

Note. Apply sqrt(1+u)=1+u/2-u^2/8+... with u=3/x.

0090 **exam-lim-008** 極限・R5

Answer. 1/3

Note. Compare leading terms.

0091 **exam-lim-009** 極限・R5

Answer. 1/3

Note. Use x-sin x ~ x^3/6 and 1-cos x ~ x^2/2.

0092 **exam-lim-010** 極限・R5

Answer. -8/3

Note. arctan u=u-u^3/3+...

0093 **exam-lim-011** 極限・R5

Answer. 1

Note. Write x^x=exp(x log x), and x log x -> 0.

0094 **exam-lim-012** 極限・R5

Answer. exp(6)

Note. This is exp(3x log(1+2/x)) -> exp(6).

0095 **exam-lim-013** 極限・R5

Answer. exp(-1/6)

Note. log(sin x/x)=-x^2/6+...

0096 **exam-lim-014** 極限・R5

Answer. -1/2

Note. log(cos x)=-x^2/2+...

0097 **exam-lim-015** 極限・R5

Answer. 1

Note. e^{sin x}-1 is asymptotic to sin x.

0098 **exam-lim-016** 極限・R5

Answer. 1/2

Note. e^{sin x}=1+x+x^2/2+...

0099 **exam-lim-017** 極限・R5

Answer. -1/12

Note. Compare x^4 coefficients: 1/24 and 1/8.

0100 **exam-lim-018** 極限・R5
Answer. `1/2`
Note. tan x-sin x has x^3 coefficient 1/3+1/6.

0101 **exam-lim-019** 極限・R5
Answer. `1`
Note. Rewrite as x log(1+1/x).

0102 **exam-lim-020** 極限・R5
Answer. `1/6`
Note. arcsin x=x+x^3/6+...

0103 **uni-lim-001** 極限・R6
Answer. `1/120`
Note. Keep the x^5 term of sin x.

0104 **uni-lim-002** 極限・R6
Answer. `-1/4`
Note. Use the fourth term of log(1+x).

0105 **uni-lim-003** 極限・R6
Answer. `3/2`
Note. Compare the x^2 coefficients.

0106 **uni-lim-004** 極限・R6
Answer. `1/2`
Note. Use tan x and sin x expansions.

0107 **uni-lim-005** 極限・R6
Answer. `1/8`
Note. Odd terms remain after subtracting the two radicals.

0108 **uni-lim-006** 極限・R6
Answer. `1/2`
Note. Put y=1/x and expand (1/y+1)log(1+y).

0109 **uni-lim-007** 極限・R6
Answer. `-1`
Note. The cubic term is the first nonzero term.

0110 **uni-lim-008** 極限・R6
Answer. `-9*exp(3)/2`
Note. Expand x^{-1}log(1+3x) through the linear term.

0111 **uni-lim-009** 極限・R6
Answer. `1`
Note. cosh x and cos x have opposite x^2 coefficients.

0112 **uni-lim-010** 極限・R6
Answer. `-1/2`
Note. Compare arctan x and arcsin x cubic terms.

0113 **uni-lim-011** 極限・R6
Answer. `-1/2`
Note. Only the quadratic term is needed.

0114 **uni-lim-012** 極限・R6
Answer. `1`
Note. Both numerator and denominator start at x^2/2.

0115 **uni-lim-013** 極限・R6
Answer. `exp(1/6)`
Note. Take log first.

0116 **uni-lim-014** 極限・R6
Answer. `1/3`
Note. Compare fourth-order coefficients.

0117 **uni-lim-015** 極限・R6
Answer. `1/2`
Note. Let u=x+x^2 and expand log(1+u).

0118 **depth-lim-001** 極限・R6
Answer. `-1/5040`
Note. Keep the x^7 term of sin x.

0119 **depth-lim-002** 極限・R6
Answer. `1/5`
Note. Use the fifth term of log(1+x).

0120 **depth-lim-003** 極限・R5
Answer. `1/2`
Note. The quadratic term comes from (sin x)^2/2.

0121 **depth-lim-004** 極限・R6
Answer. `1/6`
Note. Compare fourth-order coefficients.

0122 **depth-lim-005** 極限・R5
Answer. `2/3`
Note. Use tan x-x ~ x^3/3 and 1-cos x ~ x^2/2.

0123 **depth-lim-006** 極限・R6
Answer. `1/2`
Note. Put u=1/x and compare the u^2 term.

0124 **depth-lim-007** 極限・R6
Answer. `-exp(1)/2`
Note. Take log first, then expand.

0125 **depth-lim-008** 極限・R5
Answer. `1/2`
Note. Compare the cubic coefficients.

0126 **depth-lim-009** 極限・R5
Answer. `1/3`
Note. The cubic terms have opposite signs.

0127 **depth-lim-010** 極限・R5
Answer. `1/2`
Note. Use 1-cos x ~ x^2/2 and sin x ~ x.

0128 **depth-lim-011** 極限・R5
Answer. `-1/2`
Note. Use cos x=1-x^2/2+...

0129 **depth-lim-012** 極限・R6
Answer. `exp(1)/2`
Note. Expand x^{-1}log(1+x+x^2).

0130 **burst-boss-lim-001** 極限・R5
Answer. `2/15`
Note. Use tan x=x+x^3/3+2x^5/15+...

0131 **burst-boss-lim-002** 極限・R5
Answer. `1/120`
Note. Use the fifth-order term of sin x.

0132 **burst-boss-lim-003** 極限・R5
Answer. `-1/4`
Note. Use the fourth-order term of log(1+x).

0133 **burst-boss-lim-004** 極限・R6
Answer. `-1/8`
Note. Expand sin x first, then exponentiate to fourth order.

0134 **burst-boss-lim-005** 極限・R6
Answer. `1/6`
Note. Compare both cosine expansions through x^4.

0135 **burst-boss-lim-006** 極限・R6
Answer. `-1/7`
Note. Use the alternating arctan series.

0136 **burst-boss-lim-007** 極限・R6
Answer. `-e/2`

Note. Write it as exp(log(1+x)/x) and expand.

0137 **burst-boss-lim-008** 極限・R6

Answer. `-e/2`

Note. Use x log(1+1/x)=1-1/(2x)+...

0138 **burst-boss2-lim-001** 極限・R6

Answer. `exp(-1/6)`

Note. Take logs and use log(sin x/x)=-x^2/6+...

0139 **burst-boss2-lim-002** 極限・R6

Answer. `-1/45`

Note. Use x/tan x=1-x^2/3-x^4/45+...

0140 **burst-boss2-lim-003** 極限・R6

Answer. `7/360`

Note. Use the Laurent expansion of csc x.

0141 **burst-boss2-lim-004** 極限・R6

Answer. `-1/12`

Note. Expand log(cos x) through the fourth order.

0142 **burst-boss2-lim-005** 極限・R6

Answer. `1/10`

Note. Integrate the series for e^{-t^2} term by term.

0143 **burst-boss2-lim-006** 極限・R6

Answer. `1/21`

Note. Integrate the series for log(1+t^2) term by term.

0144 **burst-boss2-lim-007** 極限・R6

Answer. `-1/4`

Note. Use the Euler-Maclaurin correction for H_{2n}-H_n.

0145 **burst-boss2-lim-008** 極限・R6

Answer. `-1/4`

Note. Read the first endpoint correction in the Riemann sum.

0146 **burst-boss2-lim-009** 極限・R6

Answer. `(1/sqrt(2)-1)/2`

Note. Apply the first Euler-Maclaurin correction.

0147 **burst-boss2-lim-010** 極限・R6

Answer. `(2-e)/2`

Note. Expand n log(1+x/n) to the first correction term.

0148 **burst-boss3-lim-001** 極限・R6

Answer. `1/exp(1)`

Note. Maximize at x=n/(n+1).

0149 burst-boss3-lim-002 極限・R6
Answer. `4/exp(2)`
Note. Maximize at x=n/(n+2).

0150 burst-boss3-lim-003 極限・R6
Answer. `1/2`
Note. Only a boundary layer near x=1 contributes.

0151 burst-boss3-lim-004 極限・R6
Answer. `(e-1)/12`
Note. Use the second Euler-Maclaurin endpoint correction.

0152 burst-boss3-lim-005 極限・R6
Answer. `-1/24`
Note. Use the second Euler-Maclaurin endpoint correction.

0153 burst-boss3-lim-006 極限・R6
Answer. `-1/24`
Note. Use f'(1)-f'(0) for the Riemann-sum correction.

0154 burst-boss3-lim-007 極限・R6
Answer. `11/72`
Note. Expand the integrand before integrating.

0155 burst-boss3-lim-008 極限・R6
Answer. `-1/42`
Note. Integrate the sine series term by term.

0156 burst-boss3-lim-009 極限・R6
Answer. `-1/180`
Note. Use the logarithmic sine expansion.

0157 burst-boss3-lim-010 極限・R6
Answer. `-1/12`
Note. Expand log(cosh x).

0158 burst-boss3-lim-011 極限・R6
Answer. `1/16`
Note. Use the binomial series.

0159 burst-boss3-lim-012 極限・R6
Answer. `exp(1/6)`
Note. Take logs and use x/sin x=1+x^2/6+...

0160 der-001 微分・R1
Answer. `5*x^4-6*x`
Note. 逐項微分。

0161 der-002 微分・R1
Answer. `cos(x)-sin(x)`
Note. sin 的導數是 cos，cos 的導數是 -sin。

0162 der-003 微分・R1
Answer. `exp(x)*log(x)+exp(x)/x`
Note. 乘法法則。

0163 der-004 微分・R1
Answer. `1/(2*sqrt(x))`
Note. sqrt(x)=x^{1/2}。

0164 der-005 微分・R2
Answer. `2*x/(1+x^2)`
Note. 鏈鎖律：u=1+x^2。

0165 der-006 微分・R2
Answer. `2*x*sin(x)+x^2*cos(x)`
Note. 乘法法則。

0166 der-007 微分・R2
Answer. `1/(1+x^2)`
Note. 反三角函數基本導數。

0167 der-008 微分・R2
Answer. `1/(x+1)^2`
Note. 商法則或改寫成 1-1/(x+1)。

0168 der-009 微分・R3
Answer. `x^x*(log(x)+1)`
Note. 對數微分：x^x=e^{x ln x}。

0169 der-010 微分・R3
Answer. `exp(-x)*(2*x*cos(x^2)-sin(x^2))`
Note. 乘法法則加鏈鎖律。

0170 der-011 微分・R3
Answer. `cos(x)/sin(x)`
Note. 鏈鎖律，答案也可寫成 cot x。

0171 der-012 微分・R4
Answer. `((x^2+1)/x-2*x*log(x))/(x^2+1)^2`
Note. 商法則。

0172 der-013 微分・R4
Answer. `x^sin(x)*(cos(x)*log(x)+sin(x)/x)`
Note. 對數微分：x^{sin x}=e^{sin x ln x}。

0173 der-014 微分・R3
Answer. `1/(1+x^2)`

Note. 令 u=(1+x)/(1-x)，套 arctan 鏈鎖律後會大幅化簡。

0174 der-015 微分・R3
Answer. `1/sqrt(1+x^2)`
Note. 這是 asinh x 的導數。

0175 der-016 微分・R3
Answer. `1/(1+cos(x))`
Note. 商法則後分子化成 1+cos x。

0176 der-017 微分・R3
Answer. `exp(x^2)*(2*x*cos(3*x)-3*sin(3*x))`
Note. 乘法法則加兩個鏈鎖律。

0177 der-018 微分・R3
Answer. `1/sqrt(x^2-1)`
Note. 這是 arcosh x 的導數形式。

0178 der-019 微分・R2
Answer. `10*x*(x^2+1)^4`
Note. 鏈鎖律。

0179 der-020 微分・R3
Answer. `1/(2*sqrt(x)*(1+x))`
Note. 先對 arctan u 微分，再乘上 sqrt x 的導數。

0180 td-der-001 微分・R4
Answer. `2*(1-3*x^2)/(1+x^2)^3`
Note. 先算 y′=-1/(1+x^2)，再連續微分兩次。

0181 td-der-002 微分・R2
Answer. `(35*x^3-15*x)/2`
Note. 這是 Legendre P_4 的導數，逐項微分即可。

0182 td-der-003 微分・R3
Answer. `log(1+x)+x/(1+x)`
Note. 乘法法則。這也是展開 (1+x)^x 時會出現的核心項。

0183 td-der-004 微分・R3
Answer. `-3*sin(x)/(5+3*cos(x))`
Note. 鏈鎖律，內層導數是 -3 sin x。

0184 der-021 微分・R4
Answer. `2*x*y+y*cos(x*y)`
Note. 偏 x 微分得 2xy + cos(xy)*y。

0185 der-022 微分・R4
Answer. `x^2*exp(x*y)+3*y^2`
Note. d/dy of x e^{xy} = x*e^{xy}*x = x^2 e^{xy}，再加 3y^2。

0186 der-023 微分・R4

Answer. `exp(x*y)*(1+x*y)`

Note. 先得 f_y=x e^{xy}，再對 x 微分為 e^{xy}+xy e^{xy}=e^{xy}(1+xy)。

0187 der-024 微分・R4

Answer. `22/5`

Note. grad f=(2x,2y)，在 (1,2) 為 (2,4)。與 (3/5,4/5) 內積得 22/5。

0188 der-025 微分・R4

Answer. `2*x/(x^2+y^2)`

Note. 令 u=x^2+y^2，u_x=2x，所以 f_x=2x/(x^2+y^2)。

0189 der-026 微分・R2

Answer. `2*x*sin(3*x)+3*x^2*cos(3*x)`

Note. 乘法律給 2x sin(3x)+x^2*3cos(3x)。

0190 der-027 微分・R2

Answer. `x/(1+x^2)`

Note. 原式為 (1/2)log(1+x^2)，微分得 x/(1+x^2)。

0191 der-028 微分・R2

Answer. `exp(x)/(1+exp(2*x))`

Note. 由鏈鎖律得 e^x/(1+e^{2x})。

0192 der-029 微分・R2

Answer. `exp(x)*(x+2)`

Note. f′=e^x(x+1)，f″=e^x(x+1)+e^x=e^x(x+2)。

0193 der-030 微分・R3

Answer. `-(2*x+y)/(x+2*y)`

Note. 微分得 2x+y+xy′+2yy′=0，所以 y′=-(2x+y)/(x+2y)。

0194 der-031 微分・R4

Answer. `2*x*exp(x^2+y^2)`

Note. The derivative is e^{x^2+y^2} times 2x.

0195 der-032 微分・R4

Answer. `x/(x^2+y^2)`

Note. (1/x)/(1+y^2/x^2)=x/(x^2+y^2).

0196 der-033 微分・R4

Answer. `2*y-sin(x)`

Note. f_x=2xy+cos x, so f_xx=2y-sin x.

0197 der-034 微分・R4

Answer. `x-y`

Note. The Jacobian matrix is [[1,1],[y,x]], so det = x-y.

0198 **der-035** 微分・R4
Answer. `exp(t)*(4*t^3+t^4)`
Note. After substitution z=t^4e^t, so dz/dt=e^t(4t^3+t^4).

0199 **der-036** 微分・R4
Answer. `4`
Note. The Laplacian is 2+2=4.

0200 **der-037** 微分・R4
Answer. `27`
Note. The Hessian is [[6,-3],[-3,6]], whose determinant is 36-9=27.

0201 **der-038** 微分・R2
Answer. `(x^2-2*x-1)/(x-1)^2`
Note. Derivative is [2x(x-1)-(x^2+1)]/(x-1)^2=(x^2-2x-1)/(x-1)^2.

0202 **der-039** 微分・R2
Answer. `2*sin(x)*cos(x)`
Note. By chain rule, the derivative is 2sin(x)cos(x).

0203 **der-040** 微分・R3
Answer. `2*x^log(x)*log(x)/x`
Note. x^{log x}=exp((log x)^2), so the derivative is x^{log x}*2log(x)/x.

0204 **der-041** 微分・R4
Answer. `-0.04`
Note. df=4(0.01)+4(-0.02)=-0.04.

0205 **der-042** 微分・R4
Answer. `-0.028`
Note. df=(3/5)(0.06)+(4/5)(-0.08)=0.036-0.064=-0.028.

0206 **der-043** 微分・R4
Answer. `0`
Note. dz=0.02+(-0.04)/2=0.

0207 **der-044** 微分・R4
Answer. `y*exp(x*y)*dx+x*exp(x*y)*dy`
Note. df=y e^{xy}dx+x e^{xy}dy.

0208 **der-045** 微分・R4
Answer. `-y*dx/(x^2+y^2)+x*dy/(x^2+y^2)`
Note. df=(-y/(x^2+y^2))dx+(x/(x^2+y^2))dy.

0209 **der-046** 微分・R4
Answer. `4.92`
Note. The estimate is 5+2(0.02)+4(-0.03)=4.92.

0210 **der-047** 微分・R5
Answer. `-9/2`

Note. The critical point is (3/2,1/2). The value is -9/2.

0211 der-048 微分・R5
Answer. `-9`
Note. The critical point is (1,1), and the minimum value is -9.

0212 der-049 微分・R5
Answer. `-1`
Note. The minimum occurs at x=-1,y=2, so the x-coordinate is -1.

0213 der-050 微分・R5
Answer. `5`
Note. The closest point is (1,2), so the minimum of x^2+y^2 is 5.

0214 der-051 微分・R4
Answer. `3`
Note. The Hessian is [[2,1],[1,2]], so det=3.

0215 der-052 微分・R4
Answer. `144`
Note. The Hessian is diag(12,12), so the determinant is 144.

0216 der-053 微分・R4
Answer. `-1`
Note. At (0,0), the Hessian is [[0,1],[1,0]], whose determinant is -1.

0217 der-054 微分・R4
Answer. `128`
Note. The Hessian is [[12,-4],[-4,12]], so det=144-16=128.

0218 der-055 微分・R4
Answer. `localminimum`
Note. The Hessian is [[2,0],[0,2]], positive definite, so it is a local minimum.

0219 der-056 微分・R4
Answer. `saddlepoint`
Note. The Hessian determinant is -4, so the critical point is a saddle point.

0220 der-057 微分・R4
Answer. `0`
Note. At (0,0), the Hessian is [[0,0],[0,2]], so det=0.

0221 der-058 微分・R4
Answer. `-4`
Note. The Hessian at (1,0) is [[-2,0],[0,2]], so the determinant is -4.

0222 der-059 微分・R4
Answer. `-1`
Note. W=sin x(-sin x)-cos x cos x=-1.

0223 der-060 微分 • R4
Answer. `1`
Note. W=cos x cos x-(-sin x)sin x=1.

0224 der-061 微分 • R4
Answer. `exp(3*x)`
Note. W=e^x(2e^{2x})-e^x e^{2x}=e^{3x}.

0225 der-062 微分 • R4
Answer. `x^2`
Note. W=x(2x)-1*x^2=x^2.

0226 der-063 微分 • R4
Answer. `2`
Note. The determinant of [[1,x,x^2],[0,1,2x],[0,0,2]] is 2.

0227 der-064 微分 • R4
Answer. `exp(2*x)`
Note. W=e^x(e^x+xe^x)-e^x(xe^x)=e^{2x}.

0228 der-065 微分 • R4
Answer. `1/x`
Note. W=1*(1/x)-0*log x=1/x.

0229 der-066 微分 • R4
Answer. `x^4`
Note. W=x^2(3x^2)-(2x)(x^3)=x^4.

0230 der-067 微分 • R5
Answer. `4*x`
Note. After substitution, F=2x^2+2y^2, so dF/dx=4x.

0231 der-068 微分 • R5
Answer. `exp(sin(x))*(2*x+x^2*cos(x))`
Note. d/dx[x^2e^{sin x}]=e^{sin x}(2x+x^2cos x).

0232 der-069 微分 • R6
Answer. `4*(x^2+y^2)`
Note. The determinant is 4x^2+4y^2=4(x^2+y^2).

0233 der-070 微分 • R5
Answer. `-4*y`
Note. The product is (u-v)(-2). Since u-v=2y, the determinant is -4y.

0234 der-071 微分 • R5
Answer. `r`
Note. The Jacobian is r.

0235 der-072 微分 • R5
Answer. `-2*x/y`
Note. The determinant is y(-x/y^2)-x(1/y)=-2x/y.

0236 der-073 微分 • R6
Answer. `20*((2*x+y)^2+(x+3*y)^2)`
Note. By the chain rule, the determinant is 4(u^2+v^2)*5=20((2x+y)^2+(x+3y)^2).

0237 der-074 微分 • R5
Answer. `-4`
Note. The determinant is (-2)*2=-4.

0238 der-075 微分 • R6
Answer. `x^3-3*x*y^2`
Note. (x+iy)^3=x^3+3ix^2y-3xy^2-iy^3, so the real part is x^3-3xy^2.

0239 der-076 微分 • R6
Answer. `3*x^2*y-y^3`
Note. The imaginary part is 3x^2y-y^3.

0240 der-077 微分 • R6
Answer. `sqrt(2)`
Note. |1+i|=sqrt(1^2+1^2)=sqrt 2.

0241 der-078 微分 • R6
Answer. `-4`
Note. (1+i)^4=(2i)^2=-4, so the real part is -4.

0242 der-079 微分 • R6
Answer. `holomorphic`
Note. u_x=2x=v_y and u_y=-2y=-v_x, so the function is holomorphic.

0243 der-080 微分 • R6
Answer. `notholomorphic`
Note. The CR equations fail except at a point, so it is not holomorphic on any open set.

0244 der-081 微分 • R6
Answer. `1/2`
Note. The residue is 1/(i+i)=1/(2i), whose modulus is 1/2.

0245 der-082 微分 • R6
Answer. `0`
Note. There is no (z-2)^{-1} term, hence the residue is 0.

0246 der-083 微分 • R6
Answer. `1/2`
Note. e^z/z^3=sum z^{n-3}/n!, so z^{-1} comes from n=2 and has coefficient 1/2.

0247 der-084 微分 • R6
Answer. `1`

Note. By the residue theorem, the normalized integral is 1.

0248 der-085 微分・R6
Answer. `-8`
Note. f′(1+i)=4(-2+2i)=-8+8i, so the real part is -8.

0249 der-086 微分・R6
Answer. `4*(x^2+y^2)`
Note. The real Jacobian determinant is 4(x^2+y^2).

0250 der-087 微分・R6
Answer. `0`
Note. u=x^3-3xy^2 has u_xx=6x and u_yy=-6x, so the Laplacian is 0.

0251 der-088 微分・R6
Answer. `harmonic`
Note. The Laplacian is 6x-6x=0, so u is harmonic.

0252 gap-der-param-001 微分・R2
Answer. `3*t/2`
Note. dy/dx=3t^2/(2t)=3t/2.

0253 gap-der-param-002 微分・R3
Answer. `-cos(t)/sin(t)`
Note. dy/dx=cos t/(-sin t)=-cot t.

0254 gap-der-param-003 微分・R2
Answer. `2`
Note. dy/dx=2t and d/dx=d/dt, so d^2y/dx^2=2.

0255 gap-der-param-004 微分・R2
Answer. `3`
Note. dy/dx=3t/2, so at t=2 the slope is 3.

0256 gap-der-polar-001 微分・R4
Answer. `1`
Note. The numerator and denominator are both -1, so the slope is 1.

0257 gap-der-param-005 微分・R2
Answer. `5`
Note. The speed is sqrt(3^2+4^2)=5.

0258 gap-der-app-001 微分・R3
Answer. `72*pi`
Note. dV/dt=4pi(9)(2)=72pi.

0259 gap-der-app-002 微分・R2
Answer. `2*pi`
Note. dA/dt=2pi(5)(0.2)=2pi.

0260 gap-der-app-003 微分 • R3
Answer. `-3/2`
Note. dy/dt=-(x/y)dx/dt=-(6/8)*2=-3/2.

0261 gap-der-app-004 微分 • R2
Answer. `-1/4`
Note. The normal slope is -1/4.

0262 gap-der-app-005 微分 • R2
Answer. `2.01`
Note. sqrt(4.04)≈2+(1/4)(0.04)=2.01.

0263 gap-der-app-006 微分 • R2
Answer. `3/2`
Note. x1=1-(-1)/2=3/2.

0264 gap-der-app-007 微分 • R3
Answer. `2`
Note. The curvature is 2.

0265 gap-der-ode-001 微分 • R2
Answer. `2*exp(-x)`
Note. y=Ce^{-x}; the initial condition gives C=2.

0266 gap-der-ode-002 微分 • R2
Answer. `3*exp(2*x)`
Note. The solution is y=3e^{2x}.

0267 gap-der-ode-003 微分 • R3
Answer. `(exp(x)-exp(-x))/2`
Note. e^x y=(e^{2x}-1)/2, so y=(e^x-e^{-x})/2.

0268 gap-der-ode-004 微分 • R3
Answer. `sin(x)`
Note. The solution is y=sin x.

0269 gap-der-ode-005 微分 • R4
Answer. `1/(1+exp(-x))`
Note. With C=1, the solution is y=1/(1+e^{-x}).

0270 gap-tech-007 微分 • R2
Answer. `parametricdifferentiation`
Note. Use parametric differentiation.

0271 mob-tech-012 微分 • R2
Answer. `Lagrangemultiplier`
Note. Use Lagrange multiplier.

0272 mob-tech-014 微分 • R2
Answer. `nabla`
Note. Use nabla.

0273 mob-tech-018 微分 • R2
Answer. `Besselequation`
Note. Use Bessel equation.

0274 mob-tech-020 微分 • R2
Answer. `logarithmicdifferentiation`
Note. Use logarithmic differentiation.

0275 mob-trap-001 微分 • R2
Answer. `missingchainfactor`
Note. missing chain factor

0276 mob-trap-007 微分 • R2
Answer. `treatyconstant`
Note. treat y constant

0277 mob-trap-011 微分 • R2
Answer. `zero`
Note. zero

0278 mob-lm-001 微分 • R4
Answer. 25
Note. The optimal value is 25.

0279 mob-lm-002 微分 • R4
Answer. 18
Note. The optimal value is 18.

0280 mob-lm-003 微分 • R4
Answer. 64
Note. The optimal value is 64.

0281 mob-lm-004 微分 • R4
Answer. 16
Note. The optimal value is 16.

0282 mob-lm-005 微分 • R4
Answer. 16
Note. The optimal value is 16.

0283 mob-lm-006 微分 • R4
Answer. 3
Note. The optimal value is 3.

0284 mob-lm-007 微分 • R4
Answer. 25

Note. The optimal value is 25.

0285 **mob-lm-008** 微分 • R4
Answer. `1/3`
Note. The optimal value is 1/3.

0286 **mob-lm-009** 微分 • R4
Answer. `25`
Note. The optimal value is 25.

0287 **mob-lm-010** 微分 • R4
Answer. `2`
Note. The optimal value is 2.

0288 **mob-bessel-001** 微分 • R6
Answer. `1`
Note. Use the first terms of Bessel functions.

0289 **mob-bessel-002** 微分 • R6
Answer. `0`
Note. Use the first terms of Bessel functions.

0290 **mob-bessel-003** 微分 • R6
Answer. `-1/4`
Note. Use the first terms of Bessel functions.

0291 **mob-bessel-004** 微分 • R6
Answer. `1/2`
Note. Use the first terms of Bessel functions.

0292 **mob-bessel-005** 微分 • R6
Answer. `-J_1(x)`
Note. -J_1(x)

0293 **mob-bessel-006** 微分 • R6
Answer. `Besselequation`
Note. Bessel equation

0294 **mob-bessel-007** 微分 • R6
Answer. `regularsingular`
Note. regular singular

0295 **mob-bessel-008** 微分 • R6
Answer. `J_2(x)`
Note. J_2(x)

0296 **mob-bessel-009** 微分 • R6
Answer. `-J_1(x)`
Note. -J_1(x)

0297 mob-bessel-010 微分 • R6
Answer. `second-orderODE`
Note. second-order ODE

0298 mob-nabla-001 微分 • R4
Answer. `3`
Note. Use standard nabla identities.

0299 mob-nabla-002 微分 • R4
Answer. `-2`
Note. Use standard nabla identities.

0300 mob-nabla-003 微分 • R4
Answer. `6`
Note. Use standard nabla identities.

0301 mob-nabla-004 微分 • R4
Answer. `2*sqrt(2)`
Note. Use standard nabla identities.

0302 mob-nabla-005 微分 • R4
Answer. `0`
Note. Use standard nabla identities.

0303 mob-nabla-006 微分 • R4
Answer. `0`
Note. Use standard nabla identities.

0304 mob-nabla-007 微分 • R4
Answer. `2`
Note. Use standard nabla identities.

0305 mob-nabla-008 微分 • R4
Answer. `Laplacian`
Note. Laplacian

0306 mob-nabla-009 微分 • R4
Answer. `conservativefield`
Note. conservative field

0307 mob-nabla-010 微分 • R4
Answer. `curl`
Note. curl

0308 rel-basic-007 微分 • R1
Answer. `12`
Note. The derivative is 3x^2.

0309 **rel-basic-008** 微分 • R1
Answer. 1
Note. The derivative is cos x.

0310 **rel-basic-009** 微分 • R1
Answer. 2
Note. The derivative is 2e^{2x}.

0311 **rel-basic-010** 微分 • R1
Answer. 1/4
Note. The derivative is 1/(2sqrt x).

0312 **rel-basic-011** 微分 • R1
Answer. 1/e
Note. The derivative is 1/x.

0313 **rel-basic-012** 微分 • R1
Answer. 1
Note. The derivative is log x+1.

0314 **rel-adv-005** 微分 • R4
Answer. 20
Note. The determinant is 4(x^2+y^2).

0315 **rel-adv-006** 微分 • R4
Answer. 8/3
Note. Dot the gradient (4,4,2) with (2,-1,2)/3.

0316 **rel-adv-007** 微分 • R4
Answer. 128
Note. The Hessian is [[12,-4],[-4,12]].

0317 **rel-adv-008** 微分 • R4
Answer. 9
Note. The minimum distance squared to the plane is 9.

0318 **rel-boss-007** 微分 • R6
Answer. 1/64
Note. J0(x)=1-x^2/4+x^4/64+...

0319 **rel-hard-mv-005** 微分 • R5
Answer. 4
Note. Compute the determinant of [[1/y,-x/y^2],[y,x]].

0320 **rel-hard-mv-006** 微分 • R5
Answer. 1/27
Note. The symmetric maximum occurs at x=y=z=1/3.

0321 **rel-hard-mv-007** 微分 • R5
Answer. 2

Note. The minimum occurs at x=y=1.

0322 **rel-hard-der-001** 微分 • R5
Answer. `1`
Note. The Wronskian is e^{2x}.

0323 **rel-hard-der-002** 微分 • R5
Answer. `-1`
Note. Compute f g'-f'g.

0324 **rel-hard-der-003** 微分 • R5
Answer. `-4`
Note. At (1,0), the Hessian is diag(-2,2).

0325 **rel-hard-der-004** 微分 • R5
Answer. `1/(2*sqrt(2))`
Note. Use |y''|/(1+(y')^2)^{3/2}.

0326 **rel-hard-der-005** 微分 • R5
Answer. `5/3`
Note. Use (dy/dt)/(dx/dt).

0327 **rel-hard-der-006** 微分 • R5
Answer. `3/8`
Note. Differentiate dy/dx with respect to t, then divide by dx/dt.

0328 **rel-hard-der-007** 微分 • R5
Answer. `5/3`
Note. The gradient is (1,1,1).

0329 **rel-hard-der-008** 微分 • R5
Answer. `6`
Note. The divergence is y+z+x.

0330 **rel-hard-der-009** 微分 • R5
Answer. `0`
Note. Away from the origin, this field has zero curl.

0331 **rel-hard-der-010** 微分 • R5
Answer. `2`
Note. f_xx+f_yy=2e^{x+y}.

0332 **hc-der-001** 微分 • R6
Answer. `x^sin(x)*(cos(x)*log(x)+sin(x)/x)`
Note. Use log differentiation: log y=sin x log x.

0333 **hc-der-002** 微分 • R6
Answer. `sin(x)^cos(x)*(-sin(x)*log(sin(x))+cos(x)*cot(x))`
Note. Use log differentiation: log y=cos x log(sin x).

0334 **hc-der-003** 微分・R6
Answer. `exp(exp(2*x)+x^2)*(2*exp(2*x)+2*x)`
Note. Differentiate the outer exponential and then the exponent.

0335 **hc-der-004** 微分・R6
Answer. 360
Note. Only the fourth derivative of x^4 survives at x=0 in the Leibniz sum.

0336 **hc-der-005** 微分・R6
Answer. 42
Note. The x^7 coefficient in x^2 sin x is 1/120.

0337 **hc-der-006** 微分・R6
Answer. 1680
Note. The x^8 coefficient is 1/4!, then multiply by 8!.

0338 **hc-der-007** 微分・R6
Answer. `-1/11`
Note. Use sin x/x=sum (-1)^k x^{2k}/(2k+1)!.

0339 **hc-der-008** 微分・R6
Answer. 16
Note. From log y=xy, the series begins y=1+x+3x^2/2+8x^3/3+...

0340 **hc-der-009** 微分・R6
Answer. 4096
Note. The twelfth derivative returns cosine with factor 2^12.

0341 **hc-der-010** 微分・R6
Answer. 16
Note. Use Im((1+i)^9), or the Taylor coefficient of e^x sin x.

0342 **exam-der-001** 微分・R5
Answer. `x^x*(log(x)+1)`
Note. Use log differentiation: log y=x log x.

0343 **exam-der-002** 微分・R5
Answer. `x^sin(x)*(cos(x)*log(x)+sin(x)/x)`
Note. Differentiate sin x log x in the exponent.

0344 **exam-der-003** 微分・R5
Answer. `(1-log(x))/x^2`
Note. Use quotient rule or x^{-1} log x.

0345 **exam-der-004** 微分・R5
Answer. `1/(1+x^2)`
Note. The derivative simplifies after combining 1+u^2.

0346 exam-der-005 微分 • R5
Answer. `exp(-x)*(2*x*sin(x)-x^2*sin(x)+x^2*cos(x))`
Note. Apply product rule across x^2, e^{-x}, and sin x.

0347 exam-der-006 微分 • R5
Answer. `1/sqrt(1+x^2)`
Note. This is the derivative of asinh x.

0348 exam-der-007 微分 • R5
Answer. `(2+4*x^2)*exp(x^2)`
Note. Differentiate 2x e^{x^2}.

0349 exam-der-008 微分 • R5
Answer. `1/(1+cos(x))`
Note. Quotient rule reduces the numerator to 1+cos x.

0350 exam-der-009 微分 • R5
Answer. `-1/(1+sin(x))`
Note. The numerator becomes -(1+sin x).

0351 exam-der-010 微分 • R5
Answer. `log(1+x^2)+2*x^2/(1+x^2)`
Note. Product rule plus chain rule.

0352 exam-der-011 微分 • R5
Answer. `(x^2+1)^2*(6*x*(x-1)-(x^2+1))/(x-1)^2`
Note. Factor (x^2+1)^2 after quotient rule.

0353 exam-der-012 微分 • R5
Answer. `2/x+1`
Note. Take log y=2 log x+x, then differentiate.

0354 exam-der-013 微分 • R5
Answer. `11/4`
Note. dy/dx=(dy/dt)/(dx/dt)=(3t^2-1)/(2t).

0355 exam-der-014 微分 • R5
Answer. `2*x*exp(x^4)`
Note. Use FTC with upper limit x^2.

0356 exam-der-015 微分 • R5
Answer. `2*x*log(1+x^4)-log(1+x^2)`
Note. Differentiate both variable limits.

0357 exam-der-016 微分 • R5
Answer. `1/2`
Note. (f^{-1})'(1)=1/f'(0)=1/2.

0358 exam-der-017 微分 • R5
Answer. `1/(2*sqrt(x)*(1+x))`

Note. Chain rule with u=sqrt x.

0359 exam-der-018 微分 • R5
Answer. `sec(x)`
Note. This standard derivative equals sec x.

0360 exam-der-019 微分 • R5
Answer. `1/(1+x^2)^(3/2)`
Note. Differentiate x(1+x^2)^{-1/2}.

0361 exam-der-020 微分 • R5
Answer. `exp(x)/(1+exp(x))^2`
Note. Quotient rule cancels to one e^x term.

0362 exam-multi-001 微分 • R5
Answer. `2*x*y+y*cos(x*y)`
Note. Treat y as a constant.

0363 exam-multi-002 微分 • R5
Answer. `6*x*y^2+(1+x*y)*exp(x*y)`
Note. Differentiate first with respect to y, then x.

0364 exam-multi-003 微分 • R5
Answer. `20`
Note. The gradient is (2,4), whose squared norm is 20.

0365 exam-multi-004 微分 • R5
Answer. `27`
Note. The Hessian is [[6,-3],[-3,6]].

0366 exam-multi-005 微分 • R5
Answer. `16/5`
Note. Dot grad f(1,2)=(4,1) with the unit direction.

0367 exam-multi-008 微分 • R5
Answer. `4*(x^2+y^2)`
Note. Compute the 2 by 2 determinant.

0368 exam-multi-009 微分 • R5
Answer. `2*x*y+2*y*z+2*z*x`
Note. Divergence is the sum of matching partial derivatives.

0369 exam-multi-010 微分 • R5
Answer. `-2*y`
Note. The z-component is partial_x F_2 - partial_y F_1.

0370 uni-der-001 微分 • R6
Answer. `x^cos(x)*(-sin(x)*log(x)+cos(x)/x)`
Note. Use log differentiation.

0371 uni-der-002 微分・R6
Answer. `sin(x)^x*(log(sin(x))+x*cot(x))`
Note. Differentiate x log(sin x).

0372 uni-der-003 微分・R6
Answer. `11`
Note. Use the imaginary part of (2+i)^3.

0373 uni-der-004 微分・R6
Answer. `12`
Note. Read the x^4 coefficient of x^2 e^x.

0374 uni-der-005 微分・R6
Answer. `-x/(1+x^2)^(3/2)`
Note. First derivative is 1/sqrt(1+x^2).

0375 uni-der-006 微分・R6
Answer. `cos(x)*exp(sin(x)^2)`
Note. FTC plus chain rule.

0376 uni-der-007 微分・R6
Answer. `2*log(1+4*x^2)-log(1+x^2)`
Note. Differentiate both moving endpoints.

0377 uni-der-008 微分・R6
Answer. `2/(1+x^2)`
Note. Simplify after applying 1+u^2.

0378 uni-der-009 微分・R6
Answer. `5/3`
Note. dy/dx=(1+1/t^2)/(1-1/t^2).

0379 uni-der-010 微分・R6
Answer. `1`
Note. Use x=r cos theta, y=r sin theta.

0380 uni-der-011 微分・R6
Answer. `-1`
Note. Differentiate implicitly.

0381 uni-der-012 微分・R6
Answer. `128`
Note. The Hessian is [[12,-4],[-4,12]].

0382 uni-der-013 微分・R6
Answer. `2`
Note. The positive critical point is (2,1).

0383 **uni-der-014** 微分・R6
Answer. `6/5`
Note. grad f(0,2)=(2,0).

0384 **uni-der-015** 微分・R6
Answer. `exp(y)+exp(z)+exp(x)`
Note. Add the matching partial derivatives.

0385 **uni-der-016** 微分・R6
Answer. `-x`
Note. The z-component is partial_x F_2 - partial_y F_1.

0386 **uni-der-017** 微分・R6
Answer. `1/(x*log(x)*log(log(x)))`
Note. Three nested logs give three denominator factors.

0387 **uni-der-018** 微分・R6
Answer. `3*log(x)^2*exp(log(x)^3)/x`
Note. Differentiate the exponent.

0388 **uni-der-019** 微分・R6
Answer. `32`
Note. The fifth derivative returns a positive sine coefficient 2^5.

0389 **uni-der-020** 微分・R6
Answer. `(1+x^2)^x*(log(1+x^2)+2*x^2/(1+x^2))`
Note. Use log differentiation.

0390 **depth-der-001** 微分・R5
Answer. `x^(x^2)*(2*x*log(x)+x)`
Note. Differentiate x^2 log x.

0391 **depth-der-002** 微分・R6
Answer. `sin(x)^cos(x)*(-sin(x)*log(sin(x))+cos(x)*cot(x))`
Note. Differentiate cos x log(sin x).

0392 **depth-der-003** 微分・R5
Answer. `exp(x)*(x+2)`
Note. Differentiate e^x(x+1).

0393 **depth-der-004** 微分・R5
Answer. `2/x`
Note. After two derivatives the expression is 2 log x+3.

0394 **depth-der-005** 微分・R6
Answer. `3*x^2*cos(x^6)-2*x*cos(x^4)`
Note. Apply FTC to both moving limits.

0395 **depth-der-006** 微分・R5
Answer. `1/sqrt(x^2+4)`

Note. This is an asinh-type derivative.

0396 **depth-der-007** 微分 • R5
Answer. `-1`
Note. Differentiate implicitly and substitute (3,3).

0397 **depth-der-008** 微分 • R5
Answer. `1`
Note. dy/dx=(dy/dt)/(dx/dt)=t+1.

0398 **depth-der-009** 微分 • R6
Answer. `3/4`
Note. Differentiate dy/dx with respect to t, then divide by dx/dt.

0399 **depth-der-010** 微分 • R6
Answer. `-3`
Note. Use -(f''(0))/(f'(0))^3.

0400 **depth-der-011** 微分 • R5
Answer. `8`
Note. The Hessian is [[2,2],[2,6]].

0401 **depth-der-012** 微分 • R6
Answer. `0`
Note. Away from the origin the Laplacian is zero.

0402 **depth-der-013** 微分 • R5
Answer. `y*z`
Note. Only the first component contributes yz.

0403 **depth-der-014** 微分 • R5
Answer. `y^2-x^2`
Note. Compute partial_x Q - partial_y P.

0404 **depth-der-015** 微分 • R6
Answer. `2/(1+x^2)`
Note. For x>0 this angle is 2 arctan x.

0405 **depth-der-016** 微分 • R5
Answer. `(2*log(x)-log(x)^2)/x^2`
Note. Differentiate log(x)^2 x^{-1}.

0406 **depth-der-017** 微分 • R5
Answer. `81`
Note. The fourth derivative returns 3^4 cos(3x).

0407 **depth-der-018** 微分 • R5
Answer. `(1+1/(2*sqrt(x)))/(2*sqrt(x+sqrt(x)))`
Note. Chain rule across both radicals.

0408 **burst-der-001** 微分・R6
Answer. 1048576
Note. The value is 2^20.

0409 **burst-der-002** 微分・R6
Answer. −14348907
Note. Use 3^15 sin(15pi/2).

0410 **burst-der-003** 微分・R6
Answer. 262144
Note. Even derivatives of cosh(2x) return 2^18 at zero.

0411 **burst-der-004** 微分・R6
Answer. 720
Note. Read the x^10 coefficient of x^3e^x.

0412 **burst-der-005** 微分・R6
Answer. 56320
Note. Read the x^11 coefficient after shifting the sine series by x^2.

0413 **burst-der-006** 微分・R6
Answer. −5040
Note. The eighth derivative is (-1)^7 7!.

0414 **burst-der-007** 微分・R6
Answer. 479001600
Note. The x^12 coefficient is 1.

0415 **burst-der-008** 微分・R6
Answer. −64
Note. Use the real part of (1+i)^12.

0416 **burst-der-009** 微分・R6
Answer. 24024
Note. Read the x^14 coefficient of x^4e^{-x}.

0417 **burst-der-010** 微分・R6
Answer. −1199
Note. Use the imaginary part of (2+i)^9.

0418 **burst-boss-der-001** 微分・R6
Answer. 241087680
Note. Use [x^n](log(1+x))^2=(-1)^n*2H_{n-1}/n.

0419 **burst-boss-der-002** 微分・R6
Answer. −21257280
Note. Use [x^n](log(1+x))^2=(-1)^n*2H_{n-1}/n.

0420 **burst-boss-der-003** 微分 • R6
Answer. −22591
Note. Multiply the even series of e^{x^2} and cos x.

0421 **burst-boss-der-004** 微分 • R6
Answer. 16124
Note. Use the imaginary part of (1+2i)^14.

0422 **burst-boss-der-005** 微分 • R6
Answer. 164833
Note. Use the real part of (1+2i)^16.

0423 **burst-boss-der-006** 微分 • R6
Answer. 351261243360
Note. Read the x^18 coefficient after shifting by x^4.

0424 **burst-boss-der-007** 微分 • R6
Answer. 27525120
Note. Read the x^13 coefficient of sin(2x).

0425 **burst-boss-der-008** 微分 • R6
Answer. 13172296626
Note. Read the x^16 coefficient of cos(3x).

0426 **burst-boss-der-009** 微分 • R6
Answer. 598752000
Note. The x^10 coefficient is C(11,3).

0427 **burst-boss-der-010** 微分 • R6
Answer. 1307674368000
Note. Use x^3/(1+x^2)=x^3-x^5+x^7-...

0428 **burst-boss2-der-001** 微分 • R6
Answer. −1024
Note. Use the imaginary part of (1+i)^21.

0429 **burst-boss2-der-002** 微分 • R6
Answer. −1024
Note. Use the real part of (1+i)^20.

0430 **burst-boss2-der-003** 微分 • R6
Answer. 3041525760
Note. Read the x^12 coefficient of e^{2x}, then multiply by 17!.

0431 **burst-boss2-der-004** 微分 • R6
Answer. 54747463680
Note. Read the x^12 coefficient of cos(2x), then multiply by 18!.

0432 **burst-boss2-der-005** 微分 • R6
Answer. −1334792724768

Note. Read the x^15 coefficient of sin(3x), then multiply by 19!.

0433 **burst-boss2-der-006** 微分・R6
Answer. 239500800
Note. The x^10 coefficient is C(12,2).

0434 **burst-boss2-der-007** 微分・R6
Answer. 1796256000
Note. Read the x^8 coefficient of (1-x)^-5.

0435 **burst-boss2-der-008** 微分・R6
Answer. 271996268544000
Note. Read the x^12 coefficient of (1+x)^-2.

0436 **burst-boss3-der-001** 微分・R6
Answer. 863130293635276800
Note. Use [x^n](log(1+x))^2=(-1)^n2H_{n-1}/n.

0437 **burst-boss3-der-002** 微分・R6
Answer. 2481560048340172800000
Note. Read the x^14 coefficient of (1-x)^-7.

0438 **burst-boss3-der-003** 微分・R6
Answer. 3771971273477062656000 0
Note. Read the x^15 coefficient of (1-x)^-6.

0439 **burst-boss3-der-004** 微分・R6
Answer. 1621934672117760000
Note. Read the x^13 coefficient of (1-x)^-8.

0440 **burst-boss3-der-005** 微分・R6
Answer. 17166620433372086476800000
Note. Read the x^24 term of e^{-x^2}.

0441 **burst-boss3-der-006** 微分・R6
Answer. 5537619494636156928000000
Note. Read the x^24 term of cosh(x^2).

0442 **burst-boss3-der-007** 微分・R6
Answer. -2727891376668057600000
Note. Read the x^22 term of sin(x^2).

0443 **burst-boss3-der-008** 微分・R6
Answer. 1845873164878718976000 0
Note. Read the x^24 term of cos(x^2).

0444 **int-001** 積分・R1
Answer. 2*x^3
Note. 次方積分公式。常數 C 可省略。

0445 int-002 積分・R1
Answer. `sin(x)`
Note. sin x 的導數是 cos x。

0446 int-003 積分・R1
Answer. `log(abs(x))`
Note. 基本積分為 ln|x|。

0447 int-004 積分・R1
Answer. `exp(3*x)/3`
Note. 鏈鎖律反向使用。

0448 int-005 積分・R2
Answer. `exp(x^2)/2`
Note. 令 u=x^2，du=2x dx。

0449 int-006 積分・R2
Answer. `log(1+x^2)`
Note. 分子是分母導數。

0450 int-007 積分・R2
Answer. `sin(x^2)/2`
Note. 令 u=x^2。

0451 int-008 積分・R2
Answer. `1`
Note. 原函數 x^3，代入 0 到 1。

0452 int-009 積分・R3
Answer. `x^2*log(x)/2-x^2/4`
Note. 分部積分，u=ln x，dv=x dx。

0453 int-010 積分・R3
Answer. `-cos(x)+cos(x)^3/3`
Note. sin^3 x = sin x(1-cos^2 x)，令 u=cos x。

0454 int-011 積分・R3
Answer. `atan(x/2)/2`
Note. 套 arctan 形式。

0455 int-012 積分・R2
Answer. `pi/4`
Note. 用 sin^2 x=(1-cos 2x)/2。

0456 int-013 積分・R3
Answer. `x*(log(x)^2-2*log(x)+2)`
Note. 分部積分兩次。

0457 int-014 積分・R3
Answer. `x^2/2-log(x^2+1)/2`
Note. 先做多項式除法：x^3/(x^2+1)=x-x/(x^2+1)。

0458 int-015 積分・R4
Answer. `exp(x)*(sin(x)-cos(x))/2`
Note. 分部積分兩次，或猜 e^x(a sin x+b cos x)。

0459 int-016 積分・R3
Answer. `2*log(2)-1`
Note. 原函數是 (1+x)ln(1+x)-(1+x)。

0460 int-017 積分・R3
Answer. `1`
Note. 分部積分，邊界項留下 1。

0461 int-018 積分・R3
Answer. `-log(1+cos(x))`
Note. 令 u=1+cos x，du=-sin x dx。

0462 int-019 積分・R4
Answer. `2*sqrt(x)-2*log(1+sqrt(x))`
Note. 令 u=sqrt x，dx=2u du。

0463 int-020 積分・R3
Answer. `log(2)/3`
Note. 令 u=1+x^3。

0464 td-int-001 積分・R6
Answer. `pi*exp(-3)/2`
Note. 使用 Fourier transform 型公式：int_0^infty cos(tx)/(1+x^2) dx = pi e^{-|t|}/2。

0465 td-int-002 積分・R6
Answer. `pi*log(9/2)`
Note. 套公式 int_0^pi log(a+b cos x) dx = pi log((a+sqrt(a^2-b^2))/2)。

0466 td-int-003 積分・R4
Answer. `log(2/5)`
Note. Frullani 公式：int_0^infty (e^{-alpha x}-e^{-beta x})/x dx = log(beta/alpha)。

0467 td-int-004 積分・R4
Answer. `log(7/3)/2`
Note. 用 2 sin A sin B = cos(A-B)-cos(A+B)，再套 Frullani 型結果。

0468 td-int-005 積分・R6
Answer. `pi/sqrt(5)`
Note. 二次型矩陣 [[3,1],[1,2]] 的行列式為 5，面積是 pi/sqrt(det A)。

0469 td-int-006 積分・R6
Answer. `sqrt(pi)/(12*sqrt(3))`

Note. 高斯積分偶次矩：int x^{2m}e^{-a x^2} dx=(2m-1)!! sqrt(pi)/(2^m a^{m+1/2})。

0470 int-021 積分・R4
Answer. 1
Note. 內積分為 x+1/2，外積分為 1/2+1/2=1。

0471 int-022 積分・R4
Answer. 2
Note. D 是直角三角形，兩股長皆為 2，面積為 2。

0472 int-023 積分・R4
Answer. 1/8
Note. 內積分為 x*y^2/2|_0^x=x^3/2，外積分得 1/8。

0473 int-024 積分・R4
Answer. 1/3
Note. 內積分為 x(1-x)+(1-x)^2/2=(1-x^2)/2，外積分得 1/3。

0474 int-025 積分・R4
Answer. pi/2
Note. 極座標下為 int_0^{2pi} int_0^1 r^3 dr dtheta = 2pi*(1/4)=pi/2。

0475 int-026 積分・R4
Answer. 9
Note. 分離為 (int_0^2 x dx)(int_0^3 y dy)=2*(9/2)=9。

0476 int-027 積分・R2
Answer. log(1+x^2)/2
Note. u=1+x^2，積分為 (1/2)log(1+x^2)。

0477 int-028 積分・R3
Answer. x*log(x)-x
Note. 分部積分得 x log x - int 1 dx = x log x - x。

0478 int-029 積分・R2
Answer. atan(x/3)/3
Note. 由標準公式得 (1/3)atan(x/3)。

0479 int-030 積分・R2
Answer. 1/4
Note. 令 u=sin x，積分變成 int_0^1 u^3 du = 1/4。

0480 int-031 積分・R4
Answer. 1/4
Note. Swap order: integral is int_0^1 x^2*x dx = int_0^1 x^3 dx = 1/4.

0481 int-032 積分・R4
Answer. 3*pi
Note. Area = pi(4-1)=3pi.

0482 **int-033** 積分・R4
Answer. `4*pi`
Note. Integral of r^2 over disk radius 2 is int_0^{2pi}int_0^2 r^3 dr dtheta=8pi. Half is 4pi.

0483 **int-034** 積分・R4
Answer. `pi/4`
Note. The region is x>=0,y>=0,x^2+y^2<=1, so the area is pi/4.

0484 **int-035** 積分・R4
Answer. `pi/2`
Note. The integral is 2pi*(1/4)=pi/2.

0485 **int-036** 積分・R4
Answer. `1/6`
Note. Integral is (1/2)int_0^1(1-y)^2dy=1/6.

0486 **int-037** 積分・R3
Answer. `exp(x^2)/2`
Note. With u=x^2, the antiderivative is e^{x^2}/2.

0487 **int-038** 積分・R3
Answer. `x*sin(x)+cos(x)`
Note. Integration by parts gives x sin x - int sin x dx = x sin x + cos x.

0488 **int-039** 積分・R3
Answer. `pi/4`
Note. The integral is atan(1)-atan(0)=pi/4.

0489 **int-040** 積分・R5
Answer. `1`
Note. The integral equals Gamma(2)=1!=1.

0490 **int-041** 積分・R3
Answer. `log(1+x^4)/4`
Note. With u=1+x^4, the antiderivative is (1/4)log(1+x^4).

0491 **int-042** 積分・R3
Answer. `(2/9)*(1+x^3)^(3/2)`
Note. u=1+x^3 gives (1/3)int u^{1/2}du = (2/9)(1+x^3)^{3/2}.

0492 **int-043** 積分・R4
Answer. `log(1+sqrt(1+x^2))`
Note. With u=sqrt(1+x^2), the integrand becomes du/(1+u), so the answer is log(1+sqrt(1+x^2)).

0493 **int-044** 積分・R3
Answer. `log(1+exp(x^2))/2`
Note. u=1+e^{x^2} gives (1/2)int du/u = (1/2)log(1+e^{x^2}).

0494 **int-045** 積分・R3
Answer. `log(x)^3/3`
Note. u=log x changes the integral to int u^2 du = u^3/3.

0495 **int-046** 積分・R4
Answer. `3/8`
Note. The integral is int_1^2 u^{-3}du = [-1/(2u^2)]_1^2 = 3/8.

0496 **int-047** 積分・R3
Answer. `pi`
Note. The graph is the upper quarter of x^2+y^2=4, so the area is pi.

0497 **int-048** 積分・R4
Answer. `pi/4`
Note. After x=sin theta, the integral is int_0^{pi/2} sin^2 theta dtheta = pi/4.

0498 **int-049** 積分・R4
Answer. `pi`
Note. With x=2sin theta, the integral becomes 4*(pi/4)=pi.

0499 **int-050** 積分・R4
Answer. `pi/3`
Note. With x=sec theta, the integrand becomes dtheta. The bounds give pi/3.

0500 **int-051** 積分・R4
Answer. `x/(4*sqrt(x^2+4))`
Note. The standard result is int dx/(x^2+a^2)^{3/2}=x/(a^2 sqrt(x^2+a^2)); here a^2=4.

0501 **int-052** 積分・R3
Answer. `exp(x)*(x^2-2*x+2)`
Note. Two rounds of IBP give e^x(x^2-2x+2).

0502 **int-053** 積分・R4
Answer. `-x^2*cos(x)+2*x*sin(x)+2*cos(x)`
Note. IBP gives -x^2 cos x + 2int x cos x dx = -x^2 cos x + 2x sin x + 2cos x.

0503 **int-054** 積分・R4
Answer. `exp(x)*(sin(x)-cos(x))/2`
Note. Let I=int e^x sin x dx. Two IBP steps give 2I=e^x(sin x-cos x).

0504 **int-055** 積分・R4
Answer. `x*(log(x)^2-2*log(x)+2)`
Note. IBP gives x(log x)^2-2int log x dx = x(log x)^2-2(xlog x-x).

0505 **int-056** 積分・R4
Answer. `pi/4-1/2`
Note. IBP gives pi/8 - (1/2)int_0^1 x^2/(1+x^2)dx = pi/8 - 1/2 + pi/8 = pi/4-1/2.

0506 **int-057** 積分・R5
Answer. `(exp(x)-sin(x)-cos(x))/2`

Note. F=e^x int_0^x e^{-t}sin t dt = (e^x-sin x-cos x)/2.

0507 int-058 積分・R5

Answer. `(exp(2*x)-2*x-1)/4`

Note. e^{2x}int_0^x t e^{-2t}dt = e^{2x}/4-x/2-1/4.

0508 int-059 積分・R5

Answer. `x^2-2*x+2-2*exp(-x)`

Note. Solving F′+F=x^2 with F(0)=0 gives F=x^2-2x+2-2e^{-x}.

0509 int-060 積分・R5

Answer. `x*sin(x)/2`

Note. Expanding and integrating gives x sin x/2.

0510 int-061 積分・R5

Answer. `pi/4`

Note. Let I be the integral. The transformed integral has cos^3 numerator. Adding gives 2I=int_0^{pi/2}1 dx=pi/2.

0511 int-062 積分・R5

Answer. `1/2`

Note. I plus its transformed version equals int_0^1 1 dx=1, so I=1/2.

0512 int-063 積分・R5

Answer. `pi/4`

Note. Adding the integral to its transformed copy gives int_0^{pi/2}1 dx=pi/2, hence I=pi/4.

0513 int-064 積分・R5

Answer. `pi^2/4`

Note. The symmetric factor gives I=(pi/2)int_0^pi sin x/(1+cos^2x)dx. With u=cos x, the remaining integral is pi/2, so I=pi^2/4.

0514 int-065 積分・R5

Answer. `log(7/3)`

Note. Frullani gives int_0^infty (e^{-ax}-e^{-bx})/x dx = log(b/a). Here a=3,b=7.

0515 int-066 積分・R5

Answer. `log(4)`

Note. The value is log(4/1)=log 4.

0516 int-067 積分・R5

Answer. `log(4)`

Note. Frullani gives log(8/2)=log 4.

0517 int-068 積分・R5

Answer. `log(5/2)`

Note. The cosine Frullani identity gives log(5/2).

0518 int-069 積分・R5

Answer. `log(4)`

Note. int_0^infty (cos x - cos 4x)/x dx = log 4.

0519 int-070 積分・R5
Answer. `log(3)`
Note. The integral equals log(9/3)=log 3.

0520 int-071 積分・R5
Answer. `log(11/5)/2`
Note. The integral is (1/2)int_0^infty (cos 5x - cos 11x)/x dx = (1/2)log(11/5).

0521 int-072 積分・R5
Answer. `log(2)/2`
Note. The value is (1/2)log(8/4)=log(2)/2.

0522 int-073 積分・R5
Answer. `log(5/3)/2`
Note. The integral equals (1/2)log(5/3).

0523 int-074 積分・R5
Answer. `pi*log(5/2)/2`
Note. Frullani gives (pi/2)log(5/2).

0524 int-075 積分・R5
Answer. `log(5/2)`
Note. Frullani gives log(5/2).

0525 int-076 積分・R5
Answer. `log(6)`
Note. By Frullani, the value is log 6.

0526 gap-int-pf-001 積分・R3
Answer. `(log(x-1)-log(x+1))/2`
Note. Partial fractions give 1/(x^2-1)=1/2/(x-1)-1/2/(x+1).

0527 gap-int-pf-002 積分・R3
Answer. `log(x)`
Note. The integrand reduces to 1/x, so the answer is log x.

0528 gap-int-pf-003 積分・R3
Answer. `(log(x)-log(x+2))/2`
Note. 1/(x(x+2))=(1/2)/x-(1/2)/(x+2).

0529 gap-int-pf-004 積分・R3
Answer. `log(x+1)+log(x+2)`
Note. The denominator derivative is 2x+3, so the integral is log(x^2+3x+2).

0530 gap-int-pf-005 積分・R3
Answer. `-1/(x+1)`
Note. The antiderivative of (x+1)^{-2} is -(x+1)^{-1}.

0531 **gap-int-pf-006** 積分・R4
Answer. `8*log(x-1)/3+log(x+2)/3`
Note. A=8/3 and B=1/3, so integrate the two simple fractions.

0532 **gap-int-pf-007** 積分・R4
Answer. `2*log(x+2)-log(x+3)`
Note. A=2 and B=-1, giving 2log(x+2)-log(x+3).

0533 **gap-int-polar-001** 積分・R2
Answer. `pi`
Note. Area=(1/2)*4*(pi/2)=pi.

0534 **gap-int-polar-002** 積分・R3
Answer. `pi/4`
Note. The polar area is (1/2)(pi/2)=pi/4.

0535 **gap-int-triple-001** 積分・R6
Answer. `1/8`
Note. The value is (1/2)^3=1/8.

0536 **gap-int-triple-002** 積分・R6
Answer. `3/2`
Note. Each coordinate contributes 1/2, so the total is 3/2.

0537 **gap-int-triple-003** 積分・R6
Answer. `1/6`
Note. The standard simplex in R^3 has volume 1/6.

0538 **gap-int-triple-004** 積分・R6
Answer. `1/6`
Note. The integral becomes int_0^1 x^2/2 dx=1/6.

0539 **gap-int-triple-005** 積分・R6
Answer. `12*pi`
Note. The volume is pi(2^2)(3)=12pi.

0540 **gap-int-triple-006** 積分・R6
Answer. `32*pi/3`
Note. The volume is (4/3)pi(2^3)=32pi/3.

0541 **gap-int-cov-001** 積分・R6
Answer. `1/2`
Note. The inverse Jacobian determinant is -1/2, so the absolute value is 1/2.

0542 **gap-int-cov-002** 積分・R6
Answer. `1`
Note. The area is 2*(1/2)=1.

0543 **gap-tech-003** 積分・R2
Answer. `substitution`

Note. Use u=x^2.

0544 **gap-tech-004** 積分・R3
Answer. `integrationbyparts`
Note. Use integration by parts with u=log x.

0545 **gap-tech-006** 積分・R3
Answer. `partialfraction`
Note. Use partial fractions after factoring x^2-1.

0546 **mob-tech-003** 積分・R2
Answer. `u-substitution`
Note. Use u-substitution.

0547 **mob-tech-004** 積分・R2
Answer. `integrationbyparts`
Note. Use integration by parts.

0548 **mob-tech-005** 積分・R2
Answer. `partialfraction`
Note. Use partial fraction.

0549 **mob-tech-006** 積分・R2
Answer. `trigsubstitution`
Note. Use trig substitution.

0550 **mob-tech-010** 積分・R2
Answer. `King'sproperty`
Note. Use King’s property.

0551 **mob-tech-011** 積分・R2
Answer. `Frullaniintegral`
Note. Use Frullani integral.

0552 **mob-tech-013** 積分・R2
Answer. `gradienttheorem`
Note. Use gradient theorem.

0553 **mob-tech-015** 積分・R2
Answer. `Betafunction`
Note. Use Beta function.

0554 **mob-tech-016** 積分・R2
Answer. `Gammafunction`
Note. Use Gamma function.

0555 **mob-tech-017** 積分・R2
Answer. `Wallisreduction`
Note. Use Wallis reduction.

0556 **mob-trap-002** 積分・R2
Answer. `dividebyinnercoefficient`
Note. divide by inner coefficient

0557 **mob-trap-003** 積分・R2
Answer. `arctannotlog`
Note. arctan not log

0558 **mob-trap-006** 積分・R2
Answer. `requiresapositive`
Note. requires a positive

0559 **mob-trap-008** 積分・R2
Answer. `multiplybyJacobian`
Note. multiply by Jacobian

0560 **mob-trap-010** 積分・R2
Answer. `secx`
Note. sec x

0561 **mob-trap-012** 積分・R2
Answer. `denominatorGamma(a+b)`
Note. denominator Gamma(a+b)

0562 **mob-beta-001** 積分・R5
Answer. `1`
Note. Use B(a,b)=Gamma(a)Gamma(b)/Gamma(a+b).

0563 **mob-beta-002** 積分・R5
Answer. `1/2`
Note. Use B(a,b)=Gamma(a)Gamma(b)/Gamma(a+b).

0564 **mob-beta-003** 積分・R5
Answer. `1/6`
Note. Use B(a,b)=Gamma(a)Gamma(b)/Gamma(a+b).

0565 **mob-beta-004** 積分・R5
Answer. `1/12`
Note. Use B(a,b)=Gamma(a)Gamma(b)/Gamma(a+b).

0566 **mob-beta-005** 積分・R5
Answer. `pi`
Note. Use B(a,b)=Gamma(a)Gamma(b)/Gamma(a+b).

0567 **mob-beta-006** 積分・R5
Answer. `2`
Note. Use B(a,b)=Gamma(a)Gamma(b)/Gamma(a+b).

0568 **mob-beta-007** 積分 • R5
Answer. `pi/2`
Note. Use B(a,b)=Gamma(a)Gamma(b)/Gamma(a+b).

0569 **mob-beta-008** 積分 • R5
Answer. `pi/8`
Note. Use B(a,b)=Gamma(a)Gamma(b)/Gamma(a+b).

0570 **mob-beta-009** 積分 • R5
Answer. `1/30`
Note. Use B(a,b)=Gamma(a)Gamma(b)/Gamma(a+b).

0571 **mob-beta-010** 積分 • R5
Answer. `1/4`
Note. Use B(a,b)=Gamma(a)Gamma(b)/Gamma(a+b).

0572 **mob-gamma-001** 積分 • R5
Answer. `1`
Note. Use the gamma recurrence and Gamma(1/2)=sqrt(pi).

0573 **mob-gamma-002** 積分 • R5
Answer. `sqrt(pi)`
Note. Use the gamma recurrence and Gamma(1/2)=sqrt(pi).

0574 **mob-gamma-003** 積分 • R5
Answer. `sqrt(pi)/2`
Note. Use the gamma recurrence and Gamma(1/2)=sqrt(pi).

0575 **mob-gamma-004** 積分 • R5
Answer. `3*sqrt(pi)/4`
Note. Use the gamma recurrence and Gamma(1/2)=sqrt(pi).

0576 **mob-gamma-005** 積分 • R5
Answer. `120`
Note. Use the gamma recurrence and Gamma(1/2)=sqrt(pi).

0577 **mob-gamma-006** 積分 • R5
Answer. `2`
Note. Use the gamma recurrence and Gamma(1/2)=sqrt(pi).

0578 **mob-gamma-007** 積分 • R5
Answer. `sqrt(pi)/2`
Note. Use the gamma recurrence and Gamma(1/2)=sqrt(pi).

0579 **mob-gamma-008** 積分 • R5
Answer. `6`
Note. Use the gamma recurrence and Gamma(1/2)=sqrt(pi).

0580 **mob-gamma-009** 積分 • R5
Answer. `15*sqrt(pi)/8`

Note. Use the gamma recurrence and Gamma(1/2)=sqrt(pi).

0581 **mob-gamma-010** 積分・R5
Answer. `1`
Note. Use the gamma recurrence and Gamma(1/2)=sqrt(pi).

0582 **mob-wallis-001** 積分・R5
Answer. `pi/4`
Note. Evaluate by Wallis reduction or beta symmetry.

0583 **mob-wallis-002** 積分・R5
Answer. `2/3`
Note. Evaluate by Wallis reduction or beta symmetry.

0584 **mob-wallis-003** 積分・R5
Answer. `3*pi/16`
Note. Evaluate by Wallis reduction or beta symmetry.

0585 **mob-wallis-004** 積分・R5
Answer. `8/15`
Note. Evaluate by Wallis reduction or beta symmetry.

0586 **mob-wallis-005** 積分・R5
Answer. `pi/4`
Note. Evaluate by Wallis reduction or beta symmetry.

0587 **mob-wallis-006** 積分・R5
Answer. `3*pi/16`
Note. Evaluate by Wallis reduction or beta symmetry.

0588 **mob-wallis-007** 積分・R5
Answer. `pi/16`
Note. Evaluate by Wallis reduction or beta symmetry.

0589 **mob-wallis-008** 積分・R5
Answer. `1`
Note. Evaluate by Wallis reduction or beta symmetry.

0590 **mob-wallis-009** 積分・R5
Answer. `1`
Note. Evaluate by Wallis reduction or beta symmetry.

0591 **mob-wallis-010** 積分・R5
Answer. `1/2`
Note. Evaluate by Wallis reduction or beta symmetry.

0592 **rel-basic-013** 積分・R1
Answer. `1/3`
Note. Use the power rule.

0593 **rel-basic-014** 積分・R1
Answer. `2`
Note. The antiderivative is -cos x.

0594 **rel-basic-015** 積分・R1
Answer. `(e^2-1)/2`
Note. Divide by the inner coefficient 2.

0595 **rel-basic-016** 積分・R1
Answer. `log(2)`
Note. The antiderivative is log(1+x).

0596 **rel-basic-017** 積分・R1
Answer. `1`
Note. The antiderivative is sin x.

0597 **rel-basic-018** 積分・R1
Answer. `6`
Note. The antiderivative is 3x^2/2.

0598 **rel-adv-011** 積分・R5
Answer. `1/9`
Note. Use Gamma(2)/3^2.

0599 **rel-adv-012** 積分・R4
Answer. `1/2`
Note. The inner integral gives 3x^2/2.

0600 **rel-adv-013** 積分・R4
Answer. `pi/2`
Note. Use polar coordinates and integrate r^3.

0601 **rel-adv-014** 積分・R4
Answer. `5/6`
Note. Expand and integrate term by term.

0602 **rel-adv-009** 積分・R3
Answer. `exp(x^2)*(x^2-1)/2`
Note. Let u=x^2. Then x^3 dx=(u/2)du.

0603 **rel-adv-010** 積分・R3
Answer. `(1+x^2)*log(1+x^2)/2-(1+x^2)/2`
Note. Let u=1+x^2 and integrate log u.

0604 **rel-boss-001** 積分・R5
Answer. `pi/2`
Note. Dirichlet integral gives pi/2 for positive frequency.

0605 **rel-boss-002** 積分・R5

Answer. `log(5/2)`

Note. Frullani gives log(5/2).

0606 **rel-boss-003** 積分・R5

Answer. `log(7/2)`

Note. Use the cosine Frullani identity.

0607 **rel-boss-004** 積分・R5

Answer. `15*sqrt(pi)/8`

Note. This is Gamma(7/2).

0608 **rel-boss-005** 積分・R5

Answer. `3*pi/8`

Note. This is B(1/2,5/2).

0609 **rel-boss-006** 積分・R5

Answer. `3*pi/512`

Note. Convert to a beta integral.

0610 **rel-boss-008** 積分・R6

Answer. `0`

Note. There is no z^{-1} term.

0611 **rel-boss-009** 積分・R6

Answer. `1/6`

Note. The residue is the z^3 coefficient of e^z.

0612 **rel-boss-010** 積分・R6

Answer. `1/18`

Note. Use u=x+y and v=x/(x+y).

0613 **rel-boss-011** 積分・R6

Answer. `1/720`

Note. Use the simplex beta integral.

0614 **rel-boss-012** 積分・R6

Answer. `128*pi/5`

Note. Use spherical coordinates: integrate 4pi r^4.

0615 **rel-boss-014** 積分・R6

Answer. `pi^2/6`

Note. This is Gamma(2) zeta(2).

0616 **rel-boss-015** 積分・R6

Answer. `pi^4/15`

Note. This is Gamma(4) zeta(4).

0617 **rel-boss-016** 積分・R5

Answer. `-pi*log(2)/2`

Note. Use the classic symmetry product integral.

0618 **rel-boss-017** 積分・R5
Answer. `2`
Note. Differentiate the beta/gamma integral or use Gamma(3).

0619 **rel-boss-020** 積分・R6
Answer. `1/2`
Note. The z^2/2 term in 1-cos z creates the residue.

0620 **rel-hard-int-001** 積分・R5
Answer. `15/8`
Note. Use Gamma(6)/2^6.

0621 **rel-hard-int-002** 積分・R5
Answer. `3*sqrt(pi)/128`
Note. This is Gamma(5/2)/4^{5/2}.

0622 **rel-hard-int-003** 積分・R5
Answer. `-1/9`
Note. Use int_0^1 x^a log x dx=-1/(a+1)^2.

0623 **rel-hard-int-004** 積分・R5
Answer. `1/32`
Note. Use int_0^1 x^a(log x)^2 dx=2/(a+1)^3.

0624 **rel-hard-int-005** 積分・R5
Answer. `35*pi/256`
Note. Use Wallis reduction.

0625 **rel-hard-int-006** 積分・R5
Answer. `16/35`
Note. Use the odd Wallis product.

0626 **rel-hard-int-007** 積分・R5
Answer. `pi/32`
Note. Convert to one half of a beta integral.

0627 **rel-hard-int-008** 積分・R5
Answer. `log(11/3)`
Note. Frullani gives log(11/3).

0628 **rel-hard-int-009** 積分・R5
Answer. `log(5/2)`
Note. Use the cosine Frullani identity.

0629 **rel-hard-int-010** 積分・R5
Answer. `pi/2`
Note. This is the classic squared sinc integral.

0630 rel-hard-int-011 積分・R5
Answer. `pi/4`
Note. Let u=x^2.

0631 rel-hard-int-012 積分・R5
Answer. `pi/(2*sqrt(2))`
Note. The standard quartic integral equals pi/(2sqrt2).

0632 rel-hard-int-013 積分・R5
Answer. `0`
Note. Use x -> 1/x symmetry.

0633 rel-hard-int-014 積分・R5
Answer. `-1*pi^2/12`
Note. Expand 1/(1+x) and integrate termwise.

0634 rel-hard-int-015 積分・R5
Answer. `-4`
Note. Use int_0^1 x^{a-1} log x dx=-1/a^2.

0635 rel-hard-int-016 積分・R5
Answer. `log(2)/3`
Note. Let u=1+x^3.

0636 rel-hard-int-017 積分・R5
Answer. `-pi*log(2)`
Note. Both log sine and log cosine integrals equal -pi log2 /2.

0637 rel-hard-int-018 積分・R5
Answer. `pi/32`
Note. Use int_0^infty dx/(x^2+a^2)^2=pi/(4a^3).

0638 rel-hard-cpx-001 積分・R6
Answer. `-1/6`
Note. Use the z^3 coefficient of sin z.

0639 rel-hard-cpx-002 積分・R6
Answer. `-1/2`
Note. Use the z^2 coefficient of cos z.

0640 rel-hard-cpx-003 積分・R6
Answer. `1/6`
Note. Use the z^3 coefficient of e^z-1-z.

0641 rel-hard-cpx-004 積分・R6
Answer. `-1/2`
Note. Use the z^2 coefficient of log(1+z).

0642 rel-hard-cpx-005 積分・R6
Answer. `1`
Note. Expand 1/(1-z).

0643 rel-hard-cpx-006 積分・R6
Answer. `e`
Note. For a second-order pole, take the derivative of e^z.

0644 rel-hard-cpx-007 積分・R6
Answer. `1`
Note. Use csc^2 z=1/z^2+...

0645 rel-hard-cpx-008 積分・R6
Answer. `1`
Note. cot z has residue 1 at 0.

0646 rel-hard-cpx-009 積分・R6
Answer. `2`
Note. Use the z^2 coefficient of e^{2z}-1.

0647 rel-hard-cpx-010 積分・R6
Answer. `-1/24`
Note. Use the z^4 coefficient of 1-cos z-z^2/2.

0648 rel-hard-mv-001 積分・R5
Answer. `1/60`
Note. Use the two-dimensional simplex formula.

0649 rel-hard-mv-002 積分・R5
Answer. `pi/24`
Note. Use polar coordinates and integrate cos^2 theta sin^2 theta.

0650 rel-hard-mv-003 積分・R6
Answer. `4*pi/5`
Note. Use spherical coordinates.

0651 rel-hard-mv-004 積分・R6
Answer. `36*pi`
Note. Volume of a ball of radius 3.

0652 rel-hard-mv-008 積分・R6
Answer. `1/2`
Note. Use u=x+y, v=x-y, with Jacobian 1/2.

0653 rel-hard-mv-009 積分・R5
Answer. `64*pi/3`
Note. Use polar coordinates and integrate r^5.

0654 rel-hard-mv-010 積分・R6
Answer. `1/2520`

Note. Use the three-dimensional simplex beta formula.

0655 **hc-usub-001** 積分・R6
Answer. `(1+x^2)^(3/2)/3-sqrt(1+x^2)`
Note. Let u=1+x^2, so x^3 dx=(u-1)du/2.

0656 **hc-usub-002** 積分・R6
Answer. `exp(x^2)*(x^4-2*x^2+2)/2`
Note. Let u=x^2, then integrate u^2 e^u by parts.

0657 **hc-usub-003** 積分・R6
Answer. `(1+x^2)*(log(1+x^2)^2-2*log(1+x^2)+2)/2`
Note. Let u=1+x^2, then use the antiderivative of (log u)^2.

0658 **hc-usub-004** 積分・R6
Answer. `atan(x^3)/3`
Note. Let u=x^3; the integral becomes one third of int du/(1+u^2).

0659 **hc-usub-005** 積分・R6
Answer. `log(1+sqrt(1+x^4))/2`
Note. Let u=sqrt(1+x^4); the remaining integral is du/(2(1+u)).

0660 **hc-usub-006** 積分・R6
Answer. `2*(1+exp(x))^(3/2)/3`
Note. Let u=1+e^x.

0661 **hc-usub-007** 積分・R6
Answer. `pi/6`
Note. Use u=x^3 to reduce the integral to one third of arcsin u from 0 to 1.

0662 **hc-usub-008** 積分・R6
Answer. `1/16`
Note. Let u=1+x^2; then x^3 dx=(u-1)du/2.

0663 **hc-usub-009** 積分・R6
Answer. `1/4`
Note. Let u=1+sin^2 x; the bounds are 1 and 2.

0664 **hc-usub-010** 積分・R6
Answer. `log((1+e)/2)/3`
Note. Let u=e^{x^3}; then du=3x^2e^{x^3}dx.

0665 **hc-ibp-001** 積分・R6
Answer. `exp(2*x)*(4*x^3-6*x^2+6*x-3)/8`
Note. Repeated integration by parts lowers the cubic one degree at a time.

0666 **hc-ibp-002** 積分・R6
Answer. `x^2*sin(3*x)/3+2*x*cos(3*x)/9-2*sin(3*x)/27`
Note. Use integration by parts twice and keep the factor 3 through each step.

0667 hc-ibp-003 積分 • R6
Answer. `exp(x)*((x^2-1)*sin(x)+(2*x-1-x^2)*cos(x))/2`
Note. Use the complex shortcut or cyclic integration by parts.

0668 hc-ibp-004 積分 • R6
Answer. `x*(log(x)^3-3*log(x)^2+6*log(x)-6)`
Note. Each integration by parts lowers the power of log x.

0669 hc-ibp-005 積分 • R6
Answer. `(x^2+1)*atan(x)^2/2-x*atan(x)+log(1+x^2)/2`
Note. After the first IBP, split x^2/(1+x^2)=1-1/(1+x^2).

0670 hc-ibp-006 積分 • R6
Answer. `exp(2*x)*(2*sin(3*x)-3*cos(3*x))/13`
Note. Two IBP rounds return the original integral; solve algebraically.

0671 hc-ibp-007 積分 • R6
Answer. `x^4*log(x)/4-x^4/16`
Note. Use u=log x and dv=x^3 dx.

0672 hc-ibp-008 積分 • R6
Answer. `2*log(2)/5-47/300`
Note. IBP leaves int_0^1 x^5/(1+x) dx; divide the polynomial before integrating.

0673 hc-ibp-009 積分 • R6
Answer. `pi^2/4-2`
Note. Two IBP rounds give x^2 sin x+2x cos x-2sin x.

0674 hc-ibp-010 積分 • R6
Answer. `1/6`
Note. IBP reduces the problem to int_0^1 x^4/(1+x^2) dx.

0675 hc-rad-001 積分 • R6
Answer. `sqrt(x^2-1)/x`
Note. Differentiate sqrt(x^2-1)/x, or use x=sec theta.

0676 hc-rad-002 積分 • R6
Answer. `atan((x+1)/2)/2`
Note. Complete the square: x^2+2x+5=(x+1)^2+4.

0677 hc-rad-003 積分 • R6
Answer. `-log(abs(x))+log(abs(x-1))+log(abs(x+1))`
Note. Decompose into -1/x+1/(x-1)+1/(x+1).

0678 hc-rad-004 積分 • R6
Answer. `(log(abs(x-1))-log(abs(x+1)))/4-atan(x)/2`
Note. Split x^4-1 into (x-1)(x+1)(x^2+1).

0679 hc-rad-005 積分・R6
Answer. `2*sqrt(3)-2*pi/3`
Note. Use the antiderivative sqrt(x^2-4)-2 arccos(2/x).

0680 hc-rad-006 積分・R6
Answer. `1`
Note. Rationalize to (1-sin x)/cos^2 x, whose antiderivative is tan x-sec x.

0681 hc-rad-007 積分・R6
Answer. `8/315`
Note. Use the beta integral or peel off one sine factor.

0682 hc-rad-008 積分・R6
Answer. `pi/3`
Note. Use int_0^infty dx/(1+x^a)=pi/a*csc(pi/a).

0683 hc-rad-009 積分・R6
Answer. `pi/6`
Note. Use int_0^infty x^{m-1}/(1+x^n) dx=pi/n*csc(m*pi/n).

0684 hc-rad-010 積分・R6
Answer. `pi/4`
Note. Let u=x^2 to reduce the integral to half of arcsin u.

0685 hc-spec-001 積分・R6
Answer. `8/81`
Note. This is Gamma(5)/3^5.

0686 hc-spec-002 積分・R6
Answer. `pi/8`
Note. This is B(3/2,3/2).

0687 hc-spec-003 積分・R6
Answer. `1/280`
Note. This is B(4,5)=3!4!/8!.

0688 hc-spec-004 積分・R6
Answer. `63*pi/512`
Note. Use the even Wallis product.

0689 hc-spec-005 積分・R6
Answer. `8*pi^6/63`
Note. Use Gamma(6)zeta(6).

0690 hc-spec-006 積分・R6
Answer. `3*sqrt(pi)/(32*sqrt(2))`
Note. Use int_0^infty x^m e^{-a x^2} dx=Gamma((m+1)/2)/(2a^{(m+1)/2}).

0691 exam-int-001 積分・R5
Answer. `-1/(2*(1+x^2))`

Note. Let u=1+x^2.

0692 exam-int-002 積分・R5
Answer. `x^2/2-log(1+x^2)/2`
Note. Divide x^3 by 1+x^2 as x - x/(1+x^2).

0693 exam-int-003 積分・R5
Answer. `exp(x)*(x^2-2*x+2)`
Note. Use integration by parts twice.

0694 exam-int-004 積分・R5
Answer. `x^2*log(x)/2-x^2/4`
Note. Let u=log x and dv=x dx.

0695 exam-int-005 積分・R5
Answer. `log(x)^2/2`
Note. Let u=log x.

0696 exam-int-006 積分・R5
Answer. `-cos(x)+cos(x)^3/3`
Note. Write sin^3 x=(1-cos^2 x) sin x.

0697 exam-int-007 積分・R5
Answer. `tan(x)^2/2`
Note. Let u=tan x.

0698 exam-int-008 積分・R5
Answer. `atan(x/2)/2`
Note. Use the arctangent form with a=2.

0699 exam-int-009 積分・R5
Answer. `log(abs((x-1)/(x+1)))/2`
Note. Decompose 1/(x^2-1).

0700 exam-int-010 積分・R5
Answer. `2*log(1+x^(3/2))/3`
Note. Let u=1+x^{3/2}.

0701 exam-int-011 積分・R5
Answer. `-sqrt(4-x^2)`
Note. Let u=4-x^2.

0702 exam-int-012 積分・R5
Answer. `exp(2*x)*(2*sin(x)-cos(x))/5`
Note. Use the standard exponential-trig formula.

0703 exam-int-013 積分・R5
Answer. `x*atan(x)-log(1+x^2)/2`
Note. Let u=arctan x and dv=dx.

0704 **exam-int-014** 積分・R5
Answer. `x-atan(x)`
Note. Rewrite as 1-1/(x^2+1).

0705 **exam-int-015** 積分・R5
Answer. `log(log(x))`
Note. Let u=log x.

0706 **exam-int-016** 積分・R5
Answer. `sin(log(x))`
Note. Let u=log x.

0707 **exam-int-017** 積分・R5
Answer. `log(x^2+3*x+5)`
Note. The numerator is the derivative of the denominator.

0708 **exam-int-018** 積分・R5
Answer. `atan(x^2)/2`
Note. Let u=x^2.

0709 **exam-int-019** 積分・R5
Answer. `2*atan(sqrt(x))`
Note. Let u=sqrt x.

0710 **exam-int-020** 積分・R5
Answer. `2*sqrt(1+x^3)/3`
Note. Let u=1+x^3.

0711 **exam-int-021** 積分・R5
Answer. `log(2)/2`
Note. Let u=1+x^2.

0712 **exam-int-022** 積分・R5
Answer. `log(2)/3`
Note. Let u=1+x^3.

0713 **exam-int-023** 積分・R5
Answer. `pi/4`
Note. Use the average value of sin^2 on [0,pi/2].

0714 **exam-int-024** 積分・R5
Answer. `log(2)`
Note. Let u=1+cos x.

0715 **exam-int-025** 積分・R5
Answer. `-1`
Note. Integrate x log x - x and take the limit at 0.

0716 **exam-int-026** 積分・R5
Answer. `1/2`
Note. The exponential tail gives 1/2.

0717 **exam-int-027** 積分・R5
Answer. `pi/4`
Note. This is arctan x from 0 to 1.

0718 **exam-int-028** 積分・R5
Answer. `log(2)-1/2`
Note. Let u=1+x^2 and integrate log u.

0719 **exam-int-029** 積分・R5
Answer. `pi`
Note. Integration by parts leaves pi.

0720 **exam-int-030** 積分・R5
Answer. `1/6`
Note. Use u=1-x or the beta integral.

0721 **exam-multi-006** 積分・R5
Answer. `1/2`
Note. The inner integral gives 3x^2/2.

0722 **exam-multi-007** 積分・R5
Answer. `pi`
Note. Use one-half integral of r^2 from -pi/2 to pi/2.

0723 **uni-int-001** 積分・R6
Answer. `x^4/4-x^2/2+log(1+x^2)/2`
Note. Divide first: x^5/(1+x^2)=x^3-x+x/(1+x^2).

0724 **uni-int-002** 積分・R6
Answer. `exp(x^2)*(x^2-1)/2`
Note. Let u=x^2.

0725 **uni-int-003** 積分・R6
Answer. `-(log(x)+1)/x`
Note. Integrate by parts with dv=x^{-2}dx.

0726 **uni-int-004** 積分・R6
Answer. `-atan(x)/x+log(x)-log(1+x^2)/2`
Note. After parts, decompose 1/(x(1+x^2)).

0727 **uni-int-005** 積分・R6
Answer. `(1+x^2)^(3/2)/3`
Note. Let u=1+x^2.

0728 **uni-int-006** 積分・R6
Answer. `sin(x)^6/6`

Note. Let u=sin x.

0729 uni-int-007 積分・R6

Answer. `tan(x)+tan(x)^3/3`

Note. Write sec^4 x=(1+tan^2 x)sec^2 x.

0730 uni-int-008 積分・R6

Answer. `atan((x+1)/2)/2`

Note. Complete the square.

0731 uni-int-009 積分・R6

Answer. `log(x^2+x+1)`

Note. The numerator is the derivative of the denominator.

0732 uni-int-010 積分・R6

Answer. `exp(x)*(sin(x)+cos(x))/2`

Note. Use the standard exponential-trig integral.

0733 uni-int-011 積分・R6

Answer. `x^3*log(x)/3-x^3/9`

Note. Let u=log x.

0734 uni-int-012 積分・R6

Answer. `(x^2+1)*atan(x)/2-x/2`

Note. Integrate by parts and rewrite x^2/(1+x^2).

0735 uni-int-013 積分・R6

Answer. `1/2-log(2)/2`

Note. Rewrite as x - x/(1+x^2).

0736 uni-int-014 積分・R6

Answer. `pi/4`

Note. Use x=tan theta or the standard Wallis value.

0737 uni-int-015 積分・R6

Answer. `1/4`

Note. Integrate by parts; the remaining rational integral cancels log terms.

0738 uni-int-016 積分・R6

Answer. `2/15`

Note. Use the beta integral.

0739 uni-int-017 積分・R6

Answer. `pi/4`

Note. Let x=sin theta.

0740 uni-int-018 積分・R6

Answer. `1/2`

Note. Let u=x^2.

0741 **uni-int-019** 積分・R6
Answer. `2`
Note. This equals Gamma(3).

0742 **uni-int-020** 積分・R6
Answer. `pi/(3*sqrt(3))`
Note. Complete the square.

0743 **uni-int-021** 積分・R6
Answer. `1/8`
Note. Integrate y first.

0744 **uni-int-022** 積分・R6
Answer. `pi/2`
Note. Switch to polar coordinates.

0745 **uni-int-023** 積分・R6
Answer. `6*pi`
Note. This is an ellipse with semiaxes 2 and 3.

0746 **uni-int-024** 積分・R6
Answer. `1/6`
Note. The tetrahedron volume is 1/6.

0747 **uni-int-025** 積分・R6
Answer. `pi/4`
Note. Use the arctangent antiderivative.

0748 **depth-int-001** 積分・R5
Answer. `log(1+x^4)/4`
Note. Let u=1+x^4.

0749 **depth-int-002** 積分・R6
Answer. `x^3/3-log(1+x^3)/3`
Note. Divide as x^2-x^2/(1+x^3).

0750 **depth-int-003** 積分・R5
Answer. `-1/sqrt(1+x^2)`
Note. Let u=1+x^2.

0751 **depth-int-004** 積分・R5
Answer. `log(x)^3/3`
Note. Let u=log x.

0752 **depth-int-005** 積分・R6
Answer. `exp(x)*(x*(sin(x)-cos(x))+cos(x))/2`
Note. Use parts with the standard integral of e^x sin x.

0753 **depth-int-006** 積分・R5
Answer. `exp(2*x)*(2*cos(3*x)+3*sin(3*x))/13`
Note. Use the exponential-trig formula.

0754 **depth-int-007** 積分・R6
Answer. `2*sqrt(x)*atan(sqrt(x))-log(1+x)`
Note. Let u=sqrt x, then integrate atan u.

0755 **depth-int-008** 積分・R5
Answer. `2*sqrt(log(x))`
Note. Let u=log x.

0756 **depth-int-009** 積分・R6
Answer. `x*(sin(log(x))-cos(log(x)))/2`
Note. Set x=e^u.

0757 **depth-int-010** 積分・R5
Answer. `2*log(1+sqrt(x))`
Note. Let u=sqrt x.

0758 **depth-int-011** 積分・R6
Answer. `atan(x)/2-x/(2*(1+x^2))`
Note. Rewrite as 1/(1+x^2)-1/(1+x^2)^2.

0759 **depth-int-012** 積分・R5
Answer. `2*(1+x^3)^(3/2)/9`
Note. Let u=1+x^3.

0760 **depth-int-013** 積分・R5
Answer. `log(1+tan(x))`
Note. Let u=1+tan x.

0761 **depth-int-014** 積分・R6
Answer. `x^2*log(1+x)/2-x^2/4+x/2-log(1+x)/2`
Note. After parts, divide x^2 by x+1.

0762 **depth-int-015** 積分・R6
Answer. `log(3)/2-pi/(3*sqrt(3))`
Note. Split x as half the derivative minus a constant.

0763 **depth-int-016** 積分・R5
Answer. `pi/2-1`
Note. Integrate by parts.

0764 **depth-int-017** 積分・R6
Answer. `pi^2/12`
Note. Expand log(1+x) and integrate termwise.

0765 **depth-int-018** 積分・R6
Answer. `log(2)/2-1/4`

Note. Let u=1+x^2.

0766 **depth-int-019** 積分 • R5
Answer. `1/2`
Note. Let u=1+x^2.

0767 **depth-int-020** 積分 • R6
Answer. `pi/12`
Note. Use partial fractions in x^2.

0768 **depth-int-021** 積分 • R5
Answer. `3*pi/8`
Note. Use power reduction or Wallis.

0769 **depth-int-022** 積分 • R5
Answer. `1/3`
Note. Integrate x first, then y.

0770 **depth-int-023** 積分 • R6
Answer. `3*pi/2`
Note. Use one-half integral of r^2 from 0 to 2pi.

0771 **depth-int-024** 積分 • R6
Answer. `4*pi`
Note. Use symmetry or polar coordinates.

0772 **depth-int-025** 積分 • R5
Answer. `1/3`
Note. The triangular region is symmetric in x and y.

0773 **depth-int-026** 積分 • R5
Answer. `1/2`
Note. Let u=x^2.

0774 **depth-int-027** 積分 • R5
Answer. `pi/2`
Note. This is arcsin x from 0 to 1.

0775 **depth-int-028** 積分 • R5
Answer. `-1/9`
Note. Use the parameter integral or parts.

0776 **burst-int-001** 積分 • R6
Answer. `8/81`
Note. Repeated IBP gives 4!/3^5.

0777 **burst-int-002** 積分 • R6
Answer. `1/2`
Note. Use the imaginary part of 2!/(1-i)^3.

0778 **burst-int-003** 積分・R6
Answer. `4/125`
Note. Use the real part of 2!/(2-i)^3.

0779 **burst-int-004** 積分・R5
Answer. `-1/36`
Note. Differentiate the beta integral or integrate by parts.

0780 **burst-int-005** 積分・R6
Answer. `2/125`
Note. Use 2!/(5^3).

0781 **burst-int-006** 積分・R6
Answer. `-2/27`
Note. Use -3!/(3^4).

0782 **burst-int-007** 積分・R5
Answer. `1/280`
Note. This is B(4,5).

0783 **burst-int-008** 積分・R6
Answer. `pi/8`
Note. This is B(3/2,3/2).

0784 **burst-int-009** 積分・R6
Answer. `3*sqrt(pi)/8`
Note. Set u=x^2 and get one half of Gamma(5/2).

0785 **burst-int-010** 積分・R5
Answer. `1`
Note. Set u=x^2 and get one half of Gamma(3).

0786 **burst-int-011** 積分・R6
Answer. `-pi*log(2)/2`
Note. Use Wallis' product or pair with log(cos x).

0787 **burst-int-012** 積分・R6
Answer. `pi*log(2)^2/2+pi^3/24`
Note. Differentiate the beta integral twice.

0788 **burst-int-013** 積分・R6
Answer. `35*pi^2/256`
Note. Symmetry gives pi/2 times the integral of sin^8 on [0,pi].

0789 **burst-int-014** 積分・R5
Answer. `63*pi/512`
Note. Apply the even Wallis reduction.

0790 **burst-int-015** 積分・R5
Answer. `256/693`
Note. Apply the odd Wallis reduction.

0791 **burst-int-016** 積分・R6
Answer. `3*pi/512`
Note. Use one half of B(7/2,5/2).

0792 **burst-int-017** 積分・R5
Answer. `3*pi/256`
Note. Use one half of B(5/2,5/2).

0793 **burst-int-018** 積分・R5
Answer. `1/24`
Note. Use one half of B(3,2).

0794 **burst-int-019** 積分・R6
Answer. `7*pi/512`
Note. Use one half of B(9/2,3/2).

0795 **burst-int-020** 積分・R5
Answer. `35*pi/128`
Note. Double the half-interval Wallis value.

0796 **burst-int-021** 積分・R5
Answer. `16/35`
Note. Apply the odd Wallis reduction.

0797 **burst-int-022** 積分・R5
Answer. `35*pi/256`
Note. Apply the even Wallis reduction.

0798 **burst-int-023** 積分・R6
Answer. `2/63`
Note. Use one half of B(2,7/2).

0799 **burst-int-024** 積分・R6
Answer. `5*pi/256`
Note. Use one half of B(3/2,7/2).

0800 **burst-int-025** 積分・R6
Answer. `pi^4/15`
Note. Expand 1/(e^x-1) and use Gamma(4) times zeta(4).

0801 **burst-int-026** 積分・R6
Answer. `pi^2/6`
Note. Expand the Planck kernel and sum 1/n^2.

0802 **burst-int-027** 積分・R6
Answer. `pi/(2*sqrt(2))`

Note. Use the beta integral or factor x^4+1.

0803 **burst-int-028** 積分 • R6
Answer. `0`
Note. Substitute x=1/u; the integral equals its negative.

0804 **burst-int-029** 積分 • R6
Answer. `pi/sqrt(2)`
Note. Use the standard x^(a-1)/(1+x^b) beta form.

0805 **burst-int-030** 積分 • R6
Answer. `-pi*pi/12`
Note. Expand 1/(1+x) and integrate termwise.

0806 **burst-int-031** 積分 • R5
Answer. `3*pi/16`
Note. Set u=2x and use the full-period symmetry.

0807 **burst-int-032** 積分 • R6
Answer. `5*pi/2048`
Note. Use one half of B(7/2,7/2).

0808 **burst-int-033** 積分 • R6
Answer. `231*pi/2048`
Note. Apply the even Wallis reduction through twelve powers.

0809 **burst-int-034** 積分 • R6
Answer. `1024/3003`
Note. Apply the odd Wallis reduction through thirteen powers.

0810 **burst-int-035** 積分 • R6
Answer. `1/210`
Note. Use one half of B(5,3).

0811 **burst-int-036** 積分 • R6
Answer. `exp(2*x)*(x^3/2-3*x^2/4+3*x/4-3/8)`
Note. Repeated IBP lowers the power of x each time.

0812 **burst-int-037** 積分 • R6
Answer. `-(x^2)*cos(3*x)/3+2*x*sin(3*x)/9+2*cos(3*x)/27`
Note. Integrate by parts twice.

0813 **burst-int-038** 積分 • R6
Answer. `x*(log(x)^3-3*log(x)^2+6*log(x)-6)`
Note. Repeated IBP produces the descending log polynomial.

0814 **burst-int-039** 積分 • R5
Answer. `x^5*log(x)/5-x^5/25`
Note. One IBP after integrating x^4.

0815 burst-int-040 積分・R6
Answer. `(x^4-1)*atan(x)/4-x^3/12+x/4`
Note. After parts, divide x^4 by 1+x^2.

0816 burst-int-041 積分・R6
Answer. `-exp(-x)*(x^3+3*x^2+6*x+6)`
Note. Repeated IBP gives the factorial polynomial.

0817 burst-int-042 積分・R6
Answer. `x^3*log(1+x)/3-x^3/9+x^2/6-x/3+log(1+x)/3`
Note. After parts, divide x^3 by 1+x.

0818 burst-rec-001 積分・R6
Answer. `-1/(n+1)^2`
Note. Differentiate 1/(n+1) with respect to n.

0819 burst-rec-002 積分・R6
Answer. `2/(n+1)^3`
Note. Differentiate 1/(n+1) twice.

0820 burst-rec-003 積分・R6
Answer. `2/((n+1)*(n+2)*(n+3))`
Note. This is B(n+1,3).

0821 burst-rec-004 積分・R6
Answer. `(n-1)/n`
Note. Integrate by parts to get the Wallis recurrence.

0822 burst-rec-005 積分・R6
Answer. `(2*n-1)/(2*n)`
Note. Apply the Wallis recurrence with exponent 2n.

0823 burst-rec-006 積分・R6
Answer. `(2*n)/(2*n+1)`
Note. Apply the Wallis recurrence with exponent 2n+1.

0824 burst-rec-007 積分・R6
Answer. `n`
Note. Gamma(n+1)=n Gamma(n).

0825 burst-rec-008 積分・R6
Answer. `-n`
Note. IBP gives J_n=e-nJ_{n-1}.

0826 burst-rec-009 積分・R5
Answer. `1/(n+1)+1/(n+2)`
Note. Split into two power integrals.

0827 **burst-rec-010** 積分・R5
Answer. `1/((n+1)*(n+2))`
Note. Subtract 1/(n+2) from 1/(n+1).

0828 **burst-boss-int-001** 積分・R6
Answer. `pi^3/8`
Note. Differentiate pi/2*csc(pi*a/2) twice at a=1.

0829 **burst-boss-int-002** 積分・R6
Answer. `-(pi^2)/(8*sqrt(2))`
Note. Differentiate pi/4*csc(pi*a/4) at a=1.

0830 **burst-boss-int-003** 積分・R6
Answer. `pi^2/(2*sqrt(2))`
Note. Use the derivative of pi/2*csc(pi*a/2) at a=3/2.

0831 **burst-boss-int-004** 積分・R6
Answer. `pi/3`
Note. Use the standard beta integral for x^(a-1)/(1+x^b).

0832 **burst-boss-int-005** 積分・R5
Answer. `pi/6`
Note. Use pi/6*csc(pi/2).

0833 **burst-boss-int-006** 積分・R5
Answer. `pi/3`
Note. Use pi/6*csc(pi/6).

0834 **burst-boss-int-007** 積分・R5
Answer. `pi/4`
Note. Use u=x^2, then arctan.

0835 **burst-boss-int-008** 積分・R6
Answer. `pi/(2*sqrt(2))`
Note. Use pi/4*csc(3pi/4).

0836 **burst-boss-int-009** 積分・R6
Answer. `2-pi^2/6`
Note. Expand log(1-x) and integrate x^n log x termwise.

0837 **burst-boss-int-010** 積分・R6
Answer. `-(pi^2)/6`
Note. Expand 1/(1-x) and sum -1/n^2.

0838 **burst-boss-int-011** 積分・R5
Answer. `24`
Note. Use (-1)^4*4! from the gamma integral.

0839 **burst-boss-int-012** 積分・R6
Answer. `3/128`

Note. Use 4!/4^5.

0840 **burst-boss-int-013** 積分 • R6
Answer. `-40/243`
Note. Use -5!/3^6.

0841 **burst-boss-int-014** 積分 • R6
Answer. `pi*log(2)^2/2-pi^3/48`
Note. Differentiate the beta integral with respect to both exponents.

0842 **burst-boss-int-015** 積分 • R6
Answer. `pi^3/8`
Note. Subtract the log-sine and log-cosine square identities.

0843 **burst-boss-int-016** 積分 • R6
Answer. `63*pi^2/512`
Note. Pair x with pi-x, then use the Wallis value for sin^10.

0844 **burst-boss-int-017** 積分 • R6
Answer. `35*pi/65536`
Note. Write it as 2^-8 sin^8(2x) and use Wallis.

0845 **burst-boss-int-018** 積分 • R6
Answer. `429*pi/4096`
Note. Use the even Wallis reduction through fourteen powers.

0846 **burst-boss-int-019** 積分 • R6
Answer. `2048/6435`
Note. Use the odd Wallis product through fifteen powers.

0847 **burst-boss-int-020** 積分 • R6
Answer. `5280/15625`
Note. Use the imaginary part of 5!/(2-i)^6.

0848 **burst-boss-anti-001** 積分 • R6
Answer. `exp(x)*(x^4-4*x^3+12*x^2-24*x+24)`
Note. Repeated IBP produces the alternating factorial polynomial.

0849 **burst-boss-anti-002** 積分 • R6
Answer. `-exp(-2*x)*(x^4/2+x^3+3*x^2/2+3*x/2+3/4)`
Note. Choose P so that d(-e^{-2x}P)/dx=x^4e^{-2x}.

0850 **burst-boss-anti-003** 積分 • R6
Answer. `-(x^3)*cos(2*x)/2+3*x^2*sin(2*x)/4+3*x*cos(2*x)/4-3*sin(2*x)/8`
Note. Integrate by parts until the power disappears.

0851 **burst-boss-anti-004** 積分 • R6
Answer. `x^3*sin(2*x)/2+3*x^2*cos(2*x)/4-3*x*sin(2*x)/4-3*cos(2*x)/8`
Note. Integrate by parts repeatedly and collect the trig terms.

0852 burst-boss-anti-005 積分 • R6
Answer. `x^5*(log(x)^2/5-2*log(x)/25+2/125)`
Note. IBP lowers the power of log x twice.

0853 burst-boss-anti-006 積分 • R6
Answer. `x*(log(x)^4-4*log(x)^3+12*log(x)^2-24*log(x)+24)`
Note. The descending log polynomial follows from repeated IBP.

0854 burst-boss-anti-007 積分 • R5
Answer. `x^3*atan(x)/3-x^2/6+log(1+x^2)/6`
Note. After IBP, divide x^3 by 1+x^2.

0855 burst-boss-anti-008 積分 • R6
Answer. `x^5*atan(x)/5-x^4/20+x^2/10-log(1+x^2)/10`
Note. After IBP, divide x^5 by 1+x^2.

0856 burst-boss-anti-009 積分 • R6
Answer. `x^3*log(1+x^2)/3-2*x^3/9+2*x/3-2*atan(x)/3`
Note. After IBP, reduce x^4/(1+x^2).

0857 burst-boss-anti-010 積分 • R6
Answer. `x^4*log(1+x^2)/4-x^4/8+x^2/4-log(1+x^2)/4`
Note. After IBP, reduce x^5/(1+x^2).

0858 burst-boss-param-001 積分 • R6
Answer. `1/(a^2+1)`
Note. Use the imaginary part of 1/(a-i).

0859 burst-boss-param-002 積分 • R5
Answer. `1/a^2`
Note. Differentiate the Laplace integral of e^{-ax}.

0860 burst-boss-param-003 積分 • R5
Answer. `pi/(2*sqrt(a))`
Note. Set u=sqrt(a)x.

0861 burst-boss-param-004 積分 • R6
Answer. `pi*log((1+sqrt(1+a))/2)`
Note. Differentiate in a, integrate the rational trig form, then use I(0)=0.

0862 burst-boss-param-005 積分 • R6
Answer. `pi/(2*sin(pi*(n+1)/2))`
Note. Use the beta form for x^(a-1)/(1+x^2) with a=n+1.

0863 burst-boss-param-006 積分 • R6
Answer. `-6/(n+1)^4`
Note. Differentiate 1/(n+1) three times.

0864 **burst-boss2-int-001** 積分・R6
Answer. `-(2*pi^2)/27`
Note. Differentiate the beta integral pi/3*csc(pi*a/3) at a=1.

0865 **burst-boss2-int-002** 積分・R6
Answer. `10*pi^3/(81*sqrt(3))`
Note. Differentiate pi/3*csc(pi*a/3) twice at a=1.

0866 **burst-boss2-int-003** 積分・R6
Answer. `pi/3`
Note. Use the beta form with a=3/2 and b=3.

0867 **burst-boss2-int-004** 積分・R6
Answer. `pi/sqrt(3)`
Note. Use the beta form with a=4/3 and b=2.

0868 **burst-boss2-int-005** 積分・R6
Answer. `pi*log(2)/2`
Note. Set x=tan t and integrate t cot t by parts.

0869 **burst-boss2-int-006** 積分・R6
Answer. `pi/2`
Note. Use the Fourier/Dirichlet kernel identity.

0870 **burst-boss2-int-007** 積分・R6
Answer. `pi/2`
Note. Integrate by parts and use the Dirichlet integral.

0871 **burst-boss2-int-008** 積分・R6
Answer. `-pi^4/15`
Note. Expand 1/(1-x) and sum zeta(4).

0872 **burst-boss2-int-009** 積分・R6
Answer. `-8*pi^6/63`
Note. Expand 1/(1-x) and sum zeta(6).

0873 **burst-boss2-int-010** 積分・R6
Answer. `-7*pi^4/120`
Note. Expand 1/(1+x) and use eta(4).

0874 **burst-boss2-int-011** 積分・R6
Answer. `-31*pi^6/252`
Note. Expand 1/(1+x) and use eta(6).

0875 **burst-boss2-int-012** 積分・R6
Answer. `pi^2/16+1/4`
Note. Write sin^2 x=(1-cos 2x)/2.

0876 **burst-boss2-int-013** 積分・R6
Answer. `pi/8`

Note. Use sin x cos x=sin 2x/2.

0877 **burst-boss2-int-014** 積分・R6
Answer. `pi^2-4`
Note. Integrate by parts twice.

0878 **burst-boss2-param-001** 積分・R6
Answer. `b/(a^2+b^2)`
Note. Use the imaginary part of 1/(a-bi).

0879 **burst-boss2-param-002** 積分・R6
Answer. `a/(a^2+b^2)`
Note. Use the real part of 1/(a-bi).

0880 **burst-boss2-param-003** 積分・R6
Answer. `2*a/(a^2+1)^2`
Note. Differentiate the Laplace sine transform with respect to a.

0881 **burst-boss2-param-004** 積分・R6
Answer. `(a^2-1)/(a^2+1)^2`
Note. Differentiate the Laplace cosine transform with respect to a.

0882 **burst-boss2-param-005** 積分・R6
Answer. `log(a+1)`
Note. Differentiate with respect to a, then use F(0)=0.

0883 **burst-boss2-param-006** 積分・R6
Answer. `-120/(a+1)^6`
Note. Differentiate 1/(a+1) five times.

0884 **burst-boss2-anti-001** 積分・R6
Answer. `exp(2*x)*(x^5/2-5*x^4/4+5*x^3/2-15*x^2/4+15*x/4-15/8)`
Note. Repeated IBP produces the factorial polynomial.

0885 **burst-boss2-anti-002** 積分・R6
Answer. `-(x^4)*cos(x)+4*x^3*sin(x)+12*x^2*cos(x)-24*x*sin(x)-24*cos(x)`
Note. Integrate by parts until the polynomial disappears.

0886 **burst-boss2-anti-003** 積分・R6
Answer. `x^4*sin(x)+4*x^3*cos(x)-12*x^2*sin(x)-24*x*cos(x)+24*sin(x)`
Note. Integrate by parts until the polynomial disappears.

0887 **burst-boss2-anti-004** 積分・R6
Answer. `x*(log(x)^5-5*log(x)^4+20*log(x)^3-60*log(x)^2+120*log(x)-120)`
Note. Repeated IBP lowers the log power.

0888 **burst-boss2-anti-005** 積分・R6
Answer. `x^6*atan(x)/6-x^5/30+x^3/18-x/6+atan(x)/6`
Note. After parts, divide x^6 by 1+x^2.

0889 burst-boss2-anti-006 積分・R6
Answer. `x^4/4-x^2/2+log(1+x^2)/2`
Note. Divide x^5 by 1+x^2 before integrating.

0890 burst-boss3-int-001 積分・R6
Answer. `1/27`
Note. Separate the two Gamma-type integrals.

0891 burst-boss3-int-002 積分・R6
Answer. `-1/54`
Note. Separate variables and use derivatives of 1/(a+1).

0892 burst-boss3-int-003 積分・R6
Answer. `8-pi^2/3`
Note. Expand the square and use the log-log beta derivative.

0893 burst-boss3-int-004 積分・R6
Answer. `pi^2/3`
Note. The cross term cancels the elementary log-square pieces.

0894 burst-boss3-int-005 積分・R6
Answer. `-pi^2*log(2)/2`
Note. Pair x with pi-x and use the Wallis log-sine integral.

0895 burst-boss3-int-006 積分・R6
Answer. `pi^3-6*pi`
Note. Integrate by parts until only the standard sine moment remains.

0896 burst-boss3-int-007 積分・R6
Answer. `pi^4-12*pi^2+48`
Note. Repeated IBP reduces the polynomial power.

0897 burst-boss3-int-008 積分・R6
Answer. `-714/28561`
Note. Take the real part of 3!/(2-3i)^4.

0898 burst-boss3-int-009 積分・R6
Answer. `-720/28561`
Note. Take the imaginary part of 3!/(2-3i)^4.

0899 burst-boss3-int-010 積分・R6
Answer. `984/3125`
Note. Take the real part of 4!/(1-2i)^5.

0900 burst-boss3-int-011 積分・R6
Answer. `-912/3125`
Note. Take the imaginary part of 4!/(1-2i)^5.

0901 **burst-boss3-int-012** 積分 • R6
Answer. `31*pi^6/252`
Note. Use the eta factor (1-2^{-5}) Gamma(6) zeta(6).

0902 **burst-boss3-int-013** 積分 • R6
Answer. `7*pi^4/120`
Note. Use the eta factor (1-2^{-3}) Gamma(4) zeta(4).

0903 **burst-boss3-param-001** 積分 • R6
Answer. `18*(a^2-3)/(a^2+9)^3`
Note. Differentiate 3/(a^2+9) twice.

0904 **burst-boss3-param-002** 積分 • R6
Answer. `2*(a^3-3*a)/(a^2+1)^3`
Note. Take the real part of 2!/(a-i)^3.

0905 **burst-boss3-param-003** 積分 • R6
Answer. `24*a*(a^2-1)/(a^2+1)^4`
Note. Take the imaginary part of 3!/(a-i)^4.

0906 **burst-boss3-param-004** 積分 • R6
Answer. `24/(a+1)^5`
Note. Differentiate 1/(a+1) four times.

0907 **burst-boss3-param-005** 積分 • R6
Answer. `720/(a+1)^7`
Note. Differentiate 1/(a+1) six times.

0908 **burst-boss3-param-006** 積分 • R6
Answer. `2/(a+1)^3`
Note. The integral is log(a+1).

0909 **burst-boss3-anti-001** 積分 • R6
Answer. `-exp(-3*x)*(x^6/3+2*x^5/3+10*x^4/9+40*x^3/27+40*x^2/27+80*x/81+80/243)`
Note. Repeated IBP produces the descending factorial polynomial.

0910 **burst-boss3-anti-002** 積分 • R6
Answer. `-x^5*cos(2*x)/2+5*x^4*sin(2*x)/4+5*x^3*cos(2*x)/2-15*x^2*sin(2*x)/4-15*x*cos(2*x)/4+15*sin(2*x)/8`
Note. Repeated IBP alternates sine and cosine.

0911 **burst-boss3-anti-003** 積分 • R6
Answer. `x^5*sin(2*x)/2+5*x^4*cos(2*x)/4-5*x^3*sin(2*x)/2-15*x^2*cos(2*x)/4+15*x*sin(2*x)/4+15*cos(2*x)/8`
Note. Repeated IBP alternates sine and cosine.

0912 **burst-boss3-anti-004** 積分 • R6
Answer. `x^7*atan(x)/7-x^6/42+x^4/28-x^2/14+log(1+x^2)/14`
Note. After parts, divide x^7 by 1+x^2.

0913 **burst-boss3-anti-005** 積分 • R6
Answer. `x^7*log(1+x)/7-x^7/49+x^6/42-x^5/35+x^4/28-x^3/21+x^2/14-x/7+log(1+x)/7`

Note. After parts, divide x^7 by 1+x.

0914 **burst-boss3-anti-006** 積分・R6
Answer. `x^6/6-x^4/4+x^2/2-log(1+x^2)/2`
Note. Divide x^7 by 1+x^2 before integrating.

0915 **ser-001** 級數・R1
Answer. 收斂
Note. p-series，p=2>1，所以收斂。

0916 **ser-002** 級數・R1
Answer. 發散
Note. 調和級數發散。

0917 **ser-003** 級數・R1
Answer. 3/2
Note. 等比級數，和為 1/(1-1/3)=3/2。

0918 **ser-004** 級數・R1
Answer. 條件收斂
Note. 交錯調和級數條件收斂。

0919 **ser-005** 級數・R2
Answer. 發散
Note. 項行為像 1/n，和調和級數比較會發散。

0920 **ser-006** 級數・R2
Answer. 收斂
Note. 比值測試極限趨近 e^{-1}<1。

0921 **ser-007** 級數・R2
Answer. `exp(3)-1`
Note. e^3 的展開式少了 n=0 項。

0922 **ser-008** 級數・R2
Answer. 0
Note. n! 成長太快，半徑為 0。

0923 **ser-009** 級數・R3
Answer. 1
Note. 與 -ln(1-x) 的冪級數相同，半徑 1。

0924 **ser-010** 級數・R3
Answer. 條件收斂
Note. 交錯級數收斂，但絕對值級數 p=1/2 發散。

0925 **ser-011** 級數・R3
Answer. 9/4
Note. 拆成 sum (2/3)^n 加 sum n(1/3)^n，分別是 2 和 3/4。

0926 ser-012 級數・R2
Answer. 1
Note. 裂項：1/(n(n+1))=1/n-1/(n+1)。

0927 ser-013 級數・R2
Answer. 發散
Note. 通項不趨近 0，所以級數發散。

0928 ser-014 級數・R3
Answer. 2
Note. 公式 sum n r^n = r/(1-r)^2，取 r=1/2。

0929 ser-015 級數・R3
Answer. 6
Note. 公式 sum n^2 r^n = r(1+r)/(1-r)^3，取 r=1/2。

0930 ser-016 級數・R4
Answer. 收斂
Note. 這是中央二項係數倒數量級，指數衰減所以收斂。

0931 ser-017 級數・R3
Answer. 3
Note. n^2 不影響收斂半徑，核心比例是 |x|/3。

0932 ser-018 級數・R2
Answer. 絕對收斂
Note. 取絕對值後是 p-series，p=2>1。

0933 ser-019 級數・R2
Answer. 3/4
Note. 裂項成 1/2(1/n-1/(n+2))。

0934 ser-020 級數・R3
Answer. 發散
Note. 用積分測試，int (ln x)/x dx 發散。

0935 td-ser-001 級數・R6
Answer. 33/40
Note. 令 (1+x)^x=exp(x log(1+x))，展開到 x^6。

0936 td-ser-002 級數・R6
Answer. 13/12
Note. arcsin x=x+x^3/6+3x^5/40+...，再取 exp。

0937 td-ser-003 級數・R6
Answer. 2*log(2)
Note. 用 int_0^{pi/2} sin^{2n}x dx 表示雙階乘比，再交換求和。

0938 td-ser-004 級數・R4
Answer. 3/2
Note. 公式 sum n^2 r^n = r(1+r)/(1-r)^3，取 r=1/3。

0939 ser-021 級數・R2
Answer. 1
Note. 部分分式後相消，部分和為 1-1/(N+1)，極限為 1。

0940 ser-022 級數・R2
Answer. 2
Note. sum n r^n = r/(1-r)^2，代 r=1/2 得 2。

0941 ser-023 級數・R2
Answer. 條件收斂
Note. 交錯調和級數收斂，但絕對值級數 sum 1/n 發散，因此條件收斂。

0942 ser-024 級數・R2
Answer. 發散
Note. 由積分判別，int_2^infty dx/(x log x)=infty，所以級數發散。

0943 ser-025 級數・R3
Answer. 收斂
Note. 比值為 (n/(n+1))^n，極限為 e^{-1}<1，因此收斂。

0944 ser-026 級數・R2
Answer. 3
Note. 根值判別給 |x|/3<1，因此收斂半徑為 3。

0945 ser-027 級數・R2
Answer. 2/3
Note. 係數為 2^4/4! = 16/24 = 2/3。

0946 ser-028 級數・R1
Answer. 收斂
Note. p 級數 sum 1/n^p 在 p>1 時收斂，因此此級數收斂。

0947 ser-029 級數・R2
Answer. `exp(3)-1`
Note. e^3 的級數含 n=0 項 1，所以原級數為 e^3-1。

0948 ser-030 級數・R2
Answer. 5
Note. 根值判別得到 |x-2|/5<1，因此收斂半徑為 5。

0949 ser-031 級數・R3
Answer. 發散
Note. Since p=1/2<=1, the p-series diverges.

0950 ser-032 級數・R3
Answer. 發散

Note. The terms tend to e^{-1}, not 0, so the series diverges.

0951 **ser-033** 級數・R3
Answer. 條件收斂
Note. The alternating series converges, but the absolute series diverges, so it is conditionally convergent.

0952 **ser-034** 級數・R3
Answer. `0`
Note. Only x=0 converges, so the radius is 0.

0953 **ser-035** 級數・R3
Answer. `Infinity`
Note. The series converges for all x, so the radius is Infinity.

0954 **ser-036** 級數・R3
Answer. 收斂
Note. 0 <= 2^n/(3^n+n) <= (2/3)^n, so the series converges by comparison.

0955 **ser-037** 級數・R3
Answer. 收斂
Note. The integral test shows convergence because p=2>1.

0956 **ser-038** 級數・R2
Answer. `1/3`
Note. The coefficient of x^3 in log(1+x) is 1/3.

0957 **ser-039** 級數・R3
Answer. `1/2`
Note. The radius condition is |2x|<1, so R=1/2.

0958 **ser-040** 級數・R3
Answer. `6`
Note. For r=1/2, r(1+r)/(1-r)^3=(1/2)(3/2)/(1/2)^3=6.

0959 **gap-ser-root-001** 級數・R3
Answer. `1/3`
Note. The radius is 1/3.

0960 **gap-ser-root-002** 級數・R3
Answer. `1`
Note. The root test gives |x|<1, so R=1.

0961 **gap-ser-lc-001** 級數・R3
Answer. `divergent`
Note. Limit comparison with 1/n gives a nonzero constant, so the series diverges.

0962 **gap-ser-lc-002** 級數・R3
Answer. `convergent`
Note. Limit comparison with 1/n^2 shows convergence.

0963 gap-ser-cond-001 級數 • R3
Answer. `conditional`
Note. It converges by the alternating test but not absolutely.

0964 gap-ser-end-001 級數 • R4
Answer. `conditional`
Note. The endpoint x=-1 is conditionally convergent.

0965 gap-ser-end-002 級數 • R4
Answer. `divergent`
Note. At x=1, the series is the divergent harmonic series.

0966 gap-ser-taylor-001 級數 • R3
Answer. `1/120`
Note. The coefficient is 1/5!=1/120.

0967 gap-ser-taylor-002 級數 • R3
Answer. `1-x^2/2+x^4/24`
Note. The polynomial is 1-x^2/2+x^4/24.

0968 gap-tech-005 級數 • R3
Answer. `roottest`
Note. Use the root test.

0969 mob-tech-007 級數 • R2
Answer. `ratiotest`
Note. Use ratio test.

0970 mob-tech-008 級數 • R2
Answer. `roottest`
Note. Use root test.

0971 mob-tech-019 級數 • R2
Answer. `endpointanalysis`
Note. Use endpoint analysis.

0972 mob-trap-004 級數 • R2
Answer. `conditional`
Note. conditional

0973 mob-trap-005 級數 • R2
Answer. `checkendpoints`
Note. check endpoints

0974 mob-conv-001 級數 • R3
Answer. `divergent`
Note. divergent.

0975 **mob-conv-002** 級數・R3
Answer. `convergent`
Note. convergent.

0976 **mob-conv-003** 級數・R3
Answer. `conditional`
Note. conditional.

0977 **mob-conv-004** 級數・R3
Answer. `absolute`
Note. absolute.

0978 **mob-conv-005** 級數・R3
Answer. `convergent`
Note. convergent.

0979 **mob-conv-006** 級數・R3
Answer. `convergent`
Note. convergent.

0980 **mob-conv-007** 級數・R3
Answer. `divergent`
Note. divergent.

0981 **mob-conv-008** 級數・R3
Answer. `divergent`
Note. divergent.

0982 **mob-conv-009** 級數・R3
Answer. `convergent`
Note. convergent.

0983 **mob-conv-010** 級數・R3
Answer. `divergent`
Note. divergent.

0984 **rel-basic-019** 級數・R1
Answer. `2`
Note. Use 1/(1-r).

0985 **rel-basic-022** 級數・R1
Answer. `1/3`
Note. This is geometric with ratio 3x.

0986 **rel-basic-024** 級數・R1
Answer. `1/2`
Note. Use r/(1-r).

0987 **rel-basic-020** 級數・R1
Answer. `convergent`

Note. This is a p-series with p=2>1.

0988 **rel-basic-021** 級數・R1
Answer. `divergent`
Note. This is the harmonic series.

0989 **rel-basic-023** 級數・R2
Answer. `conditional`
Note. The alternating harmonic series converges conditionally.

0990 **rel-adv-017** 級數・R3
Answer. `1/6`
Note. Use e^{x^2}=sum x^{2k}/k!.

0991 **rel-adv-019** 級數・R3
Answer. `1`
Note. Polynomial factors do not change the radius from 1.

0992 **rel-adv-015** 級數・R3
Answer. `convergent`
Note. The root limit is 1/3<1.

0993 **rel-adv-016** 級數・R3
Answer. `conditional`
Note. It converges by alternating test but not absolutely.

0994 **rel-adv-018** 級數・R3
Answer. `divergent`
Note. The terms behave like 3/n.

0995 **rel-adv-020** 級數・R3
Answer. `conditional`
Note. At x=-1 this is the alternating harmonic series.

0996 **rel-boss-013** 級數・R5
Answer. `1/4`
Note. Partial fractions telescope.

0997 **rel-boss-018** 級數・R5
Answer. `0`
Note. The ratio grows without bound unless x=0.

0998 **rel-hard-ser-001** 級數・R5
Answer. `1/4`
Note. The central binomial coefficient grows like 4^n.

0999 **rel-hard-ser-002** 級數・R5
Answer. `4`
Note. This is the reciprocal central binomial scale.

1000 rel-hard-ser-003 級數・R5
Answer. `1/24`
Note. Use cos u=1-u^2/2+u^4/24-... with u=x^2.

1001 rel-hard-ser-004 級數・R5
Answer. `1/120`
Note. Use sin u=u-u^3/6+u^5/120-... with u=x^2.

1002 rel-hard-ser-005 級數・R5
Answer. `1/18`
Note. Write it as a telescoping difference of triple products.

1003 rel-hard-ser-006 級數・R5
Answer. `6`
Note. Use the standard generating function for sum n^2 r^n.

1004 rel-hard-ser-007 級數・R5
Answer. `pi^2/12`
Note. Use eta(2)=(1-2^{1-2})zeta(2).

1005 rel-hard-ser-008 級數・R5
Answer. `1/2`
Note. Use 1/(4n^2-1)=1/2(1/(2n-1)-1/(2n+1)).

1006 rel-hard-ser-012 級數・R5
Answer. `1/e`
Note. The nth root of n^n/n! tends to e.

1007 rel-hard-ser-009 級數・R5
Answer. `convergent`
Note. The nth root behaves like 1/e.

1008 rel-hard-ser-010 級數・R5
Answer. `divergent`
Note. The terms do not tend to 0.

1009 rel-hard-ser-011 級數・R5
Answer. `conditional`
Note. Alternating test gives convergence; p=1/2 gives no absolute convergence.

1010 hc-spec-007 級數・R6
Answer. `1/96`
Note. Write it as a telescoping difference of four-factor denominators.

1011 hc-spec-008 級數・R6
Answer. `26`
Note. Use the generating function for sum n^3 r^n at r=1/2.

1012 hc-spec-009 級數 · R6
Answer. 1/27
Note. The coefficient ratio tends to 27.

1013 hc-spec-010 級數 · R6
Answer. 1/147456
Note. J_0(x)=sum (-1)^k (x^2/4)^k/(k!)^2; use k=4.

1014 exam-ser-001 級數 · R5
Answer. 3
Note. The dominant ratio is |x|/3.

1015 exam-ser-002 級數 · R5
Answer. e
Note. Use (n!/n^n)^{1/n}->e^{-1}.

1016 exam-ser-003 級數 · R5
Answer. 1
Note. The coefficient 1/n has nth root 1.

1017 exam-ser-004 級數 · R5
Answer. 1
Note. Use 1/(n(n+1))=1/n-1/(n+1).

1018 exam-ser-005 級數 · R5
Answer. 2
Note. Use sum n r^n=r/(1-r)^2 with r=1/2.

1019 exam-ser-006 級數 · R5
Answer. 6
Note. Use sum n^2 r^n=r(1+r)/(1-r)^3.

1020 exam-ser-007 級數 · R5
Answer. cos(1)
Note. This is the Taylor series for cos 1.

1021 exam-ser-008 級數 · R5
Answer. (exp(1)-exp(-1))/2
Note. This is sinh 1.

1022 exam-ser-009 級數 · R5
Answer. 4/15
Note. The coefficient is 2^5/5!.

1023 exam-ser-010 級數 · R5
Answer. -1/6
Note. Use sin z=z-z^3/6+... with z=x^2.

1024 exam-ser-011 級數 · R5
Answer. -1/4

Note. The x^n coefficient is (-1)^{n+1}/n.

1025 exam-ser-012 級數・R5
Answer. `3/2`
Note. Geometric sum 1/(1-1/3).

1026 exam-ser-013 級數・R5
Answer. `8`
Note. This is sum n^2/2^n plus sum n/2^n.

1027 exam-ser-014 級數・R5
Answer. `1/4`
Note. The nth root behaves like 4.

1028 exam-ser-015 級數・R5
Answer. `log(2)`
Note. This is the Taylor value log(1+1).

1029 exam-ser-016 級數・R5
Answer. `pi^2/6`
Note. Euler's Basel sum.

1030 exam-ser-017 級數・R5
Answer. `exp(2)`
Note. This is e^2.

1031 exam-ser-018 級數・R5
Answer. `35`
Note. Choose 3 factors of x from 7.

1032 exam-ser-019 級數・R5
Answer. `16`
Note. The coefficient is 2^4.

1033 exam-ser-020 級數・R5
Answer. `3/4`
Note. Use r/(1-r)^2 with r=1/3.

1034 uni-ser-001 級數・R6
Answer. `4/9`
Note. Use sum n r^n=r/(1-r)^2.

1035 uni-ser-002 級數・R6
Answer. `3/2`
Note. Use sum n^2 r^n=r(1+r)/(1-r)^3.

1036 uni-ser-003 級數・R6
Answer. `3/4`
Note. Use 1/(n(n+2))=(1/2)(1/n-1/(n+2)).

1037 uni-ser-004 級數・R6
Answer. 0
Note. sin(x^2) only has powers x^{4k+2}.

1038 uni-ser-005 級數・R6
Answer. 0
Note. Use Re((1+i)^6/6!).

1039 uni-ser-006 級數・R6
Answer. 1/3
Note. This is the x^3 coefficient of log(1+x).

1040 uni-ser-007 級數・R6
Answer. 5
Note. The nth root of n^2/5^n is 1/5.

1041 uni-ser-008 級數・R6
Answer. 4
Note. Use the central binomial coefficient growth.

1042 uni-ser-009 級數・R6
Answer. 1/e
Note. Stirling gives nth root asymptotic to e.

1043 uni-ser-010 級數・R6
Answer. pi/4
Note. This is arctan(1).

1044 uni-ser-011 級數・R6
Answer. (exp(1)+exp(-1))/2
Note. This is cosh(1).

1045 uni-ser-012 級數・R6
Answer. 4
Note. Use the second derivative of the geometric series.

1046 uni-ser-013 級數・R6
Answer. 15
Note. The coefficient is C(6,2).

1047 uni-ser-014 級數・R6
Answer. 1/3
Note. The nth root of 3^n/n^2 is 3.

1048 uni-ser-015 級數・R6
Answer. 2/9
Note. Use r/(1-r)^2 with r=-1/2 and adjust the sign.

1049 uni-ser-016 級數 • R6
Answer. `-4/315`
Note. Use sin x cos x=(1/2)sin(2x).

1050 uni-ser-017 級數 • R6
Answer. `26`
Note. Use the standard cubic power-series sum at r=1/2.

1051 uni-ser-018 級數 • R6
Answer. `210`
Note. This is C(10,6).

1052 uni-ser-019 級數 • R6
Answer. `1`
Note. The nth root of 1/sqrt(n) tends to 1.

1053 uni-ser-020 級數 • R6
Answer. `pi^2/6`
Note. Euler's Basel sum.

1054 depth-ser-001 級數 • R5
Answer. `9/4`
Note. Use the sums for n r^n and n^2 r^n.

1055 depth-ser-002 級數 • R5
Answer. `20/27`
Note. Use r(1+r)/(1-r)^3.

1056 depth-ser-003 級數 • R5
Answer. `1/2`
Note. Decompose as one-half of consecutive odd reciprocals.

1057 depth-ser-004 級數 • R5
Answer. `243/560`
Note. The coefficient is 3^7/7!.

1058 depth-ser-005 級數 • R5
Answer. `1/24`
Note. Use cos z with z=x^2.

1059 depth-ser-006 級數 • R5
Answer. `-1/5040`
Note. Divide the sine series by x.

1060 depth-ser-007 級數 • R5
Answer. `(exp(1)+exp(-1))/2`
Note. This is cosh(1).

1061 depth-ser-008 級數 • R5
Answer. `1/4`

Note. It is one third of sum n/3^n.

1062 **depth-ser-009** 級數・R6
Answer. `1/4`
Note. Central binomial coefficients grow like 4^n.

1063 **depth-ser-010** 級數・R6
Answer. `4/3`
Note. Use the reciprocal central binomial growth.

1064 **depth-ser-011** 級數・R5
Answer. `192`
Note. The coefficient is (n+1)2^n with n=5.

1065 **depth-ser-012** 級數・R5
Answer. `3`
Note. This is half of sum n^2/2^n.

1066 **depth-ser-013** 級數・R5
Answer. `cos(1)`
Note. This is the cosine series at 1.

1067 **depth-ser-014** 級數・R5
Answer. `1/4`
Note. Use a telescoping split.

1068 **depth-ser-015** 級數・R6
Answer. `1/3`
Note. Convolve the two Taylor series.

1069 **depth-ser-016** 級數・R6
Answer. `33/8`
Note. Use the cubic power-series sum.

1070 **depth-ser-017** 級數・R5
Answer. `5`
Note. Polynomial factors do not affect the radius.

1071 **depth-ser-018** 級數・R5
Answer. `sin(1)`
Note. This is the sine series at 1.

1072 **depth-ser-019** 級數・R6
Answer. `11/12`
Note. Square the log series through x^4.

1073 **depth-ser-020** 級數・R5
Answer. `1/2`
Note. The nth root of 2^n/n^2 is 2.

1074 burst-boss-ser-001 級數 • R5
Answer. 26
Note. Differentiate the geometric series three times.

1075 burst-boss-ser-002 級數 • R5
Answer. 3/2
Note. Use r(1+r)/(1-r)^3 with r=1/3.

1076 burst-boss-ser-003 級數 • R6
Answer. `2*log(2)`
Note. Use sum H_n x^n=-log(1-x)/(1-x).

1077 burst-boss-ser-004 級數 • R6
Answer. `pi^2/12`
Note. Use the alternating zeta value at 2.

1078 burst-boss-ser-005 級數 • R6
Answer. 137/180
Note. The coefficient is 2H_5/6.

1079 burst-boss-ser-006 級數 • R6
Answer. 1553/80
Note. Collect the x^6 coefficient of e^{3x}/(1-x).

1080 burst-boss2-ser-001 級數 • R6
Answer. `pi^4/90`
Note. Use the Basel-type value zeta(4).

1081 burst-boss2-ser-002 級數 • R6
Answer. `pi^4/96`
Note. Remove the even fourth-power terms from zeta(4).

1082 burst-boss2-ser-003 級數 • R6
Answer. `7*pi^4/720`
Note. Use eta(4)=(1-2^{-3})zeta(4).

1083 burst-boss2-ser-004 級數 • R6
Answer. `31*pi^6/30240`
Note. Use eta(6)=(1-2^{-5})zeta(6).

1084 burst-boss2-ser-005 級數 • R6
Answer. 15
Note. Apply the fourth derivative of the geometric series at r=1/3.

1085 burst-boss2-ser-006 級數 • R6
Answer. 273/4
Note. Apply the fifth derivative of the geometric series at r=1/3.

1086 **burst-boss2-ser-007** 級數・R6
Answer. `3*log(3/2)/2`
Note. Use sum H_n x^n=-log(1-x)/(1-x).

1087 **burst-boss2-ser-008** 級數・R6
Answer. `4*log(4/3)/3`
Note. Use sum H_n x^n=-log(1-x)/(1-x).

1088 **burst-boss3-ser-001** 級數・R6
Answer. `9366`
Note. Differentiate the geometric series six times.

1089 **burst-boss3-ser-002** 級數・R6
Answer. `94586`
Note. Differentiate the geometric series seven times.

1090 **burst-boss3-ser-003** 級數・R6
Answer. `4108/243`
Note. Use the Eulerian polynomial for sum n^5 r^n at r=1/4.

1091 **burst-boss3-ser-004** 級數・R6
Answer. `1491/4`
Note. Use the Eulerian polynomial for sum n^6 r^n at r=1/3.

1092 **burst-boss3-ser-005** 級數・R6
Answer. `pi^2/12`
Note. Integrate the generating function for H_n.

1093 **burst-boss3-ser-006** 級數・R6
Answer. `275295799/775975200`
Note. Use [x^n](log(1+x))^2=(-1)^n2H_{n-1}/n.

1094 **burst-boss3-ser-007** 級數・R6
Answer. `111165691613818/97692469875`
Note. Convolve e^{2x} with the binomial series.

1095 **burst-boss3-ser-008** 級數・R6
Answer. `1094053506107/212837625`
Note. Convolve e^{2x} with the binomial series.
